

Mathematics Stage 4

Linear Relationships

Book 1 Plot linear relationships on the Cartesian plane
Solve linear equations using graphical techniques

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Feedback:
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Syllabus Content

MA4-LIN-C-01 creates and displays number patterns and finds graphical solutions to problems involving linear relationships

Plot and identify points on the Cartesian plane

- Plot and label points on the Cartesian plane of given coordinates, including those with coordinates that are not whole numbers
- Identify and record the coordinates of given points on the Cartesian plane, including those with coordinates that are not whole numbers

Plot linear relationships on the Cartesian plane

- Construct a geometric pattern and record the results in a table of values
- Represent a given number pattern (including decreasing patterns) using a table of values
- Describe a number pattern in words and generate an equation using algebraic symbols
- Apply an equation generated from a pattern to calculate the corresponding value for a smaller or larger number
- Recognise that a linear relationship can be represented by a number pattern, an equation (or a rule using algebraic symbols), a table of values, a set of pairs of coordinates and a line graphed on a Cartesian plane, and move flexibly between these representations
- Explain that there are an infinite number of ordered pairs that satisfy a given linear relationship by extending a line joining a set of points on the Cartesian plane
- Compare similarities and differences of multiple straight-line graphs on the same set of axes using graphing applications
- Describe linear relationships in real-life contexts and solve related problems

Solve linear equations using graphical techniques

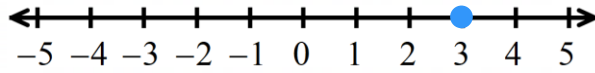
- Recognise that each point on the graph of a linear relationship satisfies the equation of a line
- Apply graphs of linear relationships to solve a corresponding linear equation using graphing applications
- Graph 2 intersecting lines on the same set of axes and identify the point of intersection using either graphing applications or a table of values
- Verify that the point of intersection satisfies the equations of both lines

Points on a Cartesian Plane

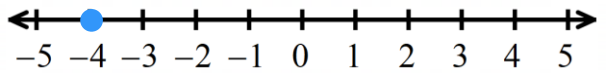
Review

☐ Read values from a number line

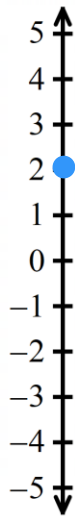
a



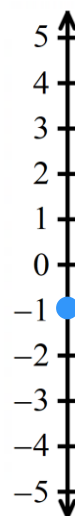
b



c



d



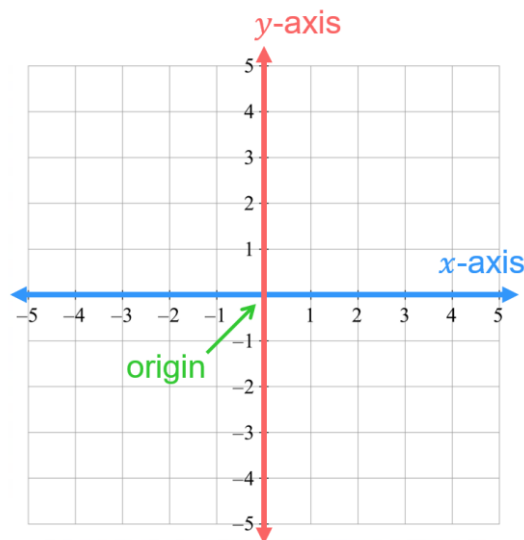
i Cartesian Plane

A **Cartesian plane**, or number plane, is two number lines at right angles to each other.

The **x-axis** is the **horizontal (left to right)** number line.

The **y-axis** is the **vertical (up and down)** number line.

The **origin (0,0)** is where the *x*-axis and *y*-axis meet.



i Coordinates

A **coordinate** (x, y) tells us the position on a Cartesian plane compared to the origin.

The first number is the *x*-coordinate. It is how much to move left or right from the origin.

The second number is the *y*-coordinate. It is how much to move up or down from the origin.

$$\begin{array}{c} (x, y) \\ \uparrow \quad \uparrow \\ \text{Left / Right} \quad \text{Down / Up} \end{array}$$

? Interpret Coordinates

a $(2, 4)$ →↑	b $(2, -4)$	c $(-4, 2)$	d $(-4, -2)$
e $(-3, 17)$	f $(0, 11)$	g $(1.5, -17)$	h $(-15, -12)$
i $\left(-26, \frac{4}{3}\right)$	j $\left(\frac{1}{2}, 16\right)$	k $\left(-\frac{3}{7}, 5\right)$	l $(-2, 0)$

i Plotting Points on a Cartesian Plane

1. Start at the origin (0,0)
2. Move along x .
3. Move along y .
4. Draw a point $\bullet$ where you end up.

(x, y)
Left / Right Down / Up

? Plot points on a Cartesian plane

a Plot these points on the plane below.

$A(1,3)$

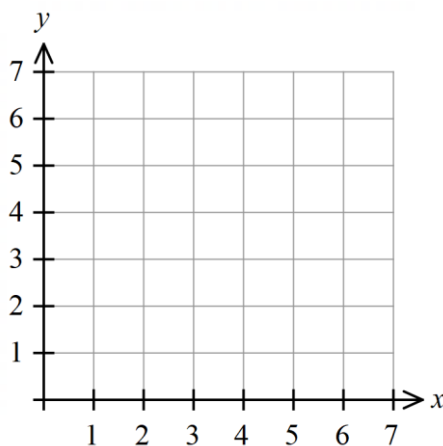
$B(1,4)$

$C(1,6)$

$D(6,1)$

$E(6,7)$

$F(7,6)$



b Plot these points on the plane below.

$A(1,3)$

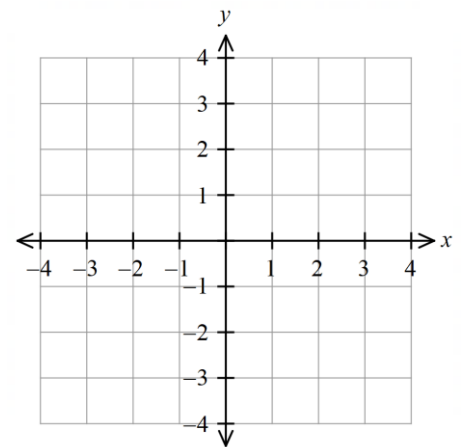
$B(-1,3)$

$C(-1, -3)$

$D(-2, -3)$

$E(-4, -3)$

$F(4, -3)$

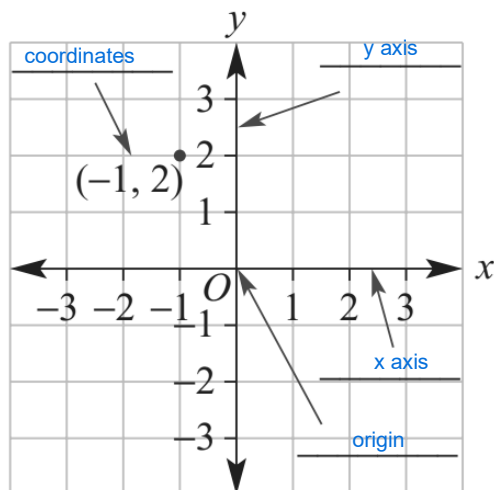


💡 Key Ideas

1. A Cartesian plane is a number line and a number line.
2. The horizontal number line is the axis. The vertical number line is the axis.
3. We can label a position on a Cartesian plane using
4. The origin (.....,) is where the two number lines
5. A coordinate is written as (x, y) ; it tells us the position of a point compared to the
6. The first number is called the x -coordinate. It tells us how much to move h....., the second number is called the it tells us how much to move

1 For the Cartesian plane shown, label these parts:

- x axis
- Origin
- y axis
- Coordinates



2

a To plot (3, 4) you would start at the origin, move 3 units and 4 units

right, up

b To plot (2, -1) you would start at the, move 2 units and 1 unit

origin, right, down

c To plot (-3, 2) you would start at the, move 3 units and 2 units

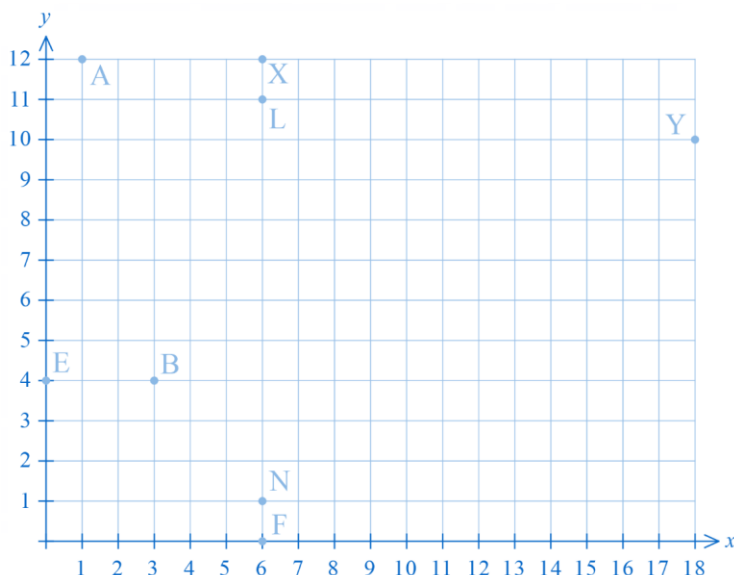
origin, left, up

d To plot (-1, -4) you would start at the, move 1 unit, and 4 units

origin, left, down

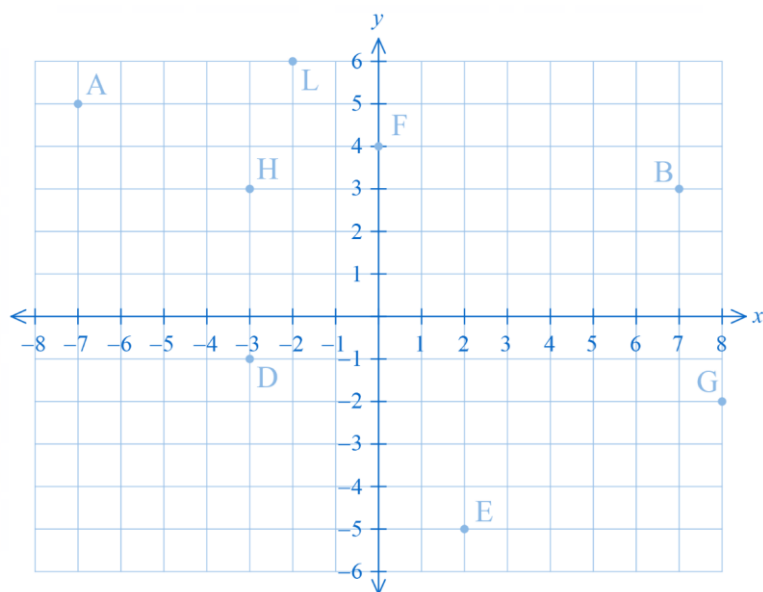
3 Plot these points on the Cartesian plane.

- $\rightarrow \uparrow$
 $A(1, 12)$
- $\rightarrow \uparrow$
 $B(3, 4)$
- $\rightarrow \uparrow$
 $Y(18, 10)$
- $\rightarrow \uparrow$
 $X(6, 12)$
- $\rightarrow \uparrow$
 $L(6, 11)$
- $\rightarrow \uparrow$
 $N(6, 1)$
- $\rightarrow \uparrow$
 $E(0, 4)$
- $\rightarrow \uparrow$
 $F(6, 0)$

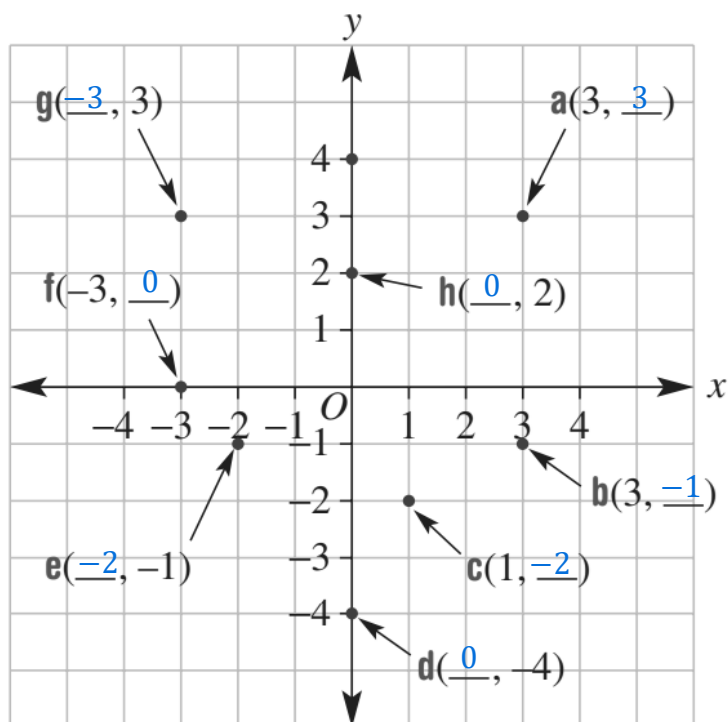


4 Plot these points on the Cartesian plane

- $A(-7, 5)$
- $H(-3, 3)$
- $G(8, -2)$
- $E(2, -5)$
- $L(-2, 6)$
- $F(0, 4)$
- $B(7, 3)$
- $D(-3, -1)$

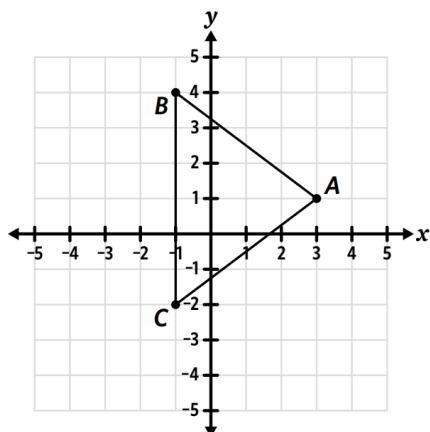


5 Write the missing number for each of these coordinates.



6 Identify the coordinates of these points.

a

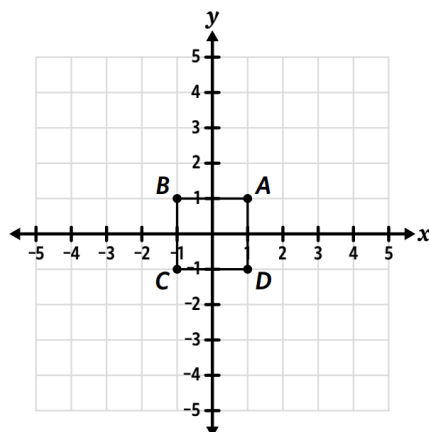


$$A \left(3, \boxed{1} \right)$$

$$B \left(\boxed{-1}, 4 \right)$$

$$C \left(\boxed{-1}, \boxed{-2} \right)$$

b



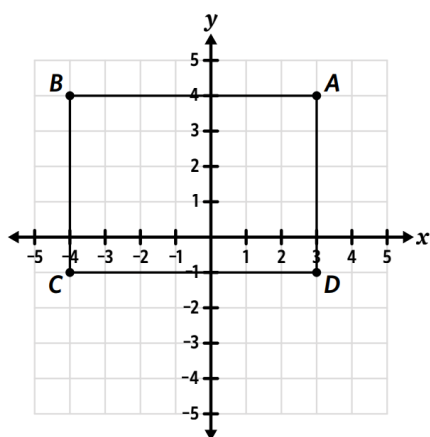
$$A \left(\boxed{1}, \boxed{1} \right)$$

$$B \left(\boxed{-1}, \boxed{1} \right)$$

$$C \left(\boxed{-1}, \boxed{-1} \right)$$

$$D \left(\boxed{1}, \boxed{-1} \right)$$

c



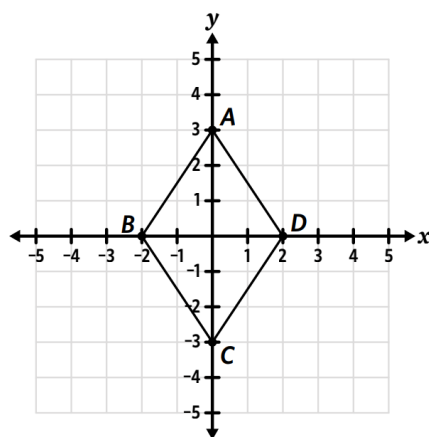
$$A \left(\boxed{3}, 4 \right)$$

$$B \left(\boxed{-4}, \boxed{4} \right)$$

$$C \left(\boxed{-4}, \boxed{-1} \right)$$

$$D \left(\boxed{3}, \boxed{-1} \right)$$

d



$$A \left(\boxed{0}, \boxed{3} \right)$$

$$B \left(\boxed{-2}, \boxed{0} \right)$$

$$C \left(\boxed{0}, \boxed{-3} \right)$$

$$D \left(\boxed{2}, \boxed{0} \right)$$

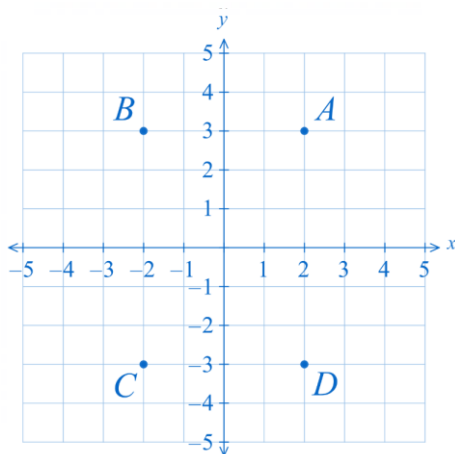
7 Plot these points and label the points with the letter.

a $A(2,3)$

b $B(-2,3)$

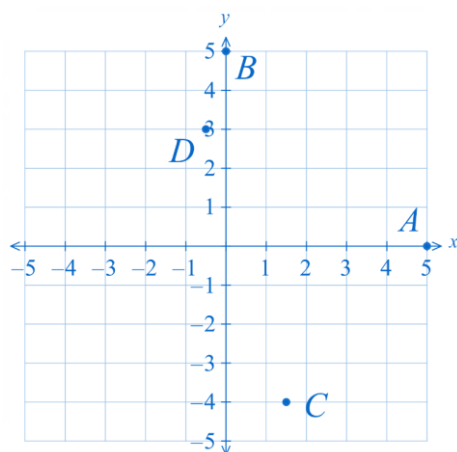
c $C(-2,-3)$

d $D(2,-3)$



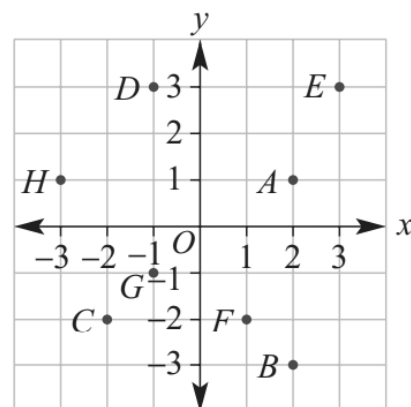
8 Plot these points and label the points with the letter.

- a $A(5,0)$
- b $B(0,5)$
- c $C(1.5, -4)$
- d $D\left(-\frac{1}{2}, 3\right)$



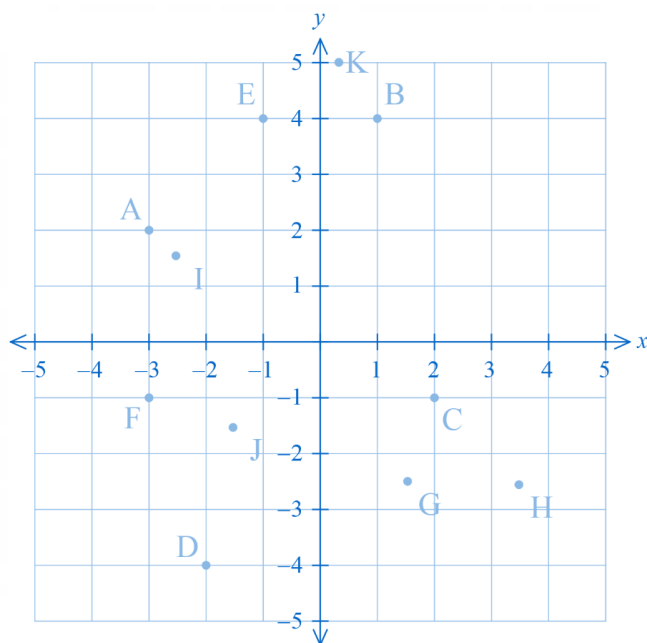
9 Match the points A, B, C, D, E, F, G and H with the given coordinates.

- | | | | |
|------------|-------|-------------|-------|
| a $(-1,3)$ | | b $(2,-3)$ | |
| | D | | B |
| c $(2,1)$ | | d $(-2,-2)$ | |
| | A | | C |
| e $(3,3)$ | | f $(-3,1)$ | |
| | E | | H |
| g $(1,-2)$ | | h $(-1,-1)$ | |
| | F | | G |



10 Plot these points on the Cartesian plane.

- a A $(-3, 2)$
- b B $(1, 4)$
- c C $(2, -1)$
- d D $(-2, -4)$
- e E $(-1, 4)$
- f F $(-3, -1)$
- g G $(1.5, -2.5)$
- h H $\left(3.5, -2\frac{1}{2}\right)$
- i I $(-2.5, 1.5)$
- j J $\left(-1\frac{1}{2}, -1.5\right)$
- k K $\left(\frac{1}{3}, 5\right)$



- 11 Chloe, Beca and Ed have each written down the coordinates of point P.

Chloe (4, -3) Beca (-3, 4) Ed (3, 4)

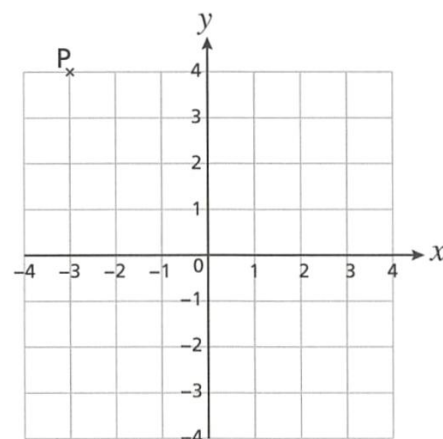
a Who is correct?

Beca

b Explain the mistakes the other two have made.

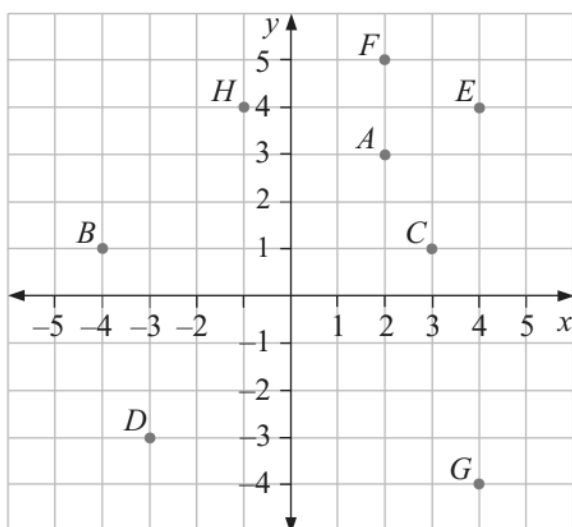
Chloe swapped x and y

Ed didn't use negative for being left of the origin



- 12 Eight points have been plotted on this number plane.

Write the coordinates of the eight points.



a (.....,)

(2,3)

b (.....,)

(-4,1)

c (.....,)

(3,1)

d (.....,)

(-3,-3)

e (.....,)

(4,4)

f (.....,)

(2,5)

g (.....,)

(4,-4)

h (.....,)

(-1,4)

- i Name two points with the same x -coordinates.

What do you notice about their positions on the number plane?

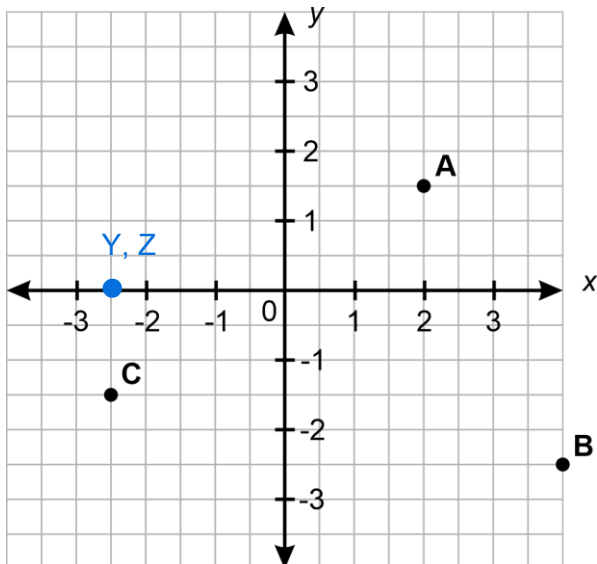
A and F. They have the same horizontal position (directly on top of each other)

- j Name two points with the same y -coordinates.

What do you notice about their positions on the number plane?

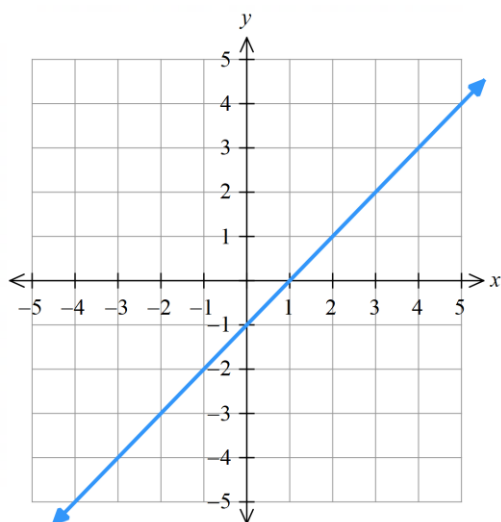
B and C. They have the same vertical position (on the same level as each other)

13



- Identify the coordinates of A
(2, 1.5)
- Identify the coordinates of B
(4, -2.5)
- Identify the coordinates of C
(-2.5, -1.5)
- Label point Y directly above C on the x-axis
- Label the point Z $2\frac{1}{2}$ units left of the origin.
- Write the coordinates of point Z
(-2.5, 0)

14 Write down the coordinates of 8 points that lie on this line.



- (.....,)
- (.....,)
- (.....,)
- (.....,)
- (.....,)
- (.....,)
- (.....,)
- (.....,)

Answers will vary.

Describe the relationship between x and y .

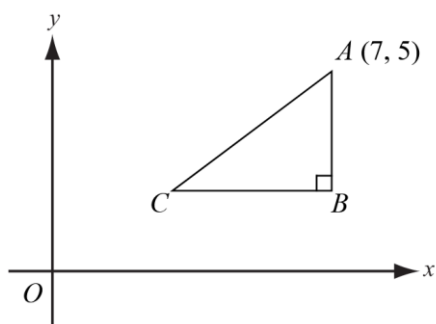
The y values are 1 less than the x values.

This line can be described by the equation $y = x - 1$. Where do you think this equation comes from?

The value of y is x subtract 1.

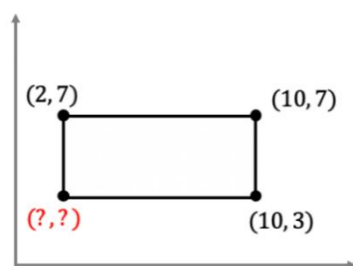
$$y = x - 1$$

15 AB is parallel to the y -axis. If $AB = 3$, $BC = 4$, and $AC = 5$, what are the coordinates of C ?



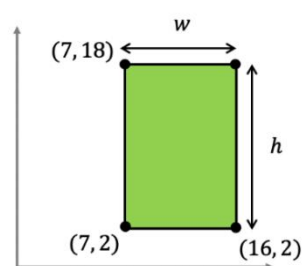
C(3, 2)

- a** What is the coordinate of the missing vertex of the rectangle?



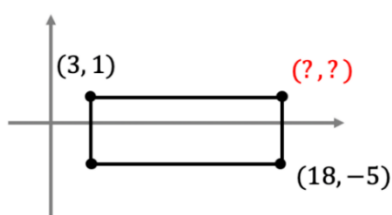
(2, 3)

- b** Determine the height and width of the rectangle.



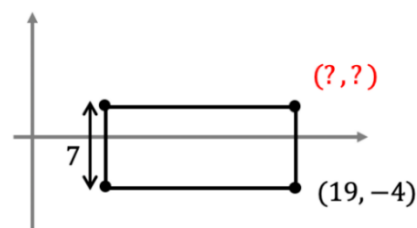
w: 9, h: 16

- c** What is the coordinate of the missing vertex of the rectangle?



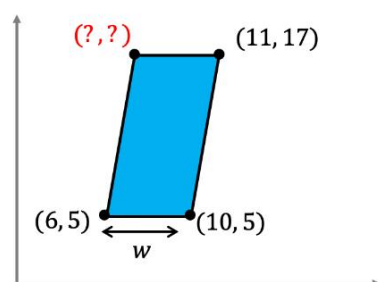
(18, 1)

- d** What is the coordinate of the missing vertex of the rectangle?



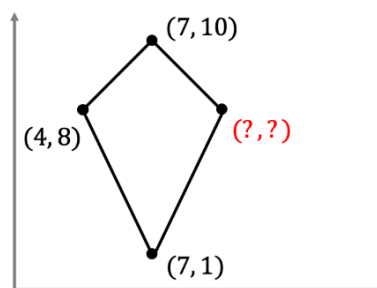
(7, 3)

- e** Find the width of the parallelogram and hence the unknown coordinates.



w: 4, (2, 17)

- f** What are the unknown coordinates on this kite?



(10, 8)

Constructing a Table of Values

Review



Substitute into formulas

a Find the value of y when $x = 3$
 $y = 4x + 1$

b Find the value of y when $x = 5$
 $y = 3x + 2$

c Find the value of y when $x = -2$
 $y = 4x + 5$

d Find the value of y when $x = 10$
 $y = -5x + 1$

Table of Values

A table of values shows a relationship between x and y using numbers.

To find a y value, substitute x in the equation with the number from the table, then work out the answer.

Example Construct a table of values

Complete the table of values for $y = 3x + 1$

x	0	1	2	3
y	1	4	7	10

$x = 0; y = 3(0) + 1 = 1$
 $x = 1; y = 3(1) + 1 = 4$
 $x = 2; y = 3(2) + 1 = 7$
 $x = 3; y = 3(3) + 1 = 10$

Guided Practice

Complete the table of values for these equations

a $y = 3x + 2$

x	0	1	2	3
y				

2, 5, 8, 11

b $y = 4x + 3$

x	0	1	2	3
y				

3, 7, 11, 15

c $y = 3x + 2$

x	-3	-2	-1	0
y				

-7, -4, -1, 2

d $y = -4x + 3$

x	-1	0	1	2
y				

7, 3, -1, -5

1 Complete these tables of values.

a $y = x + 2$

x	0	1	2	3	4
y					

$0 + 2$ $1 + 2$ $2 + 2$ $3 + 2$ $4 + 2$

2, 3, 4, 5, 6

b $y = 4x$

x	0	1	2	3	4
y					

4×0 4×1 4×2 4×3 4×4

0, 4, 8, 12, 16

c $y = 3x + 6$

x	0	1	2	3	4
y					

$3 \times 0 + 6$ $3 \times 1 + 6$ $3 \times 2 + 6$ $3 \times 3 + 6$ $3 \times 4 + 6$

6, 9, 12, 15, 18

d $y = 8 - 2x$

x	0	1	2	3	4
y					

$8 - 2 \times 0$ $8 - 2 \times 1$ $8 - 2 \times 2$ $8 - 2 \times 3$ $8 - 2 \times 4$

8, 6, 4, 2, 0

e $y = \frac{1}{2}x + 7$

x	0	1	2	3	4
y					

$\frac{1}{2} \times 0 + 7$ $\frac{1}{2} \times 1 + 7$ $\frac{1}{2} \times 2 + 7$ $\frac{1}{2} \times 3 + 7$ $\frac{1}{2} \times 4 + 7$

7, 7.5, 8, 8.5, 9

2 Construct a table of values for these equations.

a $y = 2x$

x	0	1	2	3
y				

0, 2, 4, 6

b $y = 2x + 1$

x	0	1	2	3
y				

1, 3, 5, 7

c $y = 2x + 5$

x	0	1	2	3
y				

5, 7, 9, 11

d $y = 2x - 3$

x	0	1	2	3
y				

-3, -1, 1, 3

e $y = x + 2$

x	0	1	2	3
y				

2, 3, 4, 5

f $y = 3x + 2$

x	0	1	2	3
y				

2, 5, 8, 11

g $y = 5x + 2$

x	0	1	2	3
y				

2, 7, 12, 17

h $y = -3x + 2$

x	0	1	2	3
y				

2, -1, -4, -7

i $y = 2x + 1$

x	0	1	2	3
y				

1, 3, 5, 7

j $y = \frac{x}{2} + 1$

x	0	1	2	3
y				

1, 1.5, 2, 2.5

k $y = 2x + 5$

x	-2	-1	0	1
y				

1, 3, 5, 7

l $y = -3x + 5$

x	-2	-1	0	1
y				

11, 8, 5, 2

Sketch a Line from a Table of Values

Review

- ☐ Complete the table of values.

a $y = 4x + 1$

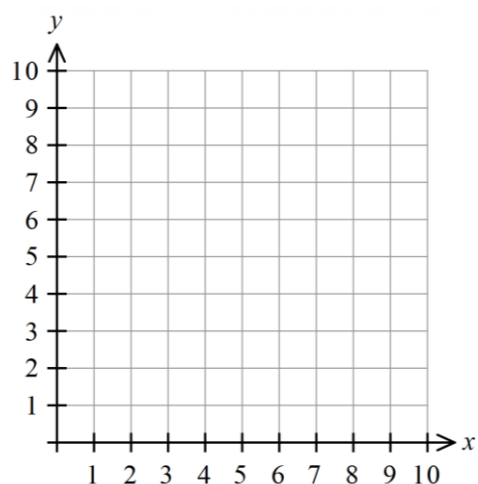
x	0	1	2	3
y				

b $y = 3x + 5$

x	0	1	2	3
y				

- ☐ Plot points on a Cartesian plane

x	0	1	2	3	4
y	9	7	5	3	1



? Identify a table column as a coordinate

We can show each (x, y) **pair** from a table of values as **coordinates**.

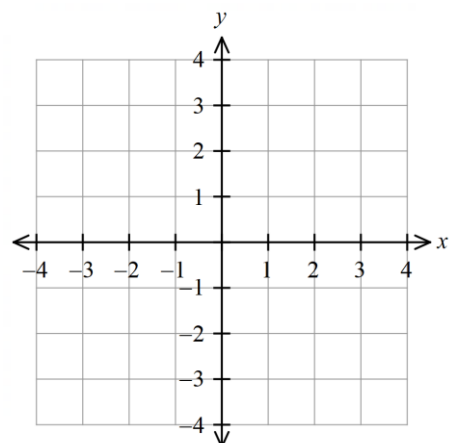
Write the coordinates from the table.

x	0	1	2
y	4	7	10

x	4	7	10
y	7	4	1

Plot this table of values on the Cartesian plane to the right.

x	-1	0	1
y	-1	1	3



① Sketch a Line from a Table of Values

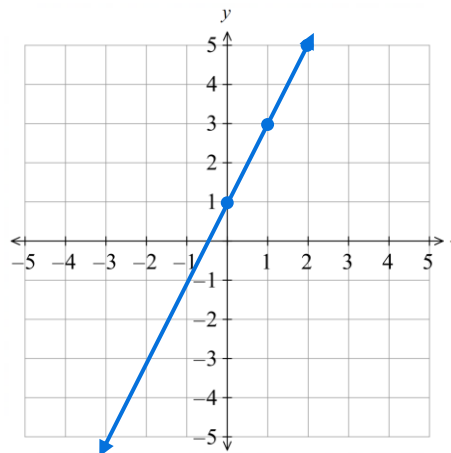
Some equations are called **linear relationships** because when we plot their (x, y) pairs, it forms a straight line.

1. Construct a table of values by substituting x values into the equation to find y .
2. Plot the (x, y) points on a Cartesian plane.
3. Draw a straight line through the points and extend it.

🔗 Example Graph a linear relationship using a table of values

Draw the graph of the equation $y = 2x + 1$

x	0	1	2	3
y	1	3	5	7

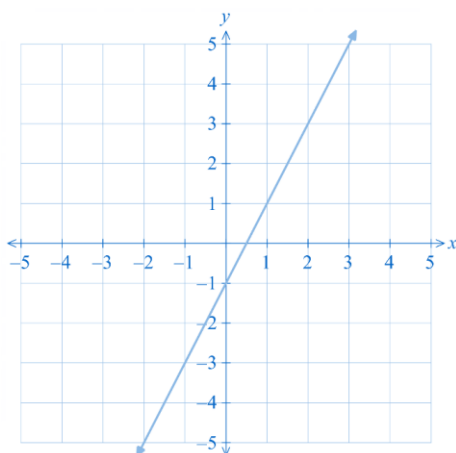


✎ Guided Practice

Draw the graph of:

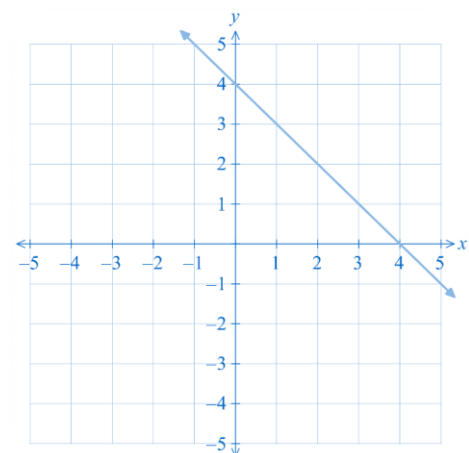
a $y = 2x - 1$

x	0	1	2	3
y				



b $y = 4 - x$

x	0	1	2	3
y				



💡 Key Ideas

1. Each column of a table of values can be written as a
2. When an equation forms points that make a straight line, we call it a relationship.

1 Identify the coordinates from the table of values.

a

x	1	2	3	4
y	2	5	8	11
Coordinate	(1,2)	(2,5)	(3,8)	(4,11)

b

x	0	1	2	3
y	1	1.5	2	2.5
Coordinate	(0,1)	(1,1.5)	(2,2)	(3,2.5)

c

x	-1	0	1	2
y	7	5	3	1
Coordinate	(-1,7)	(0,5)	(1,3)	(2,1)

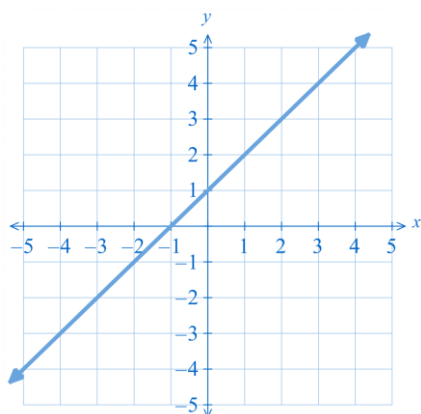
d

x	-2	-1	0	1
y	14	10	6	2
Coordinate	(-2,14)	(-1,10)	(0,6)	(1,2)

2 Complete the table of values and graph the result.

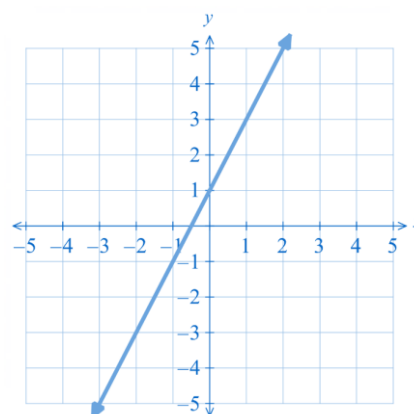
a $y = x + 1$

x	0	1	2	3
y				



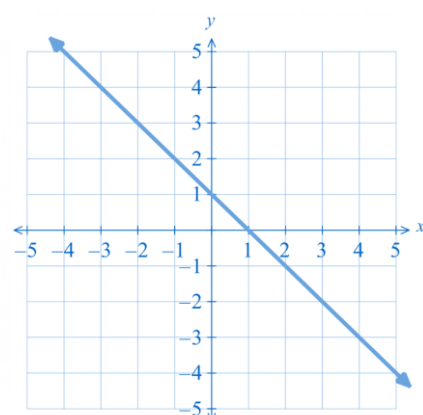
b $y = 2x + 1$

x	0	1	2	3
y				



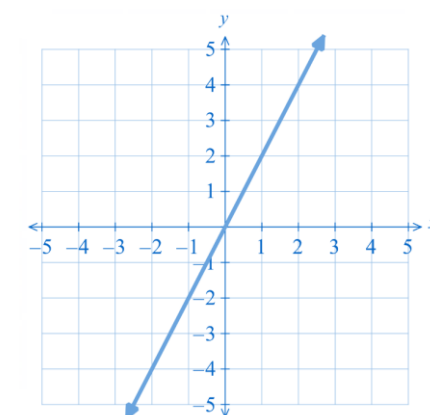
c $y = -x + 1$

x	0	1	2	3
y				



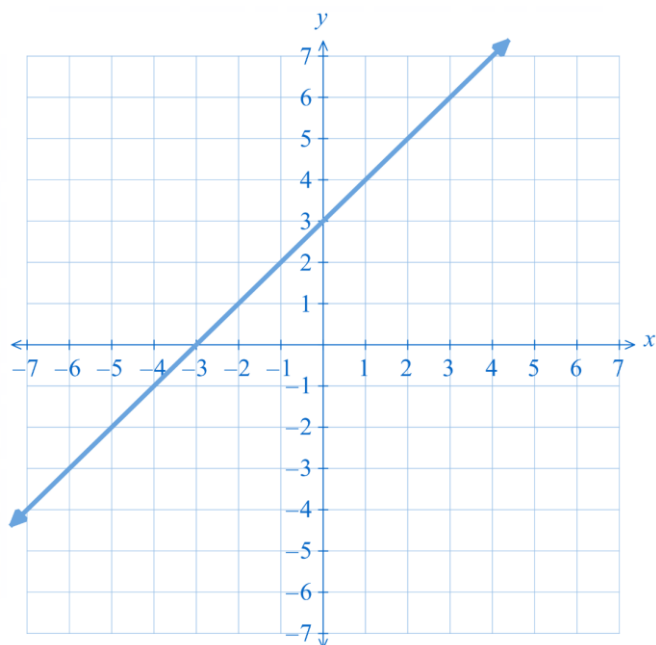
d $y = 2x$

x	0	1	2	3
y				



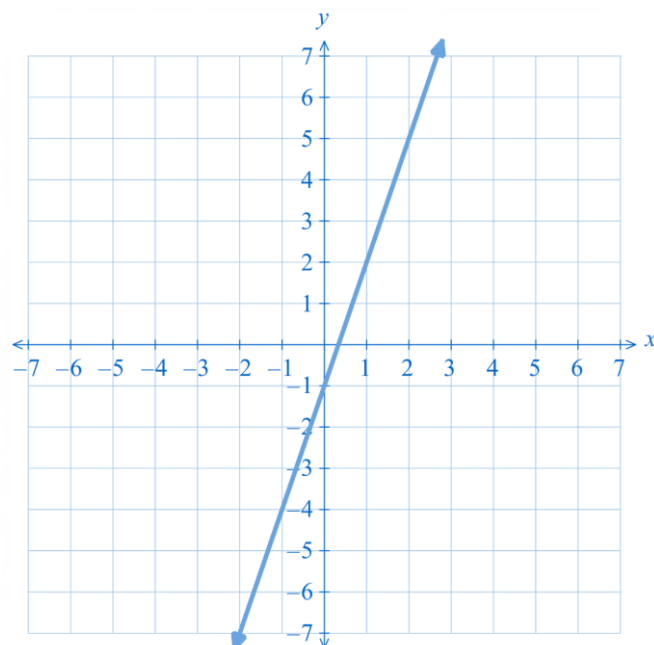
e $y = x + 3$

x	0	1	2	3
y				



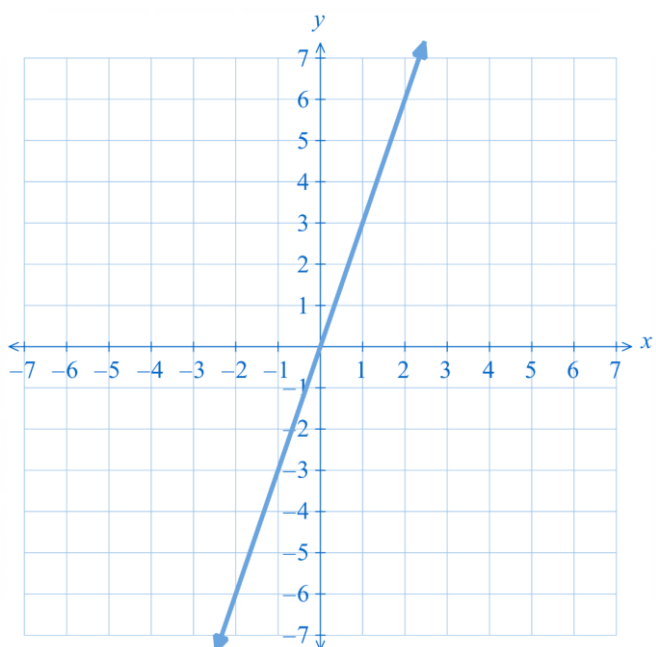
f $y = 3x - 1$

x	0	1	2	3
y				



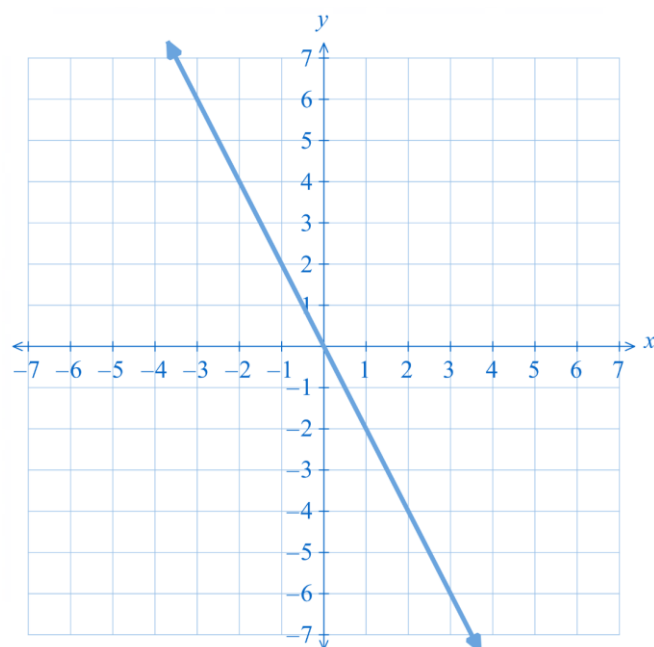
g $y = 3x$

x	0	1	2	3
y				



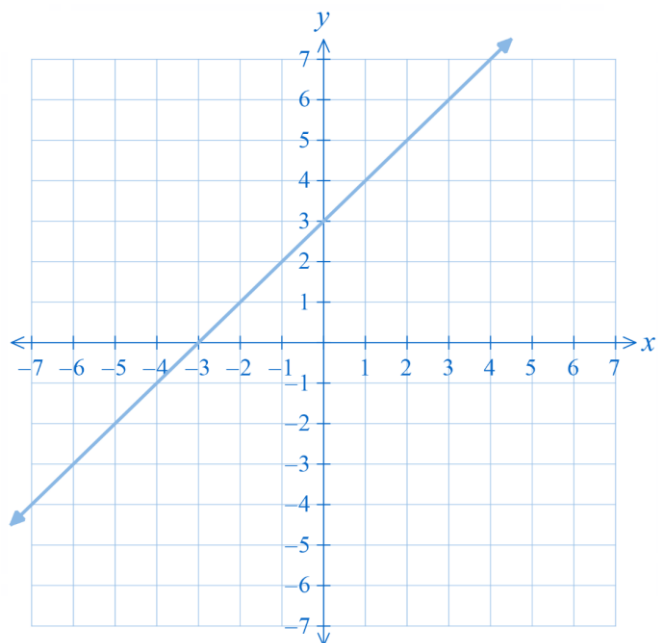
h $y = -2x$

x	0	1	2	3
y				



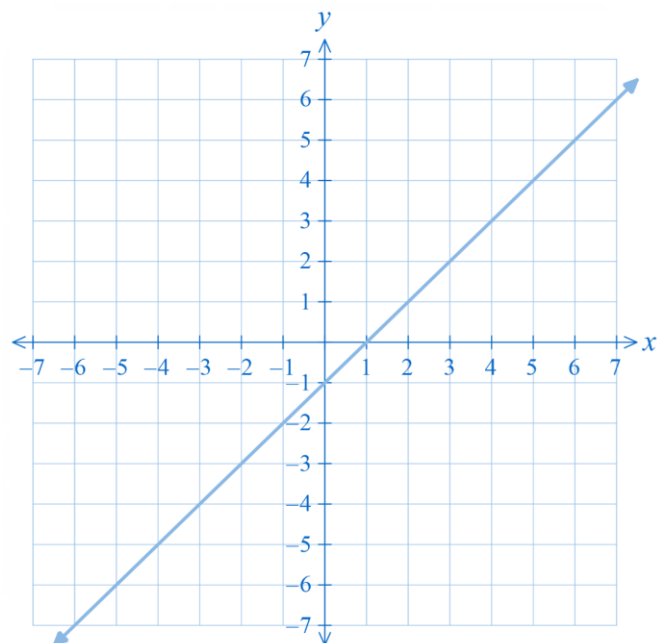
i $y = x + 3$

x	-2	-1	0	1	2
y					



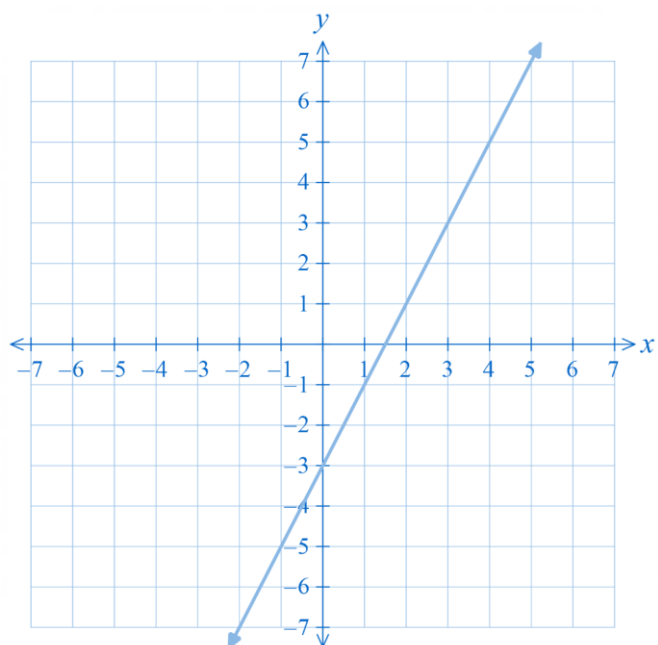
j $y = x - 1$

x	-2	-1	0	1	2
y					



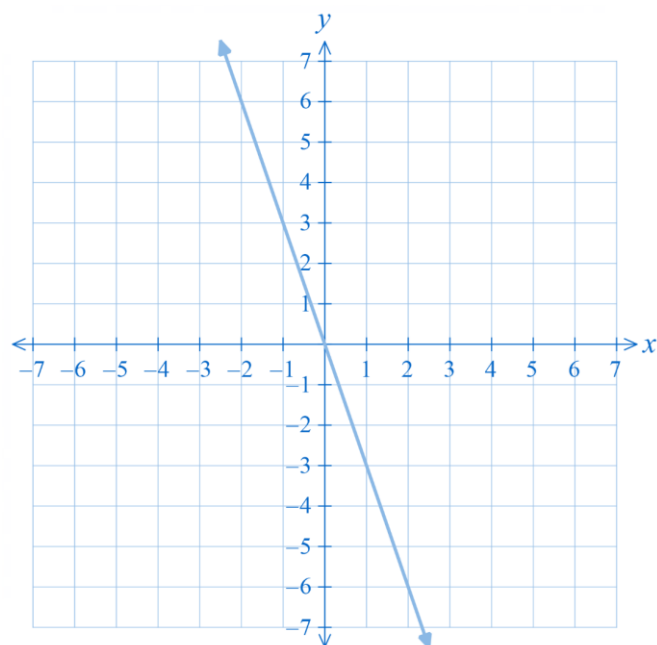
k $y = 2x - 3$

x	-2	-1	0	1	2
y					



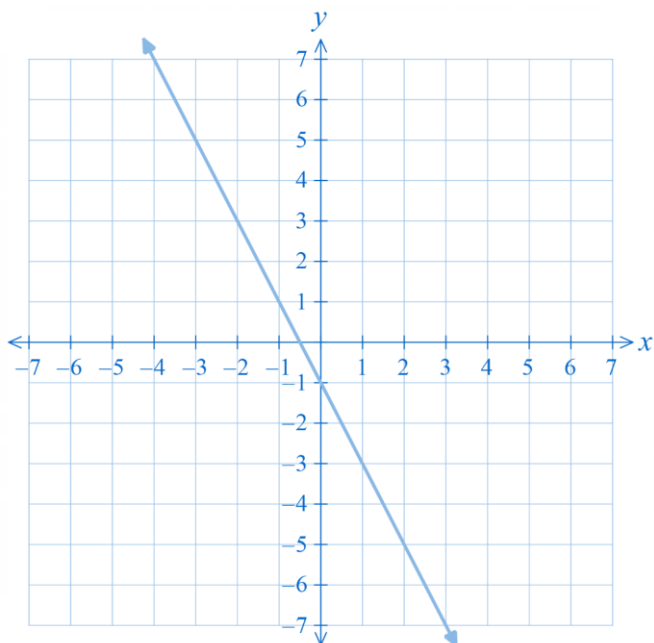
l $y = -3x$

x	-2	-1	0	1	2
y					



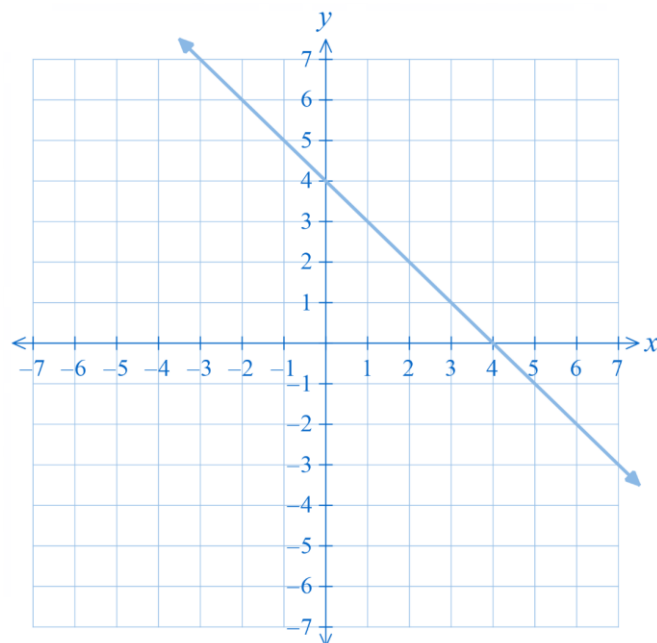
m $y = -2x - 1$

x	-2	-1	0	1	2
y					



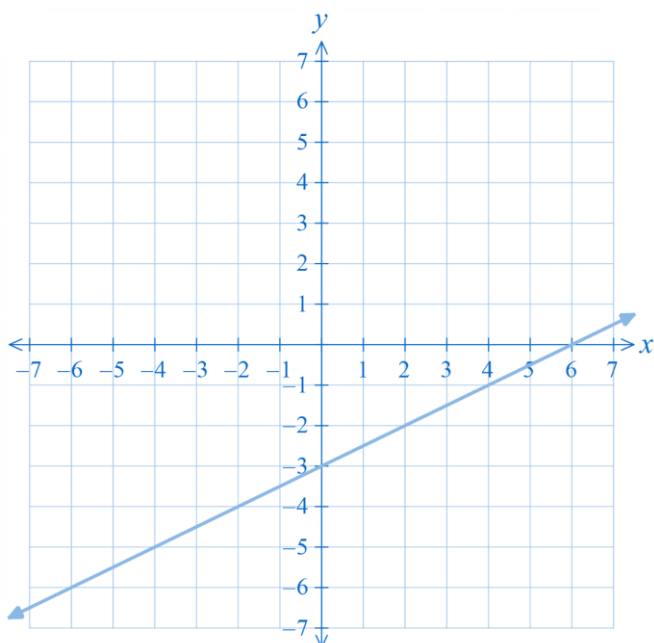
n $y = -x + 4$

x	-2	-1	0	1	2
y					



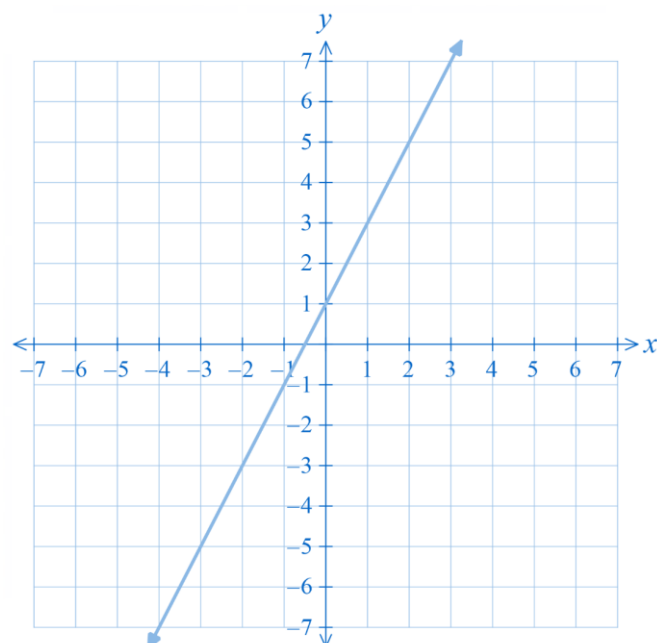
o $y = 0.5x - 3$

x	-2	-1	0	1	2
y					



p $y = 1 + 2x$

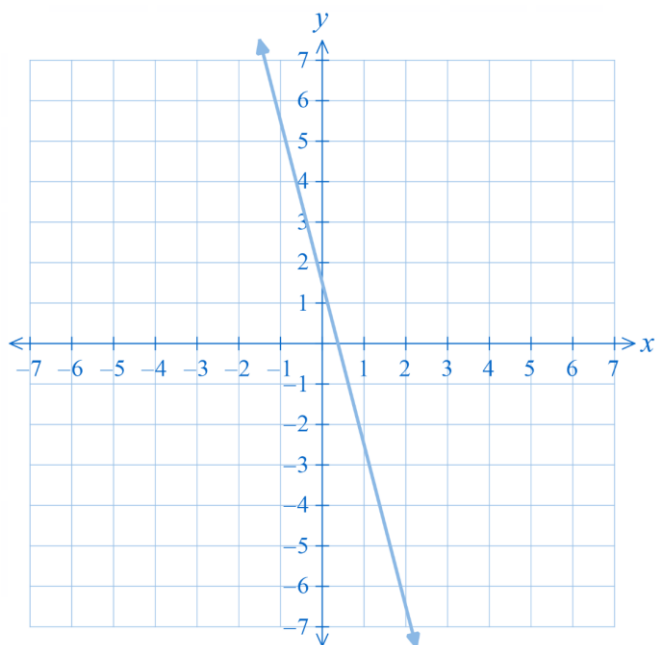
x	-2	-1	0	1	2
y					



3 Complete the table of values and graph the result.

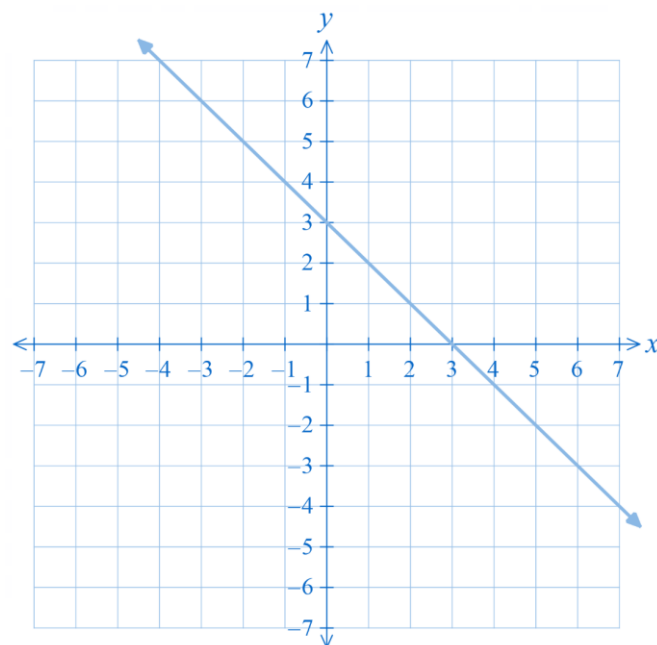
a $y = 1.5 - 4x$

x	-2	-1	0	1	2
y					



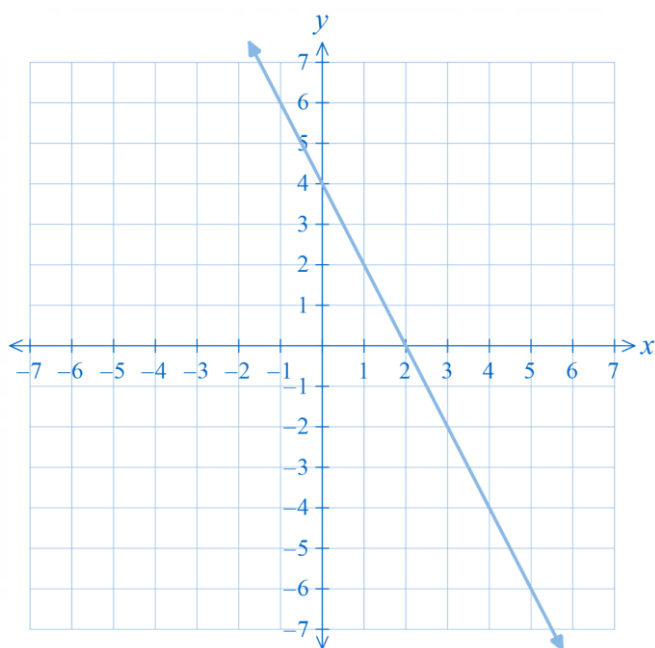
b $y = 3 - x$

x	-2	-1	0	1	2
y					



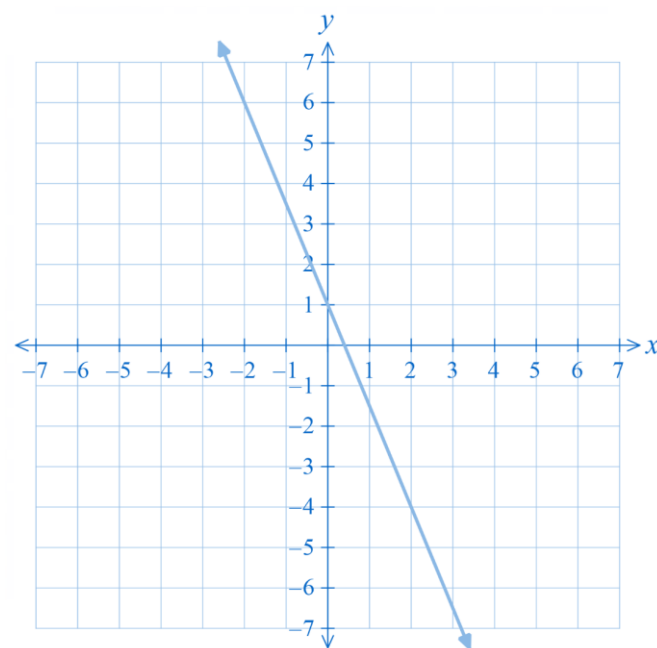
c $y = 4 - 2x$

x	-2	-1	0	1	2
y					



d $y = 1 - \frac{5}{2}x$

x	-2	-1	0	1	2
y					



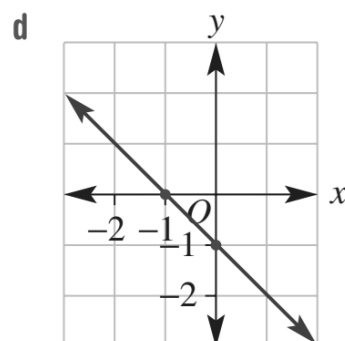
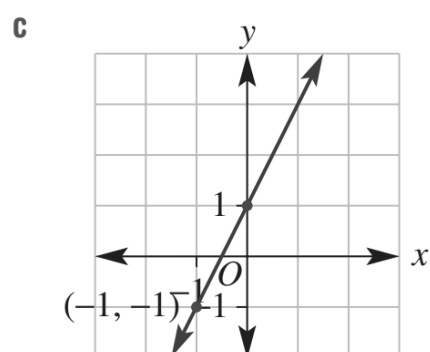
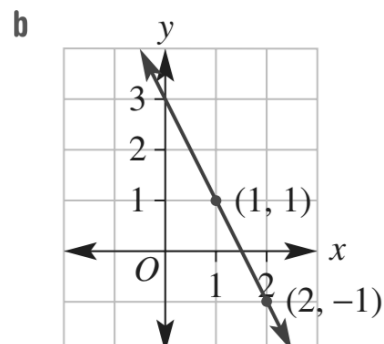
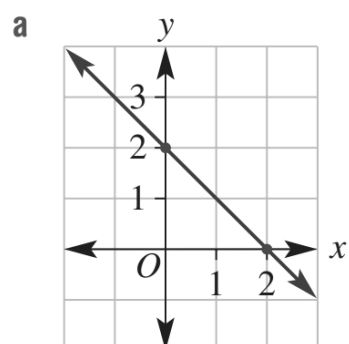
4 Match the rules A-D with the graphs a-d.

A $y = 2x + 1$

B $y = -x - 1$

C $y = -2x + 3$

D $y = -x + 2$



A – c, B – d, C – b, D – a

Finding the Rule from a Table

① Equation of a Linear Relationship

Every linear relationship is in the form:

$$y = mx + c$$

② Identify equation of a linear relationship

✗

$$y = x^2$$

✓

$$y = 2x$$

✓

$$y = 9 - 12x$$

✓

$$y = \frac{x}{6} - 22$$

✗

$$y = \frac{6}{x} - 22$$

Identify which of the following are linear relationships

a $y = 5x + 1$	b $y = 1 + 5x$	c $y = x^2 - 7$	d $y = \frac{x}{2} - 7$
e $y = \frac{2}{x}$	f $y = 3^x$	g $y = x$	h $y = 9$
i $y = 3 - 7x$	j $y = -12x + 6$	k $y = x^3 - 5$	l $y = 80 - \frac{5x}{3}$

i Equations and Tables

An equation for a table must give the correct y -value for every x -value in the table.
We say that every column of the table must **satisfy** the equation.

? Verify whether a table of values satisfies an equation.

a $y = 3x$

x	0	1	2	3
y	0	3	6	9

b $y = 3x$

x	0	1	2	3
y	0	3	5	7

c $y = x + 2$

x	0	1	2	3
y	0	3	6	9

d $y = 4x + 1$

x	0	1	2	3
y	1	5	9	13

e $y = -x + 2$

x	0	1	2	3
y	2	1	0	-1

f $y = 6x + 7$

x	0	1	2	3
y	7	6	5	4

① Finding the Equation

We can represent x y pairs in a table of values as an equation linking x and y .
The equation must work for every column of the table.

 **Example** Find the equation from a table of values

x	1	3	10
y	10	12	19

Every y value is 9 more than the x value.
 $y = x + 9$

x	0	4	10
y	1	9	21

Every y value is twice the x value plus one.
 $y = 2x + 1$

Guided Practice

Find the rule shown in the table.

a

x	2	5	6
y	6	9	10

$$y = x + 4$$

b

x	12	5	0
y	36	15	0

$$y = 3x$$

c

x	2	5	10
y	3	9	19

$$y = 2x - 1$$

d

x	10	6	3
y	5	3	1.5

$$y = \frac{x}{2}$$

Key Ideas

1. When finding the rule from a table, we are looking for an equation that every column of the table.
2. An equation for a table is correct only if it works for column in the table.

1 Determine the rule shown in the table. Every equation involves a single operation only.

a

x	5	6	7
y	15	18	21

$$y = 3x$$

b

x	10	11	12
y	13	14	15

$$y = x + 3$$

c

x	8	9	10
y	5	6	7

$$y = x - 3$$

d

x	20	14	6
y	18	12	4

$$y = x - 2$$

e

x	8	10	12
y	4	5	6

$$y = \frac{x}{2}$$

f

x	4	5	6
y	8	10	12

$$y = 2x$$

g

x	4	3	2
y	6	5	4

$$y = x + 2$$

h

x	20	14	6
y	15	9	1

$$y = x - 5$$

i

x	8	10	12
y	13	15	17

$$y = x + 5$$

j

x	1	2	3	4	5
y	4	8	12	16	20

$$y = 4x$$

k

x	10	8	3	1	14
y	21	19	14	12	25

$$y = x + 11$$

l

x	6	18	30	24	66
y	1	3	5	4	11

$$y = \frac{x}{6}$$

2 Describe the relationship between the x and y values of each point on the line.

a $y = x + 1$ The y value is **one more than** the x value.

b $y = x + 2$ The y value is the x value.

2 more than

c $y = x - 1$ The y value is the x value.

1 less than

d $y = x - 4$ The y value is the x value.

4 less than

e $y = x + 4$ The y value is the x value.

4 more than

f $y = 2x$ The y value is **two times** the x value.

g $y = 3x$ The y value is the x value.

3 times

h $y = -3x$ The y value is the x value.

-3 times

i $y = \frac{x}{2}$ The y value is the x value.

half

j $y = -\frac{x}{2}$ The y value is the x value.

Negative one half times

3 Determine the equation of the line shown by these tables of values, then complete the table.

a

x	2	5	6	1	8
y	6	9	10		

$y = x + 4$
Missing values: 5, 12

b

x	12	5	8	0	12
y	36	15	24		

$y = 3x$
Missing values: 0, 36

c

x	4	3	5	1	0
y	-8	-6	-10		

$y = -2x$
Missing values: -2, 0

d

x	6	18	30	24	66
y	1	3	5		

$y = \frac{x}{6}$
Missing values: 4, 11

e

x	8	10	12	20	-20
y	4	5	6		

$y = \frac{x}{2}$
Missing values: 10, -10

f

x	20	14	6	15	1
y	15	9	1		

$y = x - 5$
Missing values: 10, -4

4 First figure out the rule, then complete the rest of the table.

a

x	4	10	13	24			5	11
y			39		9	42	15	

$y = 3x$
12, 30, 72, 3, 14, 33

b

x	12	93	14	17		10		
y	3			8	12	1	34	0

$y = x - 9$
84, 5, 21, 43, 9

5

a In which of these coordinate pairs is the y value twice the x value?

(2, 4)	(4, 2)	(6, -3)	(3, -6)	(-3, -6)
(0, 0)	(19, 38)	(-12, -6)	(50, 25)	(-20, -40)

(2,4), (-3, -6), (0,0), (19,38), (-20, -40)

b The points you identified in part a lie on what line? Give the equation.

$y = 2x$

c Will (8, 16) lie on $y = 2x$?

Yes

d Will (9, 20) lie on $y = 2x$?

No

6 All of these points lie on the same line. (1, 10) (3, 30) (-10,-100) (-5,-50)

Complete this sentence:

a In each pair of coordinates, the y value is the x value.

10 times

b Write the equation of the straight line that passes through all four points.

$y = 10x$

7 Here are some pairs of coordinates.

(4, 19)	(19, 4)	(0, -15)	(15, 0)	(-15, 0)
(23, 38)	(3.8, 2.3)	(100, 85)	(-17, -2)	(12, -3)

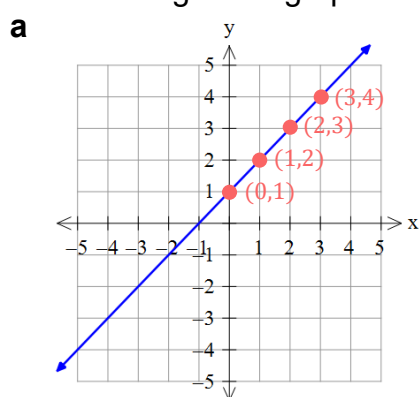
a Which of the points lie on the line $y = x + 15$?

(4, 19), (-15, 0), (23, 38), (-17, -2)

b Which of the points lie on the line $y = x - 15$?

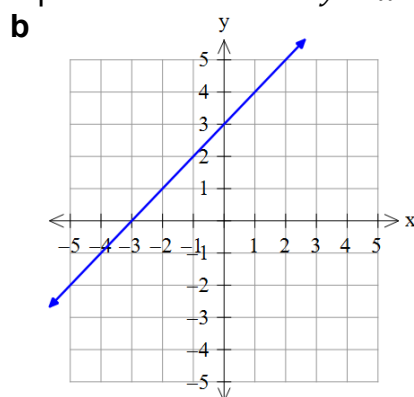
(19, 4), (0, -15), (15, 0), (100, 85), (12, -3)

8 These straight line graphs have equations of the form $y = x + c$ or $y = mx$



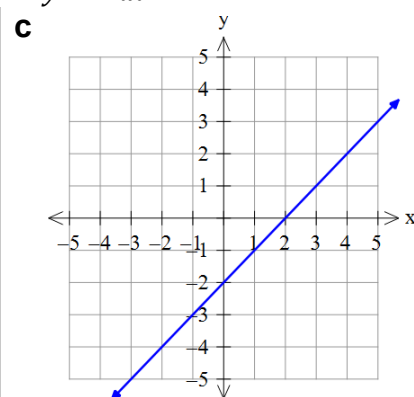
At each point on the line, the y -value is 1 more than the x -value. The equation of the line is

$y = x + 1$



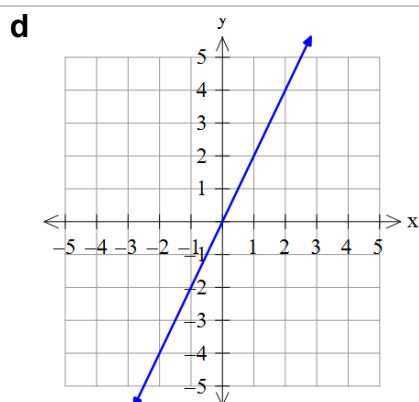
At each point on the line, the y -value is the x -value. The equation of the line is

Three more than; $y = x + 3$



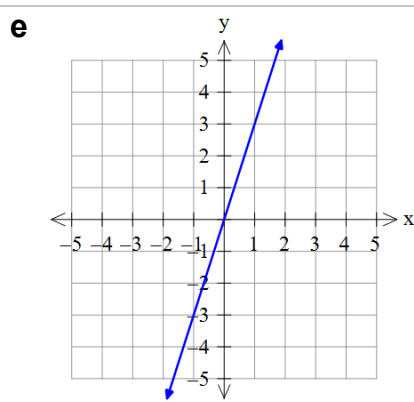
At each point on the line, the y -value is the x -value. The equation of the line is

2 less than; $y = x - 2$



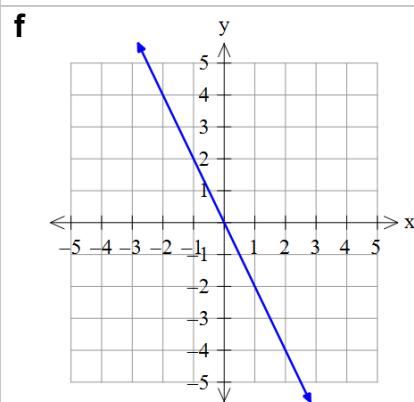
At each point on the line, the y -value is 2 times the x -value. The equation of the line is

$y = 2x$



At each point on the line, the y -value is the x -value. The equation of the line is

Three times; $y = 3x$



At each point on the line, the y -value is the x -value. The equation of the line is

Negative two times; $y = -2x$

Systematically Finding the Rule $y = mx + c$ from a table

① Equation of a Linear Relationship

Every linear relationship is in the form $y = mx + c$.

$$y = mx + c$$

↑ coefficient of x
number x is multiplied by
change in y as x increases by 1
↑ constant term
term with no variable
value of y when $x = 0$

② Identify m from a table of values

m is the coefficient of x . It is how much y changes as x increases by 1.

a

x	2	3	4
y	8	12	16

$m = \dots\dots\dots +4 \quad +4$

b

x	5	6	7
y	30	36	42

$m = \dots\dots\dots$

c

x	9	10	11
y	20	18	16

$m = \dots\dots\dots$

d

x	201	202	203
y	18	15	12

$m = \dots\dots\dots$

e

x	1	2	3
y	905	910	915

$m = \dots\dots\dots$

f

x	7	2	11
y	25	15	33

Why can't we find m ?

③ Identify c from a table of values

c is the constant term. It is the value of y when $x = 0$.

a

x	0	1	2
y	8	9	10

$c = \dots\dots\dots$

b

x	0	4	18
y	11	3	-25

$c = \dots\dots\dots$

c

x	-2	0	15
y	15	23	83

$c = \dots\dots\dots$

d

x	0	22	8
y	-5	127	43

$c = \dots\dots\dots$

e

x	5	-3	0
y	17	-15	-3

$c = \dots\dots\dots$

f

x	-7	14	8
y	51	-96	-54

Why can't we find c ?

① Finding the Rule from a Table of Values

If we have a table where the value of x increases by 1, and the value of y when $x = 0$, then we can use this method to find the equation $y = mx + c$:

1. Find m : This is how much y changes by as x increases by 1.
2. Find c : This is the value of y when $x = 0$.
3. Write the equation $y = mx + c$ and check it works for every column.

⚠ Check that your rule works for every column.

🔗 Example Find the equation of a linear relationship from a table of values

a

x	0	1	2	3	4	5
y	1	3	5	7	9	11

$$m = 2, c = 1$$

$$y = 2x + 1$$

b

x	-1	0	1	2	3	4
y	10	7	4	1	-2	-5

$$m = -3, c = 7$$

$$y = -3x + 7$$

✎ Guided Practice

Determine the equation that relates x and y from the table of values.

a

x	0	1	2	3	4	10
y	2	6	10	14	18	42

$$m = \dots \quad c = \dots$$

$$y = \dots$$

$$y = 4x + 2$$

b

x	0	1	2	3	4	10
y	3	1	-1	-3	-5	-17

$$m = \dots \quad c = \dots$$

$$\dots$$

$$y = -2x + 3$$

c

x	0	1	2	3	4	10
y	-6	-3	0	3	6	24

$$y = 3x - 6$$

d

x	-2	-1	0	1	2	3
y	-13	-8	-3	2	7	12

$$y = 5x - 3$$

💡 Key Ideas

1. Every linear relationship has the equation $\dots = \dots$
2. m is the change in y as x increases by $\dots$, c is the value of y when $x = \dots$

- 1 Determine the change in y as x increases by 1. This is m . Find the y -value at $x = 0$; this is c . Write the equation linking x and y with this information.

a

x	0	1	2	3	4
y	5	7	9	11	13

$m = 2$ $c = 5$

$y = 2x + 5$

b

x	0	1	2	3	4
y	-2	1	4	7	10

$m = \dots\dots$ $c = \dots\dots$

$y = \dots\dots x + \dots\dots$

$y = 3x - 2$

c

x	0	1	2	3	4
y	0	5	10	15	20

$m = \dots\dots$ $c = \dots\dots$

$y = \dots\dots x + \dots\dots$

$y = 5x$

d

x	0	1	2	3	4
y	-9	-10	-11	-12	-13

$m = \dots\dots$ $c = \dots\dots$

$y = \dots\dots\dots$

$y = -x - 9$

e

x	-2	-1	0	1	2
y	5	6	7	8	9

$y = x + 7$

f

x	-2	-1	0	1	2
y	-2	-1	0	1	2

$y = x$

- 2 Find the equation shown in these tables.

a

x	0	1	2	3
y	0	2	4	6

$y = 2x$

b

x	0	1	2	3
y	3	5	7	9

$y = 2x + 3$

c

x	0	1	2	3
y	-1	1	3	5

$y = 2x - 1$

d

x	0	1	2	3
y	1	2	3	4

$y = x + 1$

e

x	0	1	2	3
y	1	4	7	10

$y = 3x + 1$

f

x	0	1	2	3
y	1	-1	-3	-5

$y = -2x + 1$

3 Find the equation shown in these tables.

a

x	-2	-1	0	1	2
y	0	2	4	6	8

$$y = 2x + 4$$

b

x	-3	-2	-1	0	1
y	4	3	2	1	0

$$y = -x + 1$$

c

x	-1	0	1	2	3
y	8	6	4	2	0

$$y = -2x + 6$$

d

x	-2	-1	0	1	2
y	-7	-4	-1	2	5

$$y = 3x - 1$$

4 Find the equation for each table, then complete the table.

a

x	0	1	2	3		5	8
y	3	6	9	12			

$$y = 3x + 3$$

 Missing values: 18, 27

b

x	0	1	2	3		4	7
y	1	4	7	10			

$$y = 3x + 1$$

 Missing values: 13, 22

c

x	0	1	2	3		6	-2
y	5	8	11	14			

$$y = 3x + 5$$

 Missing values: 23, -1

d

x	0	1	2	3		40	-3
y	0	0.5	1	1.5			

$$y = x/2$$

 Missing values: 20, -1.5

e

x	0	1	2	3		23	-7
y	-1	-3	-5				

$$y = -2x - 1$$

 Missing values: -7, -47, 13

f

x	-1	0	1	2		8	22
y		13	8	3			

$$y = -5x + 13$$

 Missing values: 18, -27, -97

g

x	-1	0	1	2		9	-4
y		20	10	0			

$$y = -10x + 20$$

 Missing values: 30, -70, 60

h

x	0	1	2	3		12	10
y	-0.5	-1	-1.5				

$$y = -\frac{x}{2} - \frac{1}{2}$$

 Missing values: -2, -6.5, -5.5

- 5 Find the equation that describes this table of values. Select A, B, C, or D.

x	-1	0	1	2
y	3	1	-1	-3

- A** $y = 2x + 1$ **B** $y = 2x - 1$ **C** $y = x - 2$ **D** $y = -2x + 1$

D

- 6 Which table of values matches $y = 3x - 1$? Select A, B, C, or D.

A

x	1	2	3	4	5
y	2	3	4	5	6

B

x	1	2	3	4	5
y	2	5	8	11	14

C

x	1	2	3	4	5
y	4	7	10	13	16

D

x	1	2	3	4	5
y	0	3	6	9	12

B

- 7 Extend the table so that $x = 0$ is shown, then find the equation of the line.

a

0	1	2	x	3	4	5	6	7
3	5	7	y	9	11	13	15	17

-2 -2 -2 +2 +2 +2 +2

$m = \dots\dots$ $c = \dots\dots$

$y = \dots\dots\dots$

$y = 2x + 3$

b

		x	2	3	4	5	6
		y	2	5	8	11	14

□ □ □ □ □ □

$m = \dots\dots$ $c = \dots\dots$

$y = \dots\dots\dots$

$y = 3x - 4$

c

				x	4	5	6	7
				y	11	13	15	17

□ □ □ □ □ □ □

$y = 2x + 3$

d

x	-6	-5	-4	-3			
y	-9	-6	-3	0			

□ □ □ □ □ □

$y = 3x + 9$

8 Determine the equation shown in the table.

a

x	1	2	3	4
y	6	10	14	18

$$y = 4x + 2$$

b

x	1	2	3	4
y	0	4	8	12

$$y = 4x - 4$$

c

x	2	3	4	5
y	17	27	37	47

$$y = 10x - 3$$

d

x	4	5	6	7
y	4	6	8	10

$$y = 2x - 4$$

e

x	-5	-4	-3	-2	-1
y	-13	-11	-9	-7	-5

$$y = 2x - 3$$

f

x	-6	-5	-4	-3	-2
y	10	9	8	7	6

$$y = -x + 4$$

9 Determine the equation for each table, then complete the last two columns.

a

x	1	2	3	4		7	9
y	2	7	12	17			

$$y = 5x - 3$$

Missing values: 32, 42

b

x	3	4	5	6		9	11
y	8	10	12	14			

$$y = 2x + 2$$

Missing values: 20, 24

c

x	2	3	4	5		0	12
y	2	4	6	8			

$$y = 2x - 2$$

Missing values: -2, 22

d

x	3	4	5	6		9	50
y	31	41	51	61			

$$y = 10x + 1$$

Missing values: 91, 501

e

x	3	4	5	6		12	-2
y	14	19	24	29			

$$y = 5x - 1$$

Missing values: 59, -11

f

x	5	6	7	8		9	12
y	4	6	8	10			

$$y = 2x - 6$$

Missing values: 12, 18

g

x	1	2	3	4		7	-3
y	5	9	13	17			

$$y = 4x + 1$$

Missing values: 29, -11

h

x	4	5	6	7		0	-3
y	1	3	5	7			

$$y = 2x - 7$$

Missing values: -7, -13

10 These x values don't increase by 1. You will need to divide to find m .

a

x	-4	-2	0	2	4
y	-19	-11	-3	5	13

$$m = 8 \div 2 \quad c = \dots\dots\dots$$

$$= \dots\dots\dots$$

$$y = \dots\dots\dots$$

$$y = 4x - 3$$

b

x	0	5	10	15	20
y	-3	22	47	72	97

$$m = \dots\dots \div 5 \quad c = \dots\dots\dots$$

$$= \dots\dots\dots$$

$$y = \dots\dots\dots$$

$$y = 5x - 3$$

c

x	-6	-3	0	3	6
y	8	-1	-10	-19	-28

$$y = -3x - 10$$

d

x	0	2	4	6	8
y	-2	-14	-26	-38	-50

$$y = -6x - 2$$

e

x	-40	-30	-20	-10	0
y	195	155	115	75	35

$$y = -4x + 35$$

11 2006 HSC Maths Standard 2 Band 4

What equation represents the relationship between x and y in this table?

x	0	2	4	6	8
y	1	2	3	4	5

$$y = \frac{x}{2} + 1$$

12 These tables need to be extended and the x -value steps to be considered.

a

		x	4	6	8	10	12
		y	17	23	29	35	41

$$y = 3x + 5$$

b

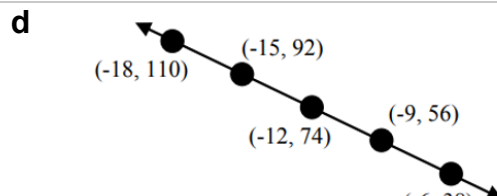
				x	12	15	18	21
				y	16	25	34	43

$$y = 3x - 20$$

c

			x	15	20	25	30	35
			y	20	30	40	50	60

$$y = 2x - 10$$



$$y = -6x + 2$$

13 Determine the rule shown in the table. Notice that the x values are not increasing by 1.

a

x	-4	-2	0	2	4
y	-5	-1	3	7	11

b

x	-3	-1	1	3	5
y	-10	-4	2	8	14

$y = 2x + 3$

$y = 3x - 1$

c

x	-6	-3	0	3	6
y	15	9	3	-3	-9

d

x	-10	-8	-6	-4	-2
y	20	12	4	-4	-12

$y = -2x + 3$

$y = -4x - 20$

14 Describe the relationship between the x and y values at each point on these lines.

a $y = 5x + 3$

To obtain the y -value, we multiply x by 5 then add 3

b $y = 2x + 1$

To obtain the y -value, we multiply x by then
 $2, \text{ add } 1$

c $y = 4x - 1$

To obtain the y -value, we multiply x by then
 $4, \text{ subtract } 1$

d $y = -3x - 9$

To obtain the y -value, we multiply x by then
 $\text{Negative } 3, \text{ subtract } 9$

e $y = \frac{x}{2} + 7$

To obtain the y -value, we divide x by then
 $2, \text{ add } 7$

f $y = \frac{x}{3} - 12$

To obtain the y -value, we x by then
 $\text{divide, } 3, \text{ subtract } 12$

- 15** Remember that every point on a line is a solution to the equation.
If you know one of either m or c , then you can find the equation by substitution.

Example 1						Example 2					
	x	7	8	9	10		x	0	8	47	-7
	y	10	13	16	19		y	5	-11	-89	19
$m = 3$ $y = 3x + c$ Sub in any point: (7,10) $10 = 3(7) + c$ $10 = 21 + c$ $c = -11$ $\therefore y = 3x - 11$						$c = 5$ $y = mx + 5$ Sub in any point, <i>except</i> (0,5): (8, -11) $-11 = m(8) + 5$ $-11 = 8m + 5$ $-16 = 8m$ $m = -2$ $\therefore y = -2x + 5$					

It is a good idea to double check your rule by using the columns.

- 16** Work out the rule for these tables.

a

x	9	10	11	12
y	1	5	9	13

b

x	-2	12	0	9
y	4	-38	-2	-29

c

x	-10	-9	-8	-7
y	13	15	17	19

d

x	10	12	14	16
y	10	11	12	13

$$y = 4x - 35$$

$$y = -3x - 2$$

$$y = 2x + 33$$

$$y = \frac{x}{2} + 5$$

17 Find the rule for these tables of values. Use any strategy you like.

a

x	-2	-1	0	1	2
y	0	2	4	6	8

$$y = 2x + 4$$

b

x	-2	-1	0	1	2
y	-7	-4	-1	2	5

$$y = 3x - 1$$

c

x	-3	-2	-1	0	1
y	4	3	2	1	0

$$y = -x + 1$$

d

x	-1	0	1	2	3
y	8	6	4	2	0

$$y = -2x + 6$$

e

x	1	2	3	4	5
y	5	9	13	17	21

$$y = 4x + 1$$

f

x	-5	-4	-3	-2	-1
y	-13	-11	-9	-7	-5

$$y = 2x - 3$$

g

x	5	6	7	8	9
y	-12	-14	-16	-18	-20

$$y = -2x - 2$$

h

x	-6	-5	-4	-3	-2
y	10	9	8	7	6

$$y = -x + 4$$

i

x	-4	-2	0	2	4
y	-5	-1	3	7	11

$$y = 2x + 3$$

j

x	-3	-1	1	3	5
y	-10	-4	2	8	14

$$y = 3x - 1$$

k

x	-6	-3	0	3	6
y	15	9	3	-3	-9

$$y = -2x + 3$$

l

x	-10	-8	-6	-4	-2
y	20	12	4	-4	-12

$$y = -4x - 20$$

m

x	0	2	4	6
y	-10	-16	-22	-28

$$y = -3x - 10$$

n

x	-5	-4	-3	-2
y	29	24	19	14

$$y = -5x + 4$$

o

x	1	3	5	7
y	1	2	3	4

$$y = \frac{x}{2} + \frac{1}{2}$$

p

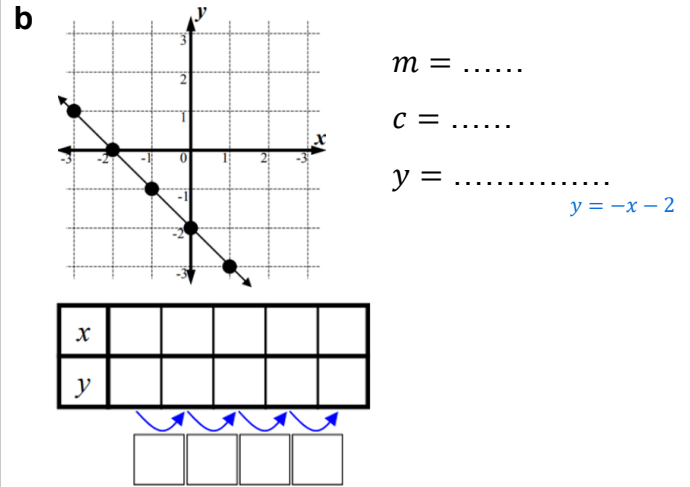
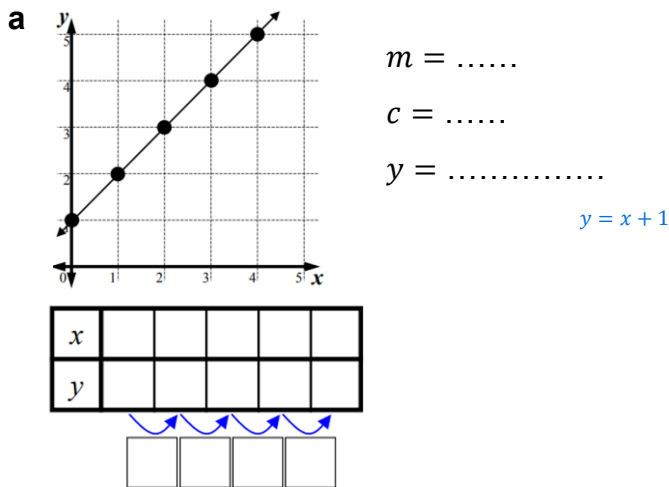
x	-14	-13	-12	-11
y	$5\frac{1}{2}$	5	$4\frac{1}{2}$	4

$$y = -\frac{x}{2} - \frac{3}{2}$$

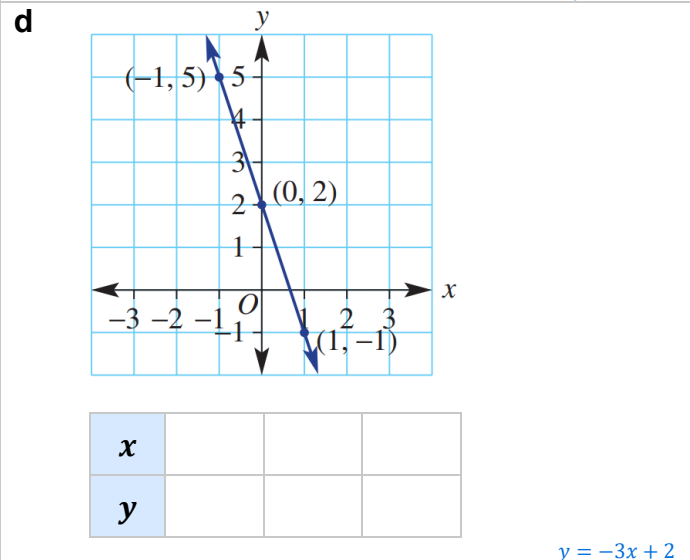
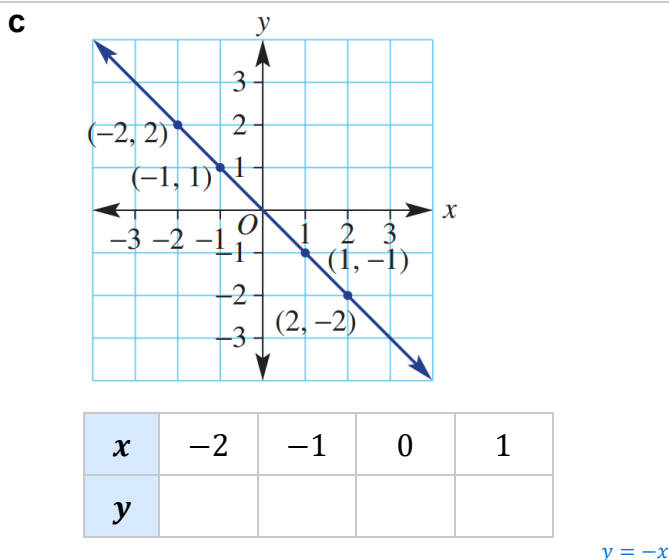
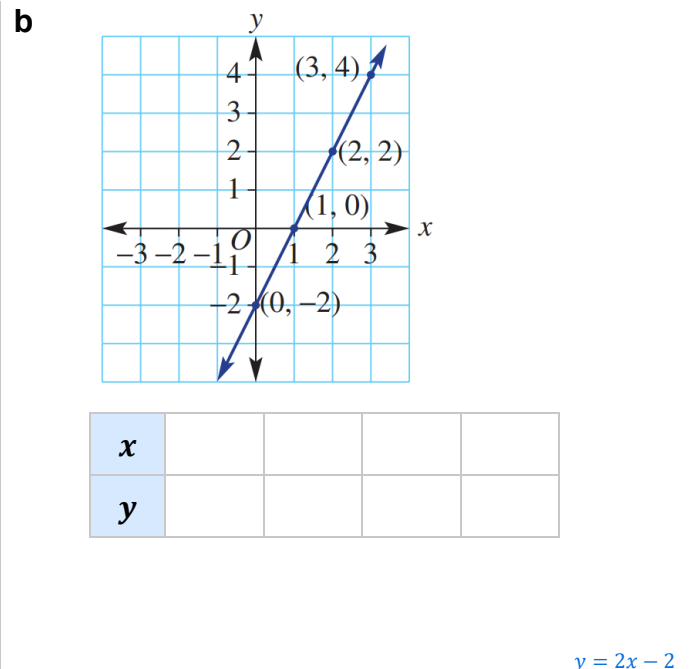
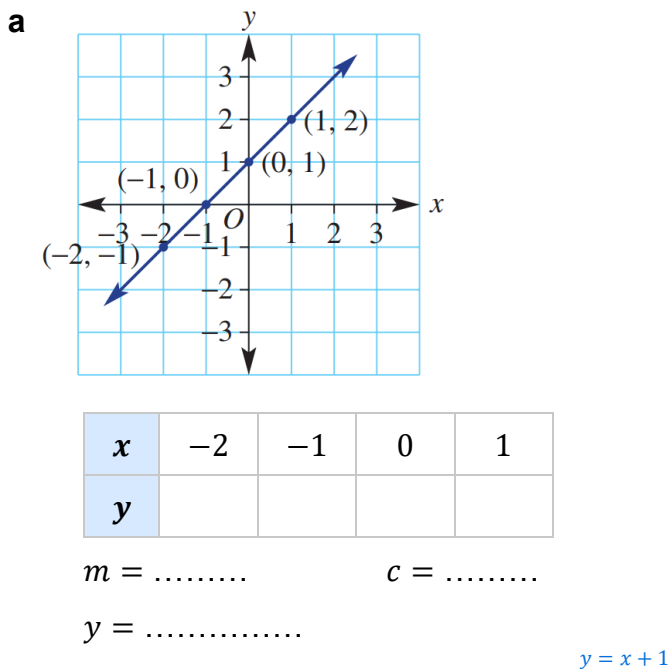
Finding the Rule from the Graph

FOUNDATION

- 1 Change the points to a table of values then find the equation of the line.

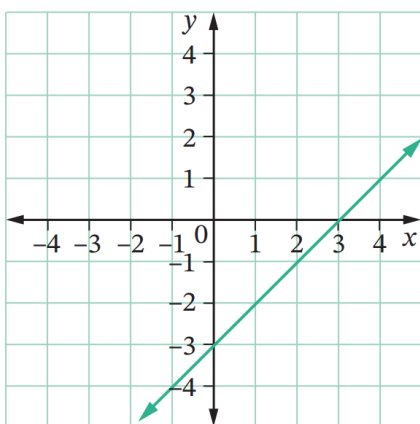


- 2 By first constructing a table of values, find the equations of these lines.
Ensure your x values are in ascending order.



3 Complete the table for each graph, then find the equation.

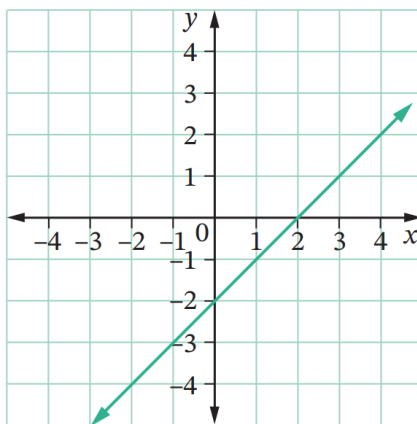
a



x	-1	0	1	2
y				

$$y = x - 3$$

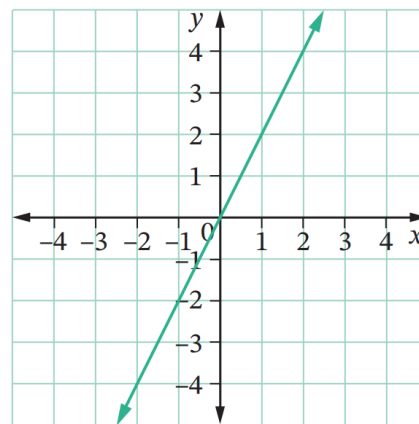
b



x	-2	-1	0	1
y				

$$y = x - 2$$

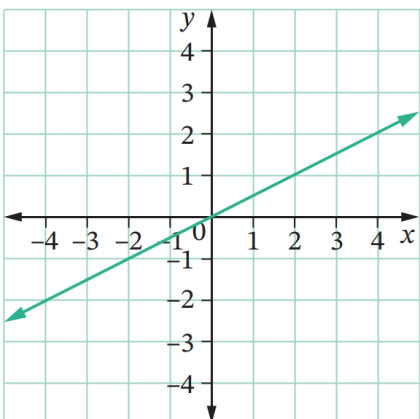
c



x	-1	0	1	2
y				

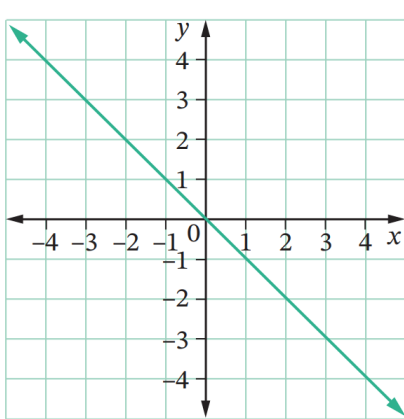
$$y = 2x$$

d



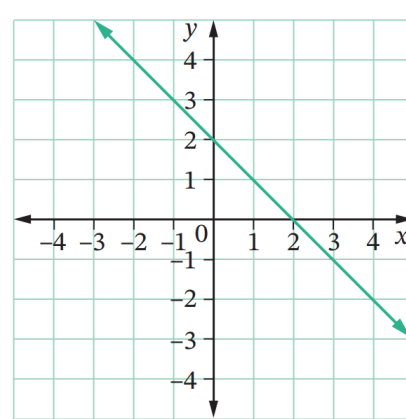
$$y = \frac{x}{2}$$

e



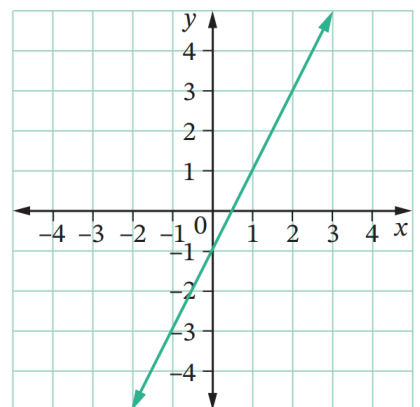
$$y = -x$$

f



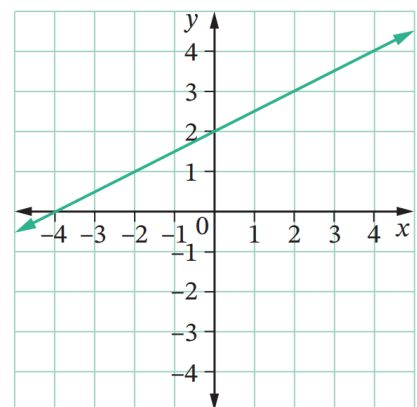
$$y = -x + 2$$

g



$$y = 2x - 1$$

h



$$y = \frac{x}{2} + 2$$

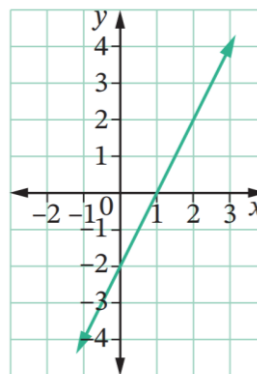
4 What is the equation of this line? Select A, B, C, or D.

A $y = -x + 2$

B $y = 2x - 2$

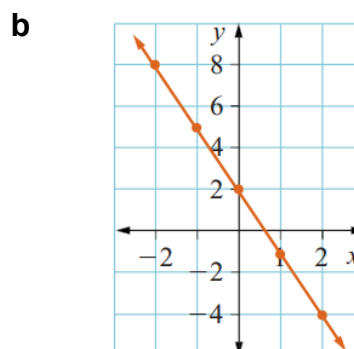
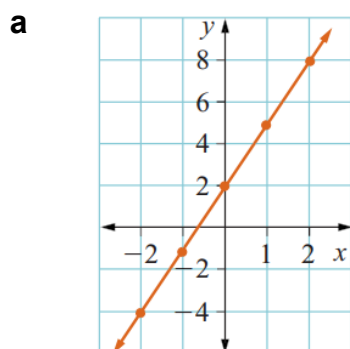
C $y = 2 - 2x$

D $y = x - 2$



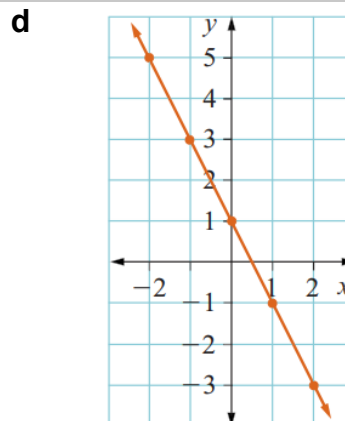
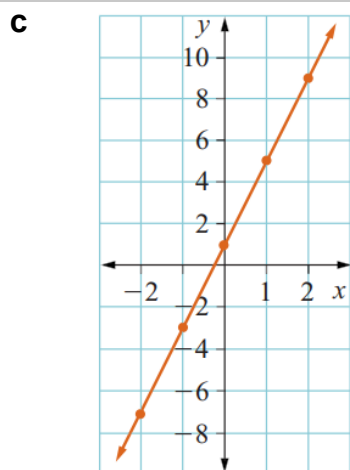
B

5 By first constructing a table of values, find the equations of these lines. Notice that the y -axis scale is different to the x -axis scale.



$y = 3x + 2$

$y = -3x + 2$



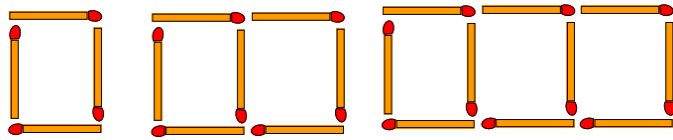
$y = 4x + 1$

$y = -2x + 1$

Review

- ☐ Complete a table of values from a spatial pattern

Draw the next term in the pattern and fill in the table.



Number of shapes	1	2	3	4	5
Number of matches					

- ☐ Identify equation relating x and y .

a

x	0	1	2	3
y	1	4	7	10

b

x	1	2	3
y	5	7	9

c

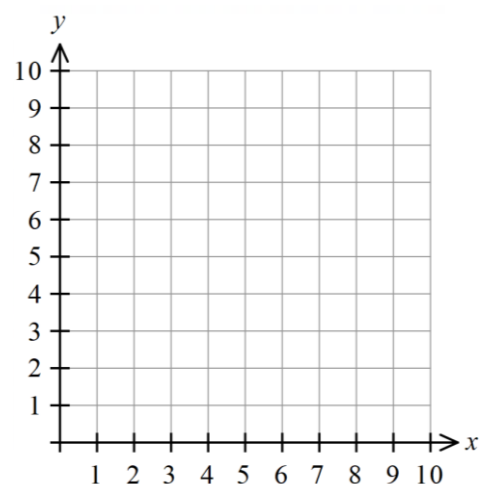
x	1	2	3
y	3	5	7

d

x	1	2	3
y	6	10	14

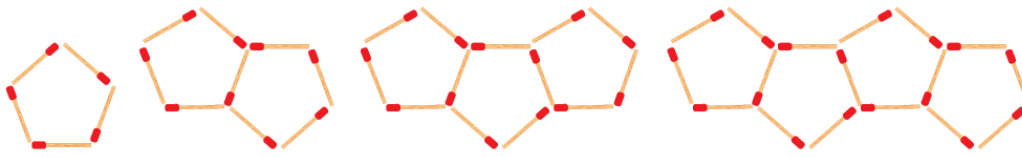
- ☐ Plot points on a Cartesian plane

x	0	1	2	3	4
y	9	7	5	3	1



? Investigation Spatial patterns and linear relationships

Linear relationships show up in simple patterns.

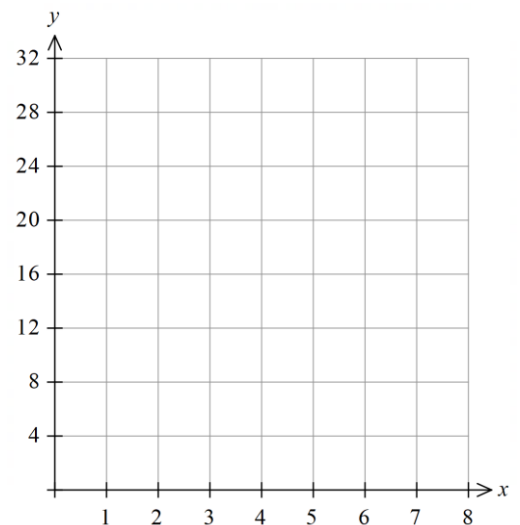


a Complete the table of values.

Number of shapes x	1	2	3	4
Number of matches y				

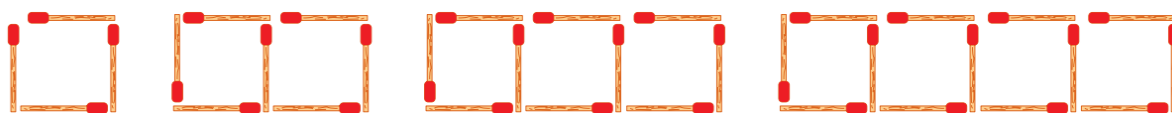
b Determine an equation that relates the two quantities.

c Graph the relationship on the Cartesian Plane



Guided Practice

1



Complete the table of values.

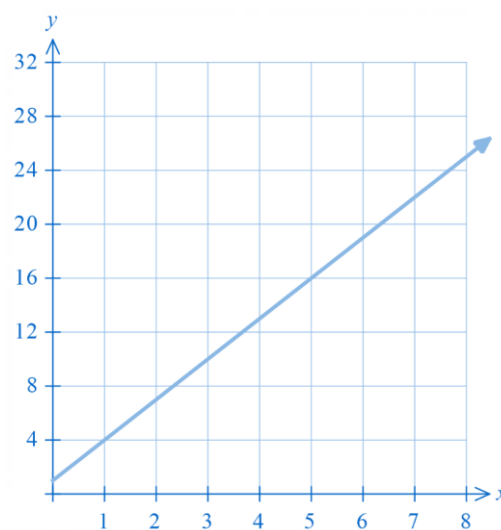
Number of shapes x	1	2	3	4
Number of matches y				

4, 7, 10, 13

Determine an equation that relates the two quantities.

$$y = 3x + 1$$

Graph the relationship on the Cartesian Plane.



2



Table:

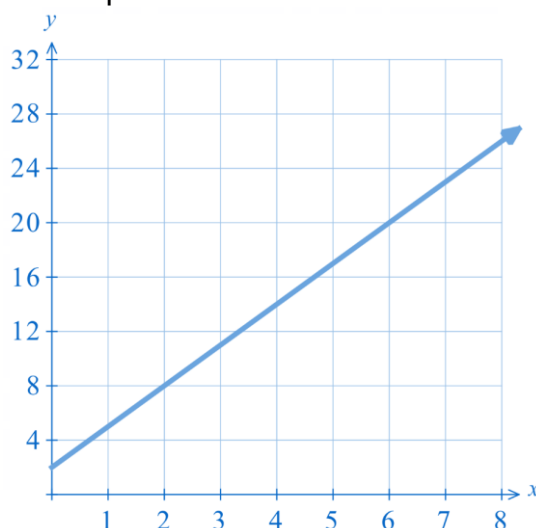
x	1	2	3	4
y				

5, 8, 11, 14

Equation:

$$y = 3x + 2$$

Graph:



Key Ideas

- Linear relationships can be represented by a matchstick pattern.
- m is the number of matches each shape, c is the number of matches for '.....' shapes.



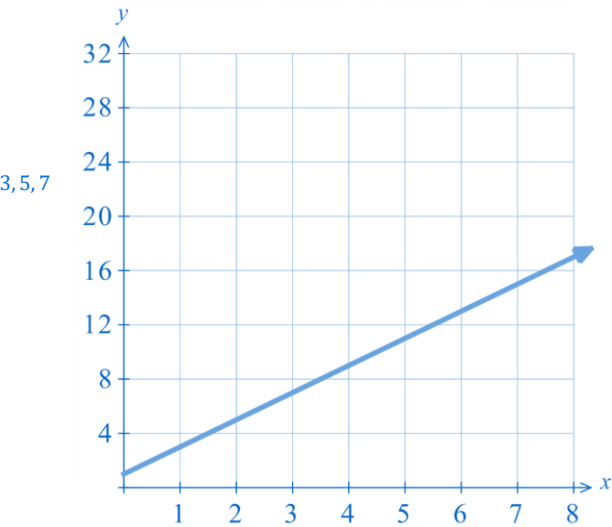
Table:

x	1	2	3
y			

Equation:

.....
 $y = 2x + 1$

Graph:



a Complete the table of values.

Number of shapes x	1	2	3	4
Number of matches y				

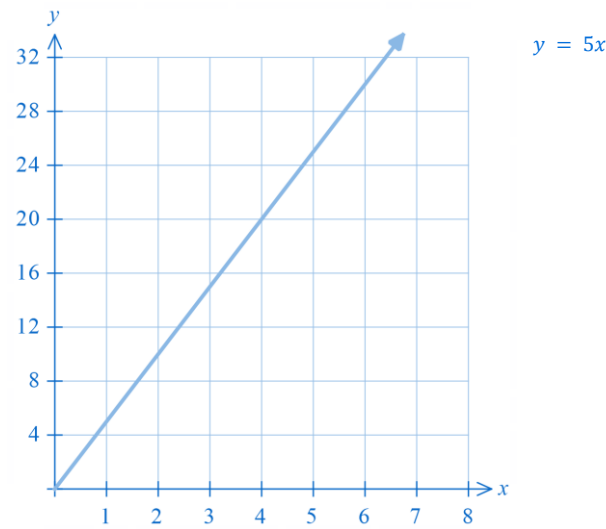
5, 10, 15, 20

b Determine an equation that relates the two quantities.

Number of matches =

$y = \dots\dots\dots$

c Graph the relationship on the Cartesian Plane.



3

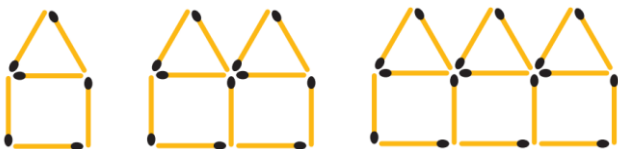


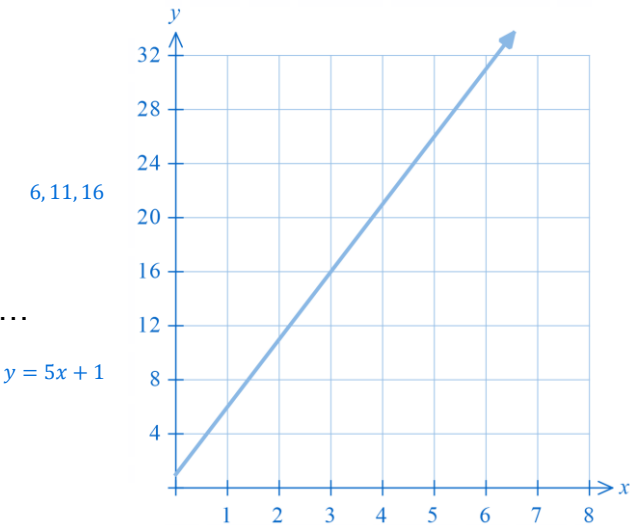
Table:

x	1	2	3
y			

Equation:

.....

Graph:



4

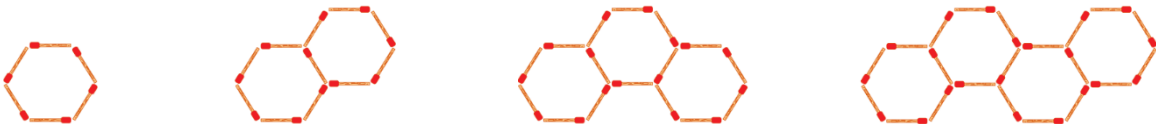


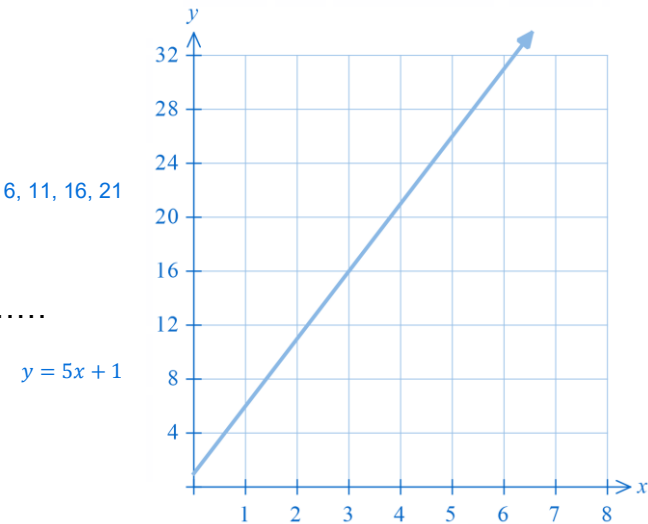
Table:

x	1	2	3	4
y				

Equation:

.....

Graph:



5 What do you notice about the equation of Q3 and Q4 above?

.....

They are the same.

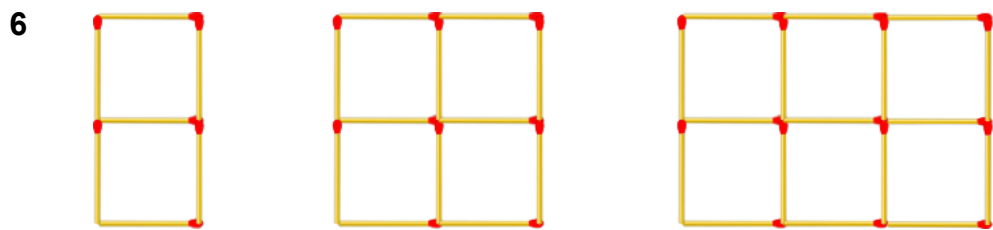


Table:

x	1	2	3
y			

Equation:

.....

Graph:

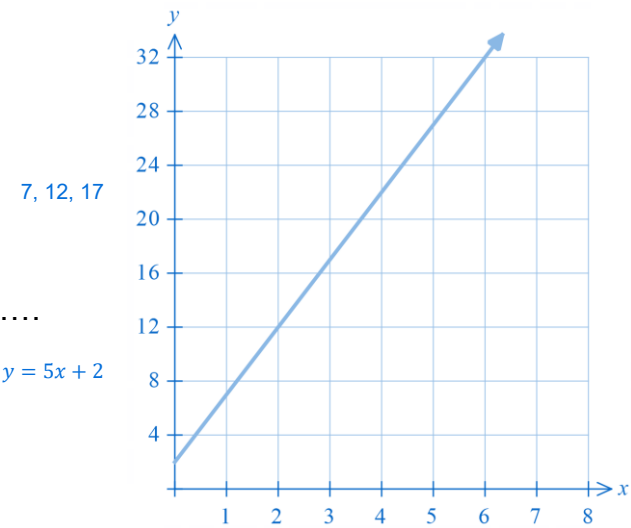


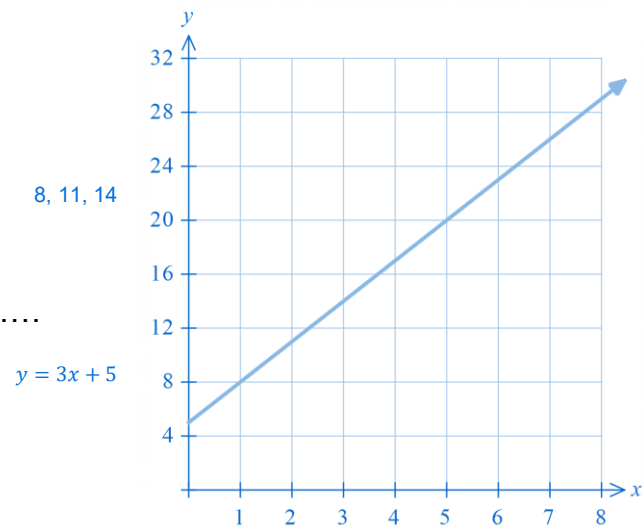
Table:

x	1	2	3
y			

Equation:

.....

Graph:



8

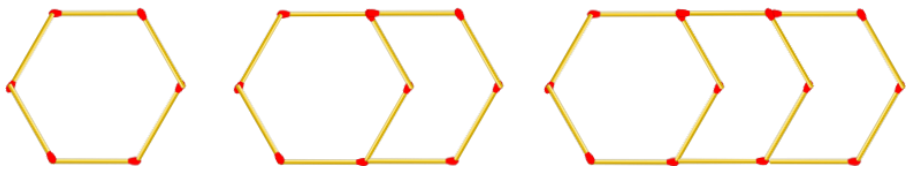


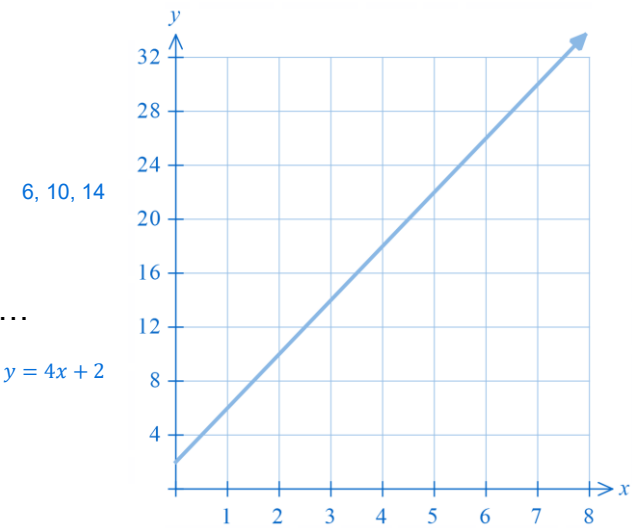
Table:

x	1	2	3
y			

Equation:

.....

Graph:



9



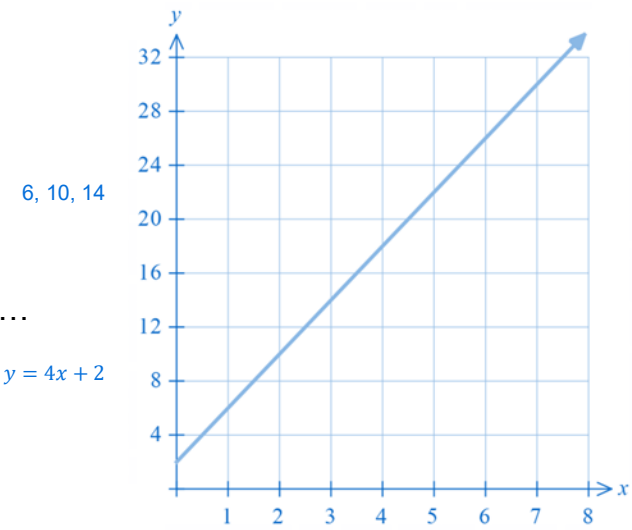
Table:

x	1	2	3
y			

Equation:

.....

Graph:



- 10 x is “number of tables”
 y is “number of chairs”

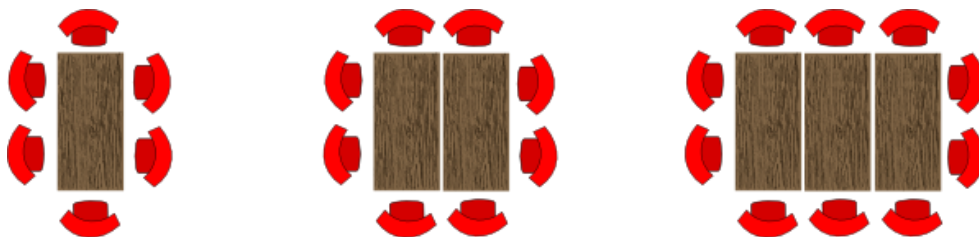


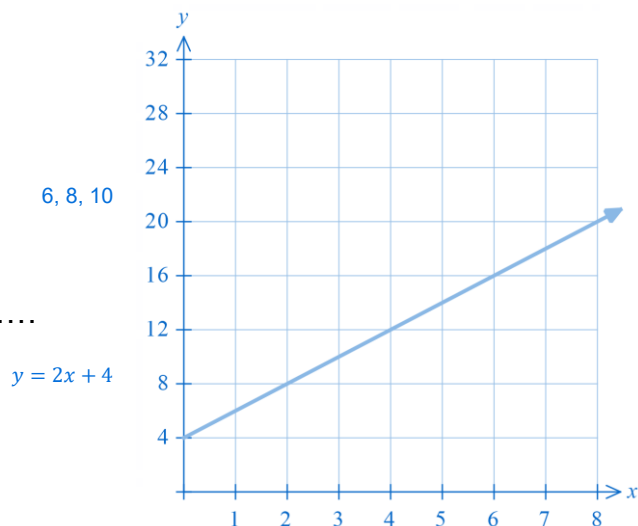
Table:

x	1	2	3
y			

Equation:

.....

Graph:



11

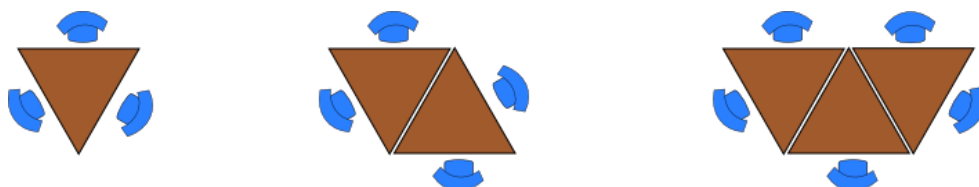


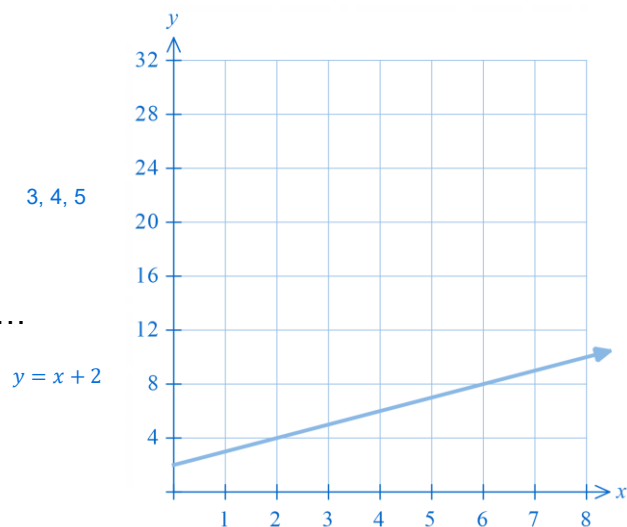
Table:

x	1	2	3
y			

Equation:

.....

Graph:



Applying the Equation

Review

☐ Substitute into a formula and solve

a $y = 3x + 5$
Find y when $x = 4$

b $y = 4x + 3$
Find y when $x = 10$

c $y = 3x + 5$
Find x when $y = 35$

d $y = 4x + 3$
Find x when $y = 23$

Using the Equation to Make Predictions

We can use the equation to make predictions without having to draw out the entire pattern.

1. Identify whether you are given x or y .
2. Substitute into the equation and solve.

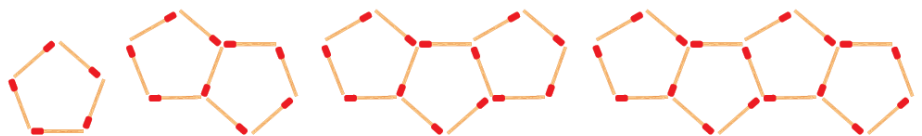
Example Apply the equation of a linear relationship to make predictions

Consider the pattern of pentagons and matches.

The equation is $y = 4x + 1$

x is number of pentagons

y is number of matches



If we want to make 25 pentagons, how many matches do we need?

$$\begin{aligned}\text{Let } x &= 25 \\ y &= 4x + 1 \\ &= 4(25) + 1 \\ &= 101\end{aligned}$$

We need 101 matches to make 25 pentagons.

If we have 73 matches, how many pentagons can we make?

$$\begin{aligned}\text{Let } y &= 73 \\ 73 &= 4x + 1 \\ 72 &= 4x \\ 18 &= x\end{aligned}$$

We can make 18 pentagons with 73 matches.

Guided Practice

The rule relating number of triangles and number of matches is

$$y = 2x + 1$$

x is number of triangles

y is number of matches



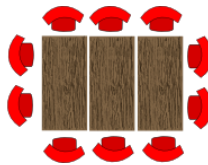
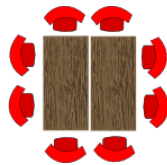
a How many matches do we need to make 36 triangles?

b How many triangles can we make with 27 matches?

73

13

Find the equation relating x and y , then use the equation to find the missing values.



a Complete the table of values.

Number of tables	x	1	2	3
Number of chairs	y			

6, 8, 10

b Determine an equation that relates the two quantities.

c If we want 36 tables,
how many chairs do we need?

Let =

d If we have 42 chairs,
how many tables can we have?

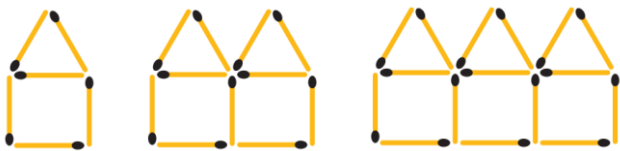
Let =

$$y = 2x + 4$$

76

19

1 This matchstick pattern has the equation $y = 5x + 1$
 x is number of 'houses'
and y is number of matchsticks



- a How many matchsticks do you need to make 10 'houses'?
- b How many houses can you make with 86 matchsticks?

51

17



- a Complete the table of values.
- | | | | |
|-----------------------|---|---|---|
| Number of shapes x | 1 | 2 | 3 |
| Number of matches y | | | |
- b Determine an equation that relates the two quantities.

$y = 3x$

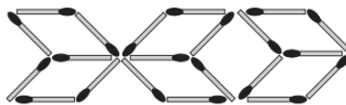
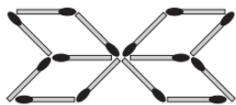
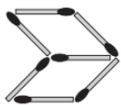
- c How many matches are needed for a pattern with 14 triangles?

Let =
.....
.....
..... 42 matches

- d How many triangles can we make with 120 matches?

Let =
.....
.....
..... 40 triangles

3



a Complete the table of values.

x	1	2	3	4	5
y					

b Determine an equation that relates the two quantities.

$$y = 7x$$

c How many matches are needed for a pattern with 10 shapes?

Let =

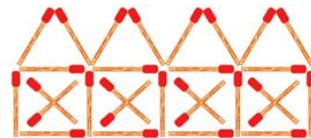
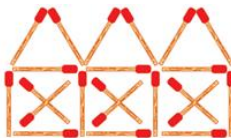
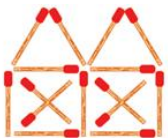
.....

.....

.....

70 matches

4



a Complete the table of values.

x	1	2	3	4	5
y					

b Determine an equation that relates the two quantities.

$$y = 7x + 1$$

c How many matches are needed for a pattern with 30 “houses”?

Let =

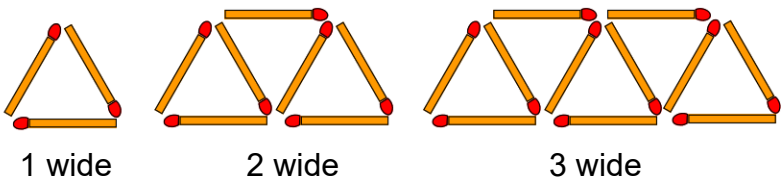
.....

.....

.....

211 matches

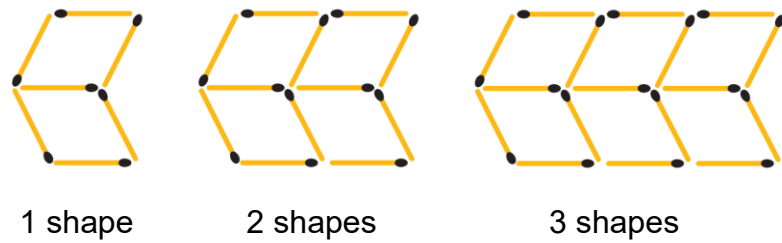
5



By first finding the equation of the pattern, determine how many matches are needed to make a shape 40 wide.

x					
y					

6

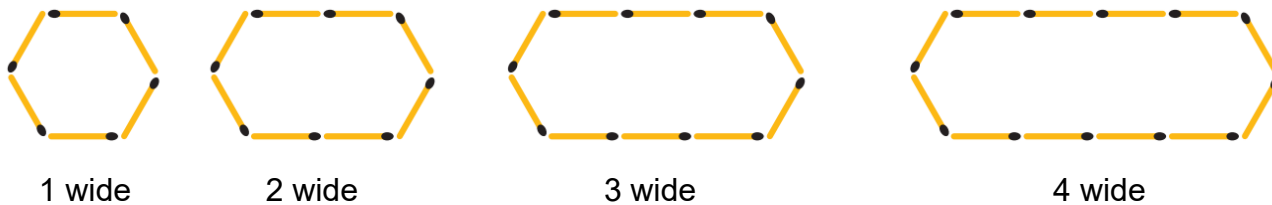


By first finding the equation of the pattern, determine how many matches are needed to make 17 shapes.

$y = 4x - 1$
159 matches

$y = 5x + 2$
87 matches

7



a Determine the equation of the pattern.

$$y = 2x + 4$$

b Determine how many matches are needed to make a shape 27 wide.

58

c Determine the width of a shape that can be made with 136 matches.

66

8 A pattern of towers is being assembled using blocks shown.



a Determine the equation of the pattern, relating number of blocks (y) to height (x).

$$y = 2x - 1$$

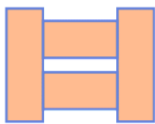
b Determine how many blocks would be needed for a tower 43 high.

85

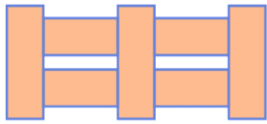
c Determine the height of a tower that can be made with 133 blocks.

67

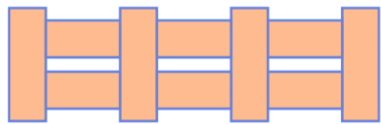
9 Consider this fence.



1 section



2 sections



3 sections

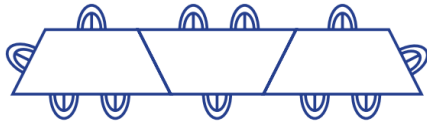
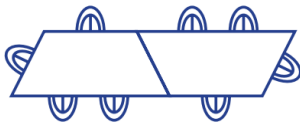
a Determine how many wooden planks are needed to make 20 fence sections.

b Determine how many fence sections can be made with 34 wooden planks.

61

11

10 Consider this table arrangement.



a Determine how many chairs are needed for 10 tables in a straight line.

b The tables are to be arranged in a single straight line for a picnic.
How many tables are needed to seat 65 people?

32

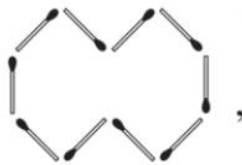
21

11 2007 HSC Maths Standard 2 Band 6

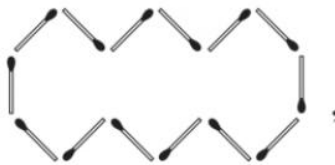
Chris started to make this pattern of shapes using matchsticks.



Shape 1



Shape 2



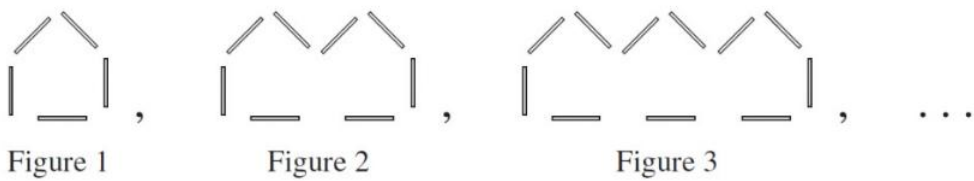
Shape 3

...

If the pattern of shapes is continued, which shape would use exactly 486 matchsticks?

121st shape

12 2011 HSC Maths Standard 2 Band 5

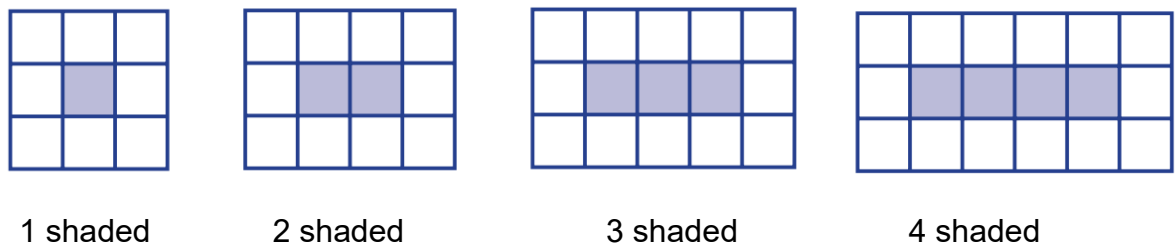


Is it possible to create a figure in this pattern using exactly 543 sticks?
Show suitable calculations to support your answer.

$y = 3x + 2$
 $543 = 3x + 2$
 $3x = 541$
 $x = 180.333 \dots$

No, as x is not a whole number.

13 A tiling pattern is made from surrounding a row of shaded tiles with unshaded tiles.



a Determine the total number of tiles needed to make a pattern with 23 shaded tiles.

75

b Determine the number of shaded tiles in a pattern with 45 total tiles.

13

c Determine how many *unshaded* tiles would be in a pattern with 17 shaded tiles.

40

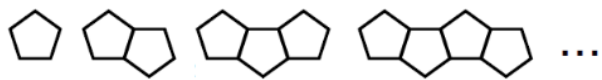
d Determine how many *unshaded* tiles would be in a pattern with 72 total tiles.

Multiple Representations of Linear Relationships

① Linear Relationships

A linear relationship can be a pattern, a table of values, an equation, and coordinates forming a straight-line graph on a Cartesian plane.

Pattern



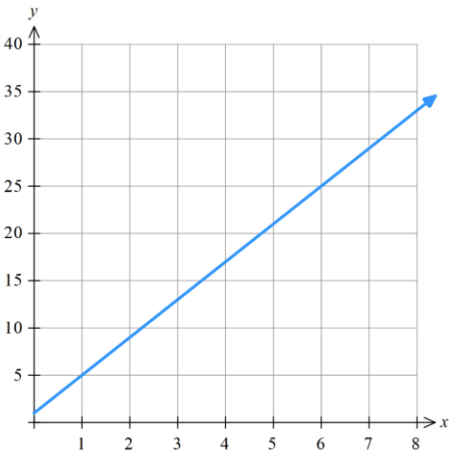
Table

Pentagons x	1	2	3	4
Matches y	5	9	13	17

Equation

$$y = 4x + 1$$

Straight-line Graph



1

Pattern:



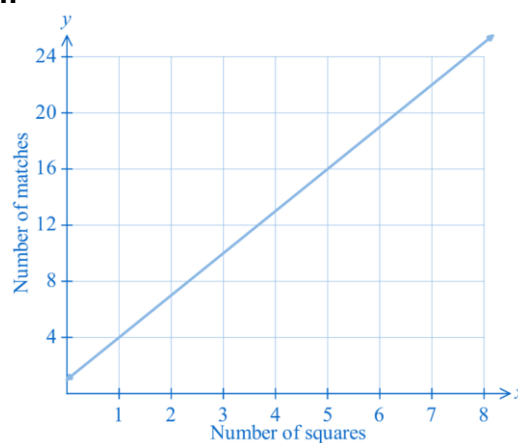
Table:

Squares	x	0	1	2	3	4
Matches	y					
		1	4	7	10	13

Equation:

$$y = 3x + 1$$

Graph:



2

Sequence:



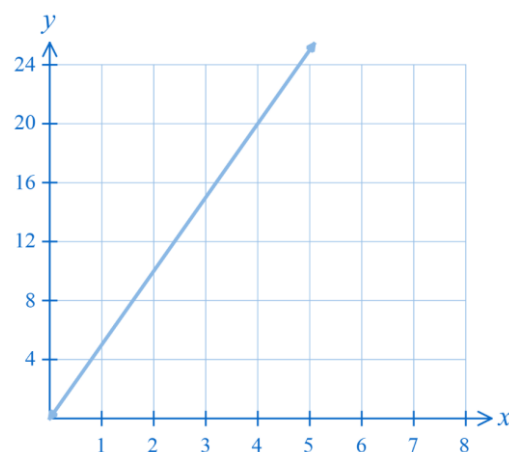
Table:

x	0	1	2	3	4
y					
	0	5	10	15	20

Equation:

$$y = 5x$$

Graph:



3

Pattern:

1st term



2nd term



3rd term



4th term



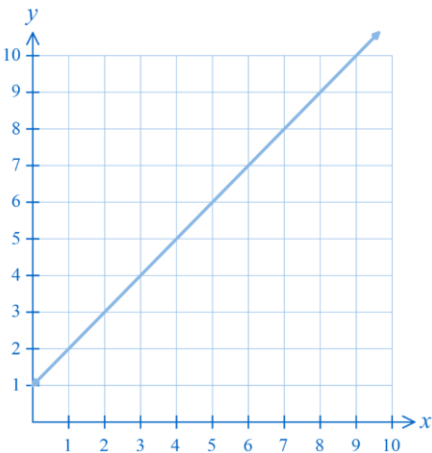
Table:

Term x	0	1	2	3	4	5	43
Value y		2	3	4	5	6	44

Equation:

$y = x + 1$

Graph:



4

Pattern:

1st term

2nd term



3rd term



4th term



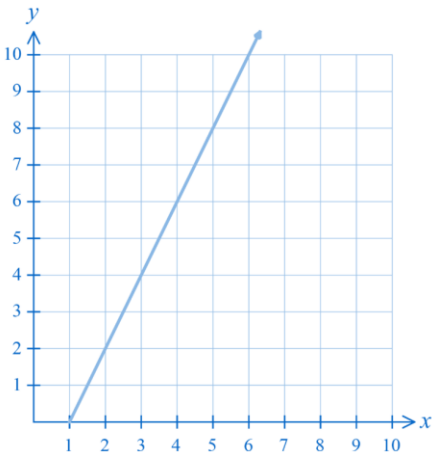
Table:

x	0	1	2	3	4	5	17
y	-2	0	2	4	6	8	32

Equation:

$y = 2x - 2$

Graph:



5

Pattern:

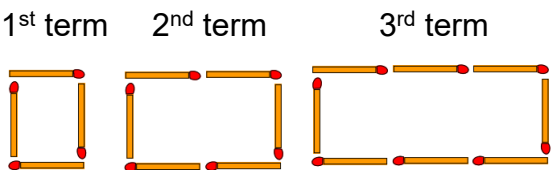


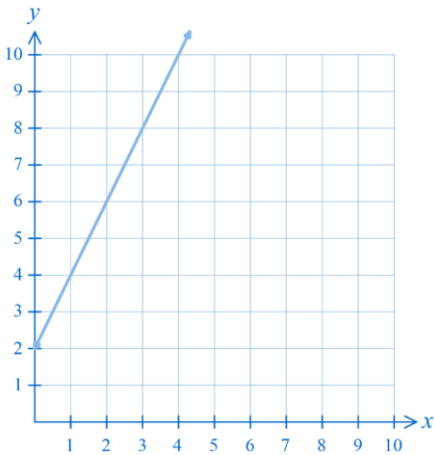
Table:

<i>x</i>	0	1	2	3	4	5	18
<i>y</i>	2	4	6	8	10	12	38

Equation:

$y = 2x + 2$

Graph:



6

Pattern:

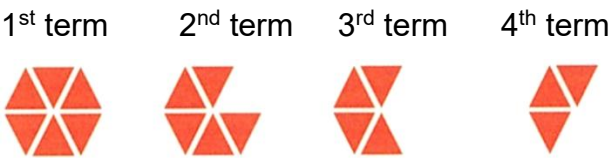


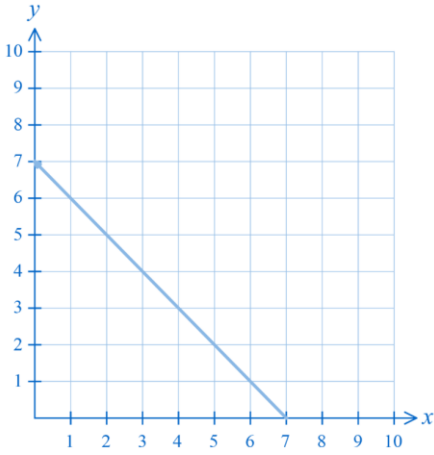
Table:

<i>x</i>	0	1	2	3	4	5	17
<i>y</i>	7	6	5	4	3	2	-10

Equation:

$y = -x + 7$

Graph:



7

Pattern:

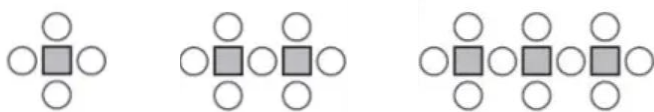


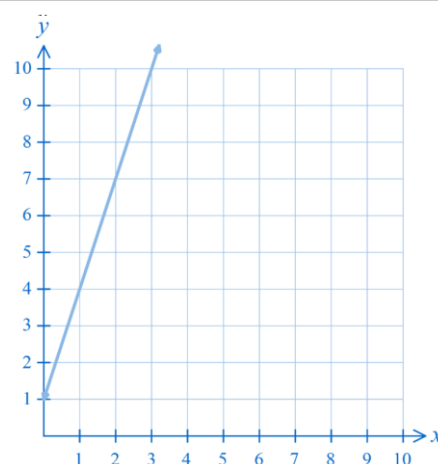
Table:

x	0	1	2	3	4	5	
							9
y		4					28
	1		7	10	13	16	

Equation:

$$y = 3x + 1$$

Graph:



8 For this question, fill in every section by first looking at the table.

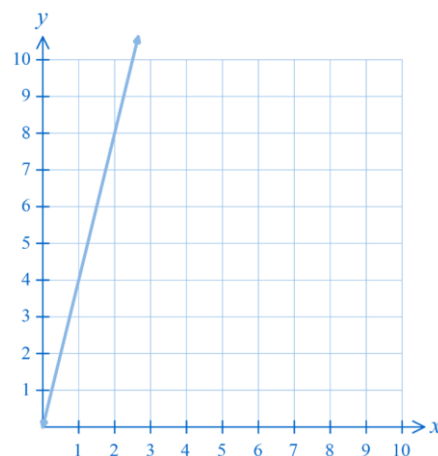
Table:

x	0	1	2	3	4	21	
							19
y		4	8	12			76
	0				16	84	

Equation:

$$y = 4x$$

Graph:



9 Fill in every section by first looking at the table.

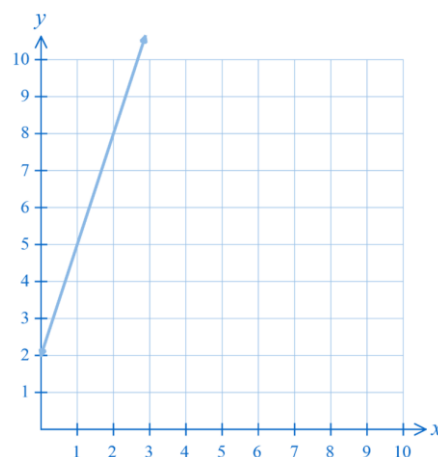
Table:

x	0	1	2	3	4	81	
							60
y		5	8	11			182
	2				14	245	

Equation:

$$y = 3x + 2$$

Graph:



10 Fill in every section by first looking at the table.

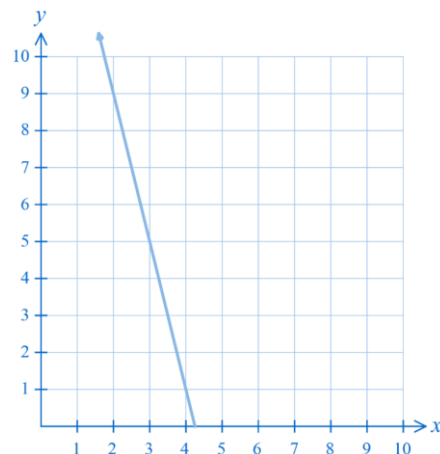
Table:

x	0	1	2	3	4	-4	
y		13	9	5	1		85

Equation:

$$y = -4x + 17$$

Graph:



11 Fill in every section by first looking at the graph.

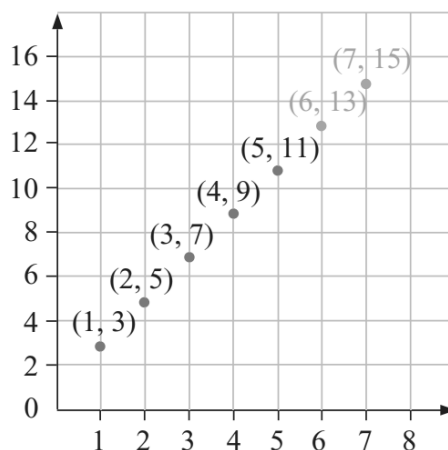
Table:

x							
y							

Equation:

$$y = 2x + 1$$

Graph:



12 Fill in every section by first looking at the graph.

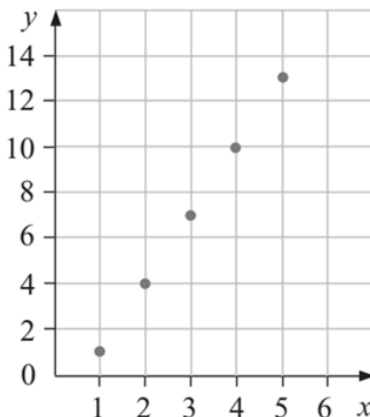
Table:

x					
y					

Equation:

$$y = 3x - 2$$

Graph:



13 Fill in every section by first looking at the graph.

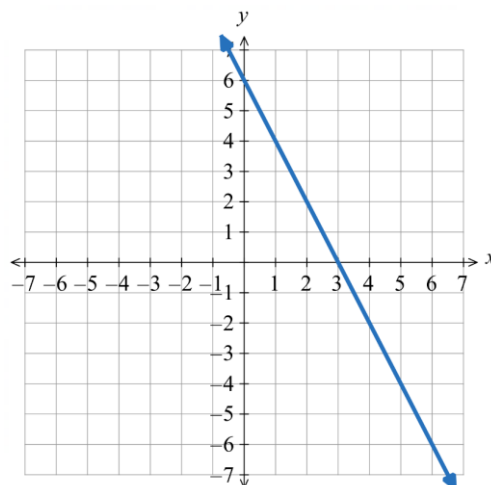
Table:

x				
	0	1	2	3
y				
	6	4	2	0

Equation:

$$y = -2x + 6$$

Graph:



14 Fill in every section by using the equation

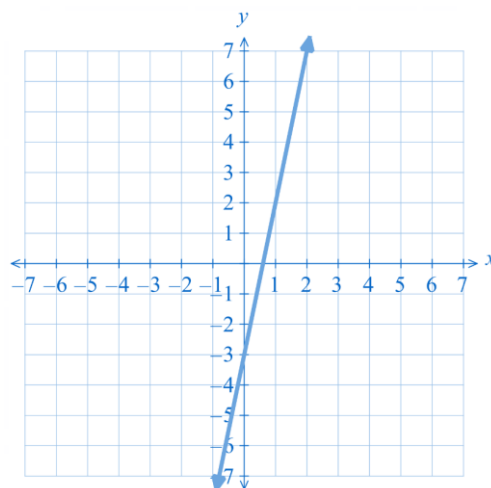
Table:

x	0	1	2	3	4
y					
	-3	2	7	12	17

Equation:

$$y = 5x - 3$$

Graph:



15 Fill in every section by using the equation

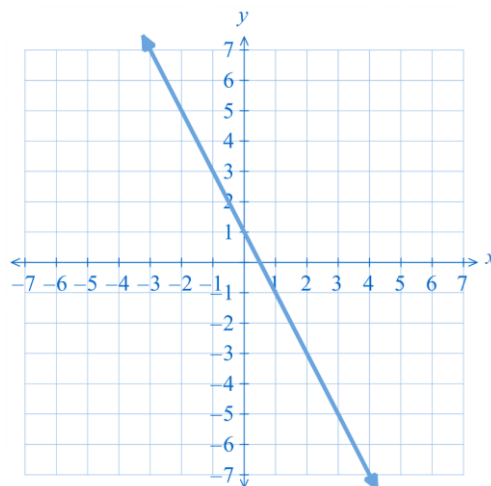
Table:

x	0	1	2	3	4
y					
	1	-1	-3	-5	-7

Equation:

$$y = -2x + 1$$

Graph:



Linear Relationships in Real Life Contexts

i Variables

A **variable** is a quantity that can change.

Examples: Weight w , cost c , time t , distance d , speed s

The **dependent variable** depends on another variable.

It is the **second row** of a table of values.

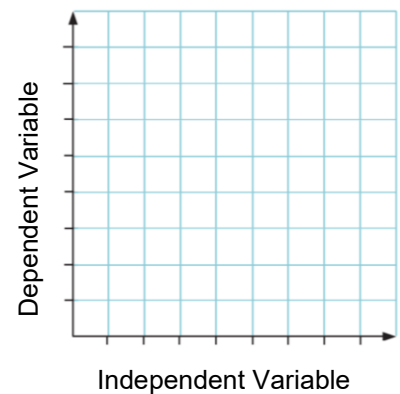
It is the **y axis** of a graph.

The **independent variable** does not depend on anything.

It is the **first row** of a table of values.

It is the **x axis** of a graph.

Independent variable x	0	1	2	3
Dependent variable y	3	5	7	9



💡 Key Ideas

1. The independent variable is the first of a table of values and always goes on the axis.
2. The dependent variable is the second of a table of values and always goes on the axis.
3. For matchstick patterns:
 - a. The number of matches was the variable.
 - b. The number of shapes was the variable.
4. For *age* and *height*:
 - a. is the dependent variable, since it changes based on
 - b. is the independent variable, since it doesn't depend on anything.
5. If you graph the dependent and independent variable, and you can draw a straight line through all the coordinates, then we say that the variables have a relationship.

② **Investigation** Real life contexts and linear relationships

A taxi has a \$10 hiring fee (also called a 'flag fall' fee) and costs \$4 per kilometre.

Independent variable x :

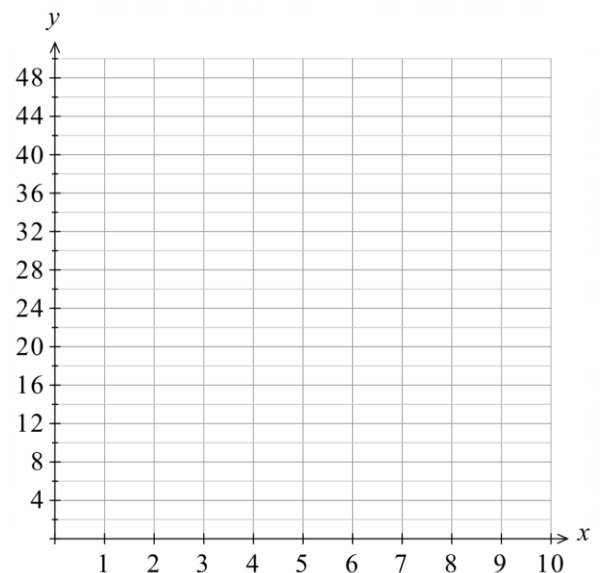
Dependent variable y :

Complete the table of values.

Distance travelled	x	0	1	2	3	4
Cost	y					

Determine an equation that relates the two quantities.

Graph the relationship on the Cartesian Plane



Find the cost of a journey of 25 km in distance.

Find how far you can travel for \$70.

Guided Practice

The cost of a taxi is \$5 to start and \$3 per kilometre.

Independent variable

Distance in km

Dependent variable

Cost in \$

..... x	0	10	20	30
..... y				

5, 35, 65, 95

Equation

Graph

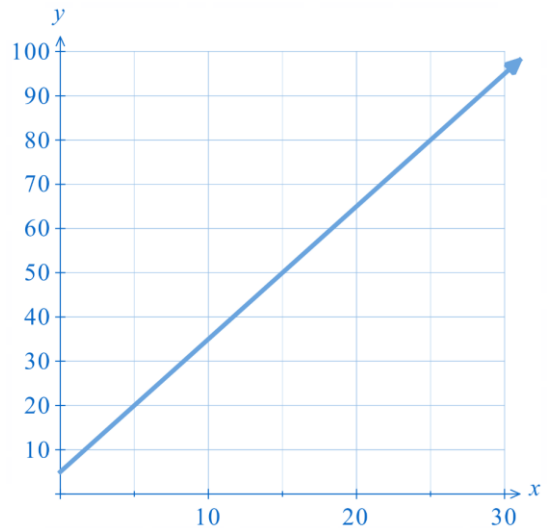
Find the cost of travelling 15 km.

$$y = 3x + 5$$

\$50

Find how far you could travel with \$80

25 km



A candle starts at 100 cm and decreases in height by 4 cm per hour.

Independent variable

Time in hours

Dependent variable

Height in cm

..... x	0	1	2	3
..... y				

100, 96, 92, 88

Equation

Graph

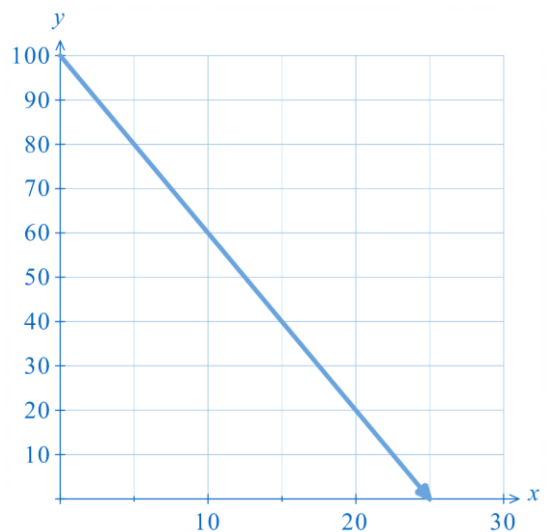
Find how long it takes for the candle to completely burn out.

$$y = -4x + 100$$

25 hours

Find the height of the candle after 15 hours.

40 cm



1

Scenario:

The cost of travelling in a hire car is \$10 booking fee plus \$6 per kilometre.

Independent variable

Distance (km)

Dependent variable

Cost (\$)

Table:

..... x Distance (km)	0	5	10	15
..... y Cost (\$)	10	40	70	100

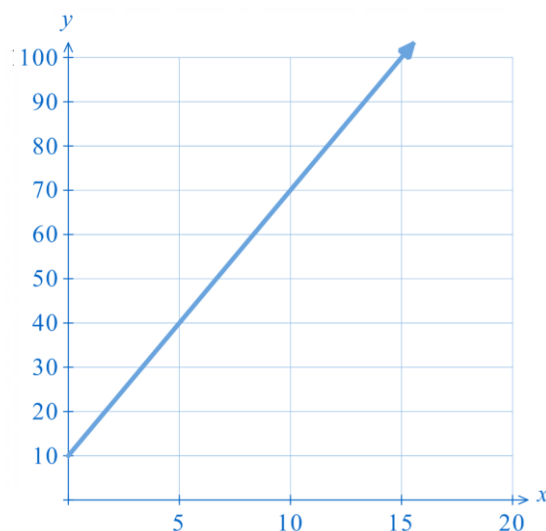
Equation:

$$y = 6x + 10$$

a Find the cost of a 25 km journey.

\$160

Graph:



b How far can the car be driven for \$60?

8.333... km

2

Scenario:

Oranges are sold for \$3.00 per kilogram

Independent variable

Mass (kg)

Dependent variable

Cost (\$)

Table:

..... x Mass (kg)	0	1	2	3	4
..... y Cost (\$)	0	3	6	9	12

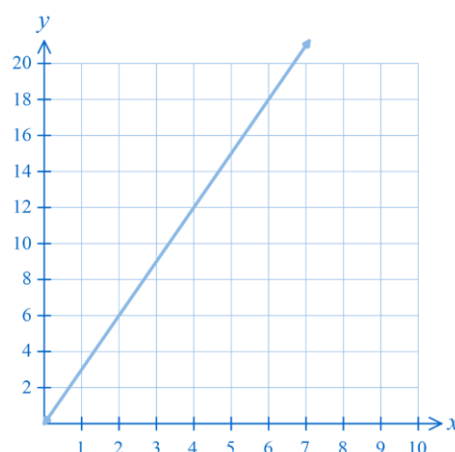
Equation:

$$y = 3x$$

a Find the cost of 17 kg of oranges.

\$51

Graph:



b Find how many kilograms of oranges could be purchased for \$21.90

7.3 kg

3

Scenario:

The cost of hiring a plumber is \$80 call out fee and \$240 per hour.

Independent variable

Time (hours)

Dependent variable

Cost (\$)

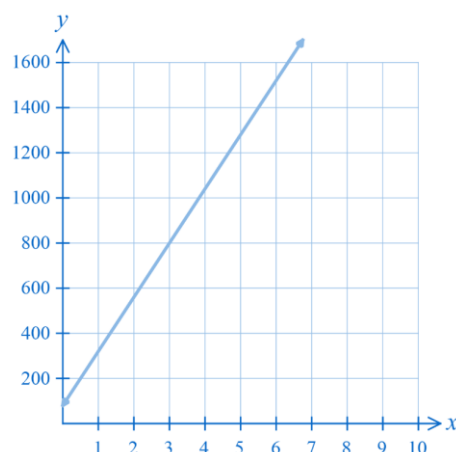
Table:

..... x Time (h)	0	1	2	3	4
..... y Cost (\$)	80	320	560	800	1040

Equation:

$$y = 240x + 80$$

- a** How much does it cost for a 3.5-hour job?

Graph:

- b** How long can you afford the plumber for if you have \$200?

\$920

30 minutes

4

Scenario:

A car uses 3 L of fuel per hour.

Its fuel tank initially holds 32 L.

Independent variable

Time (h)

Dependent variable

Fuel (L)

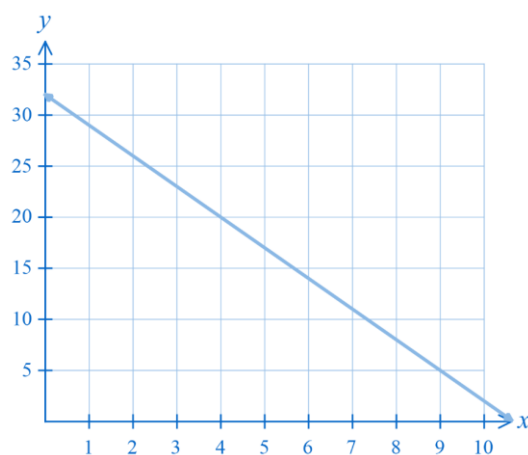
Table:

..... x Time (h)	0	1	2	3	4
..... y Fuel (L)	32	29	26	23	20

Equation:

$$y = -3x + 32 \text{ or } y = 32 - 3x$$

- a** How much fuel would be left after 4 hours 30 minutes?

Graph:

- b** When exactly, to the nearest minute, will the car run out of fuel?

18.5 L

10 h 40 min

5

Scenario:

A candle starts off 3 cm tall and decreases by half a centimetre every minute.

Independent variable

Time (min)

Dependent variable

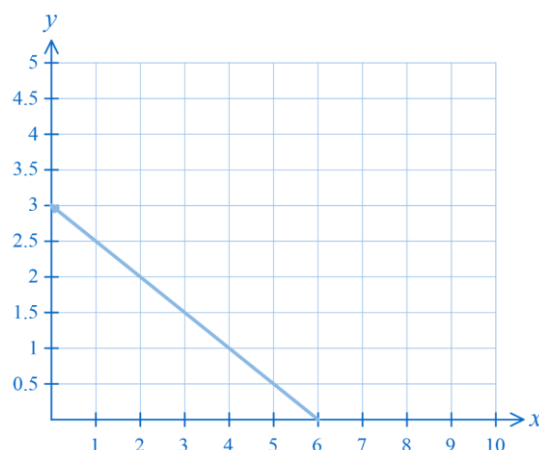
Height (cm)

Table:

..... x Time (min)	0	1	2	3	4
..... y Height (cm)	3	2.5	2	1.5	1

Equation:

$$y = -\frac{x}{2} + 3 \text{ or } y = 3 - \frac{x}{2}$$

Graph:

a How tall is the candle after 15 minutes?

-4.5 cm

b Based on your answer to part **a**, what might be a limitation of this model?

The formula predicts a negative height, which is not possible.
This model is only valid for positive values of x and y

6

Scenario:

A water tank initially holds 20 L of water.
During a rainstorm, its volume increases by 4 L per hour.

Independent variable

Time (hr)

Dependent variable

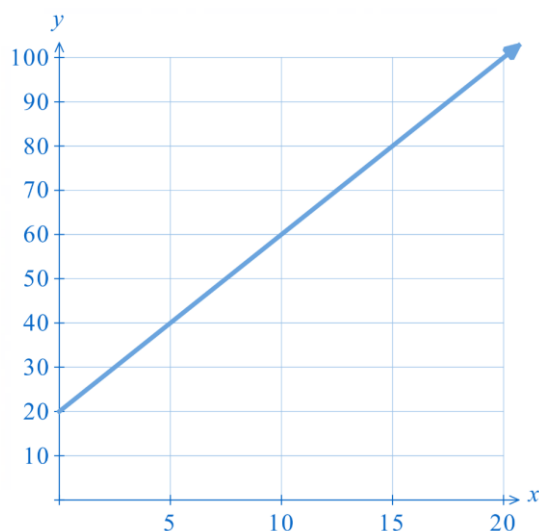
Volume (L)

Table:

..... x Time (hr)	0	5	10	15	20
..... y Volume (L)	20	40	60	80	100

Equation:

$$y = 4x + 20$$

Graph:

a How long would it take to fill to 100 L?

20 hours

b Why might your answer to part **a** not be accurate in real life?

The water tank may not hold 100 L, and the rain may not last that long.

7 Write an equation for the following situations. Make the first listed pronumeral the subject.

a $\$D$ given c cents

$$D = \frac{c}{100}$$

b d cm given e metres

$$d = 100e$$

c The discounted price $\$D$ that is 30% off the marked price $\$M$

$$D = 0.7M$$

d The value of an investment $\$V$ that is 15% more than the initial amount $\$P$

$$V = 1.15P$$

e The cost $\$C$ of hiring a car at $\$50$ upfront plus $\$18$ per hour for t hours

$$C = 50 + 18t$$

f The distance d km remaining in a 42 km marathon after t hours if the running speed is 14 km/h

$$d = 42 - 14t$$

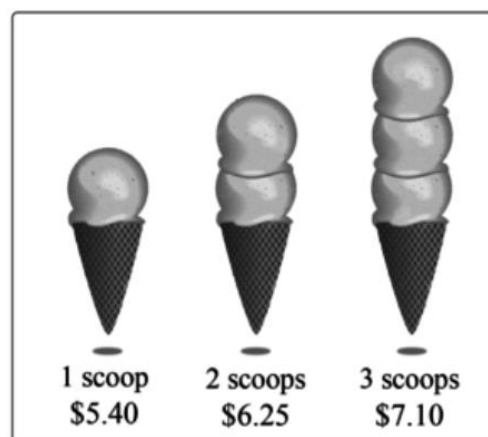
g The price $\$P$ of a bottle of soft drink if b bottles cost $\$c$

$$P = \frac{c}{b}$$

8 NAPLAN A

Each extra scoop of ice cream costs the same amount of money.

How much will an ice cream with 7 scoops cost?



each scoop costs $\$0.85$

The equation is $y = 0.85x + 4.55$

$\therefore \$10.50$

9 NAPLAN A

Sabre is saving to buy a new skateboard. After one week she has saved \$11. She then saves the same amount of money each week.

Week	1	2	3	4	5
Total amoun saved	\$11	\$18	\$25	\$32	\$39

How much money will she have saved by the end of week 10?

$$y = 7x + 4$$

$$\$74$$

10 NAPLAN A

A plumber calculates the price of a job using a call out fee and an amount per hour. This table shows some of the job prices. Determine the call out fee and amount per hour.

Hours	2	4	6	8
Job price	\$130	\$202	\$274	\$346

$$\text{Amount per hour: } \$36$$

$$\text{Call out fee: } \$58$$

11 2020 HSC Maths Standard 2 Band 5

A plumber charges a call-out fee of \$90 as well as \$2 per minute while working.

Suppose the plumber works for t hours.

Which equation expresses the amount the plumber charges ($\$C$) related to time (t hours)?

A. $C = 2 + 90t$

B. $C = 90 + 2t$

C. $C = 120 + 90t$

D. $C = 90 + 120t$

D

Comparing Linear Relationships

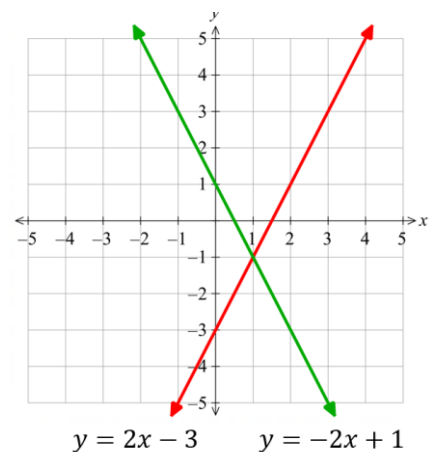
i Equation of a Linear Relationship

We have special words to describe parts of a linear equation.

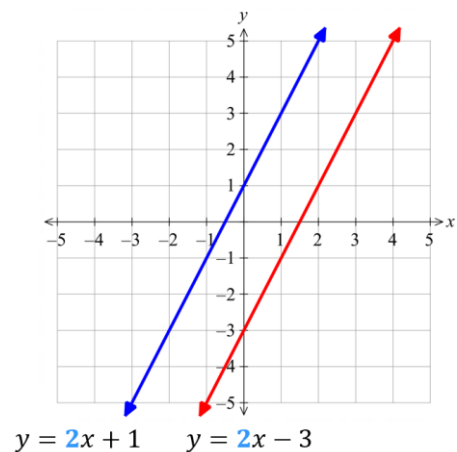
$$y = \underset{\substack{\uparrow \\ \text{coefficient of } x \\ \text{gradient} \\ \text{(steepness)}}}{mx} + \underset{\substack{\uparrow \\ \text{constant term} \\ \text{y-intercept} \\ \text{(where the line crosses the y-axis)}}}{c}$$

If the **coefficient of x is positive**,
the graph **slopes upwards** from left to right;
increasing graph.

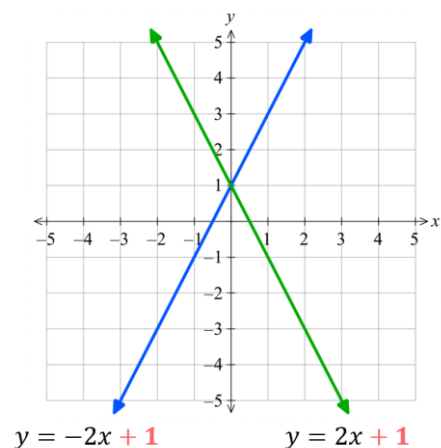
If the **coefficient of x is negative**,
the graph **slopes downwards** from left to right;
decreasing graph.



If two equations have the **same coefficient**,
then the lines are **parallel**.



If two equations have the **same constant term**,
then the **y-intercepts** are the same.



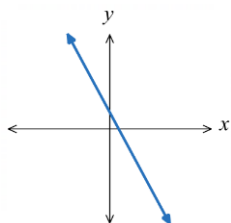
? Identify Increasing and decreasing lines

When m is **positive**, the line is **increasing**; it slopes up from left to right.

When m is **negative**, the line is **decreasing**; it slopes down from left to right.

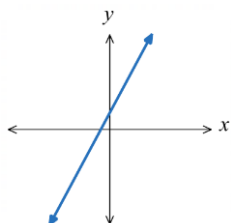
Decreasing

$$y = -2x + 1$$



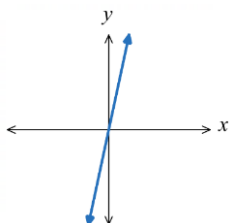
Increasing

$$y = 2x + 1$$



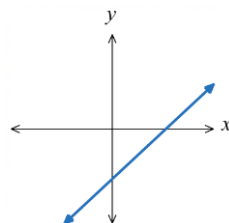
Increasing

$$y = 5x$$



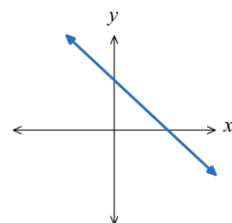
Increasing

$$y = x - 3$$



Decreasing

$$y = 3 - x$$



Identify whether these equations have increasing or decreasing graphs.

a $y = 5x - 12$

b $y = -5x + 12$

c $y = 12 - 5x$

d $y = -3x + 11$

e $y = \frac{x}{2} + 4$

f $y = -6 + 7x$

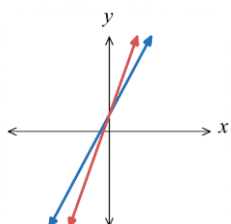
? Identify Parallel lines

Two lines with the **same gradient** m are **parallel**; same direction and never meet.

Not parallel

$$y = 2x + 1$$

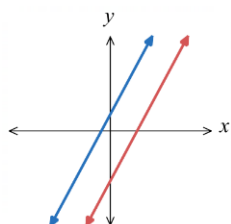
$$y = 3x + 1$$



Parallel

$$y = 2x + 1$$

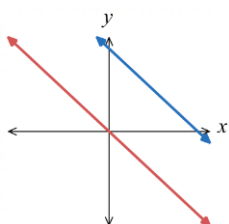
$$y = 2x - 3$$



Parallel

$$y = 5 - x$$

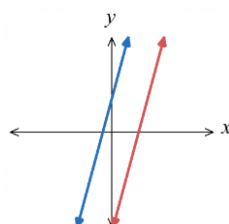
$$y = -x$$



Parallel

$$y = 4x + 2$$

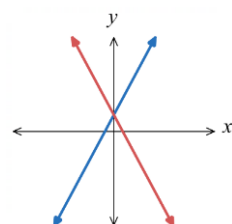
$$y = -6 + 4x$$



Not parallel

$$y = 2x + 1$$

$$y = -2x + 1$$



Decide whether these lines are parallel.

a $y = 5x - 12$
 $y = 5x + 48$

b $y = -5x + 1$
 $y = 5x + 1$

c $y = \frac{x}{2} + 11$
 $y = x + 11$

d $y = 6 - 2x$
 $y = -2x$

e $y = 4x + 1$
 $y = 4x$

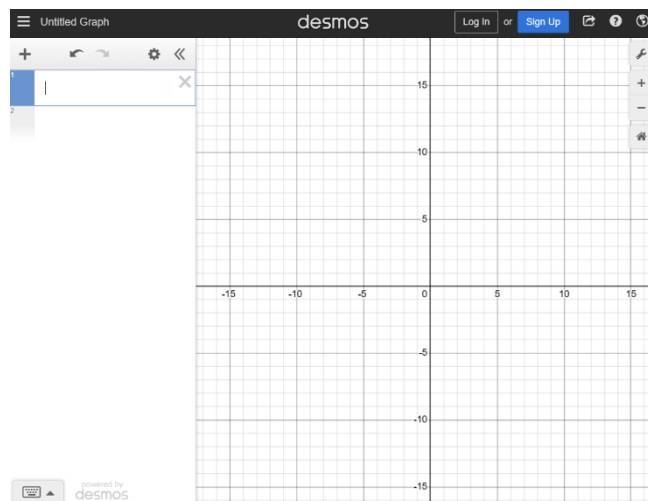
f $y = 12x - 1$
 $y = 21x + 45$

? Investigation Graphing software

Go to

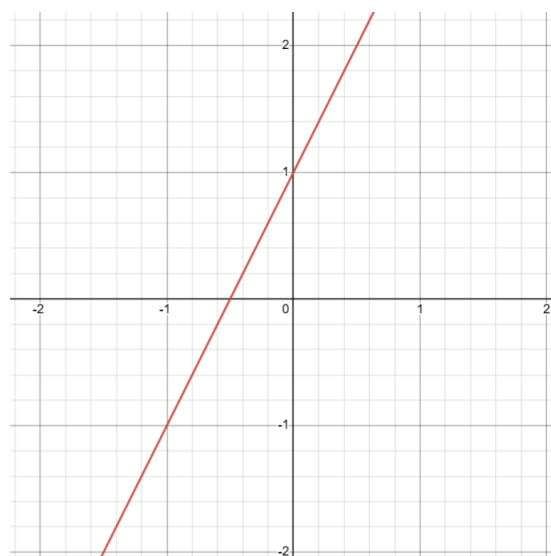
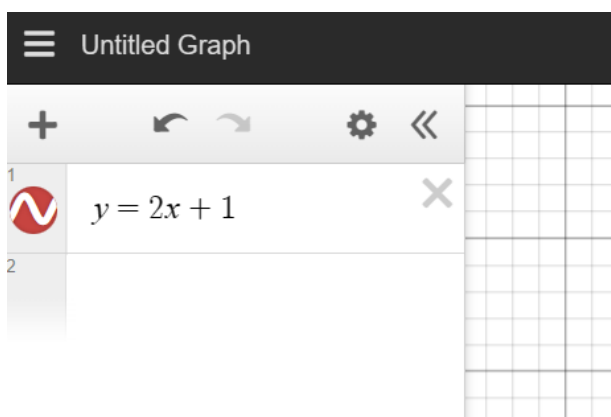
 www.desmos.com/calculator

your screen should look like the image.



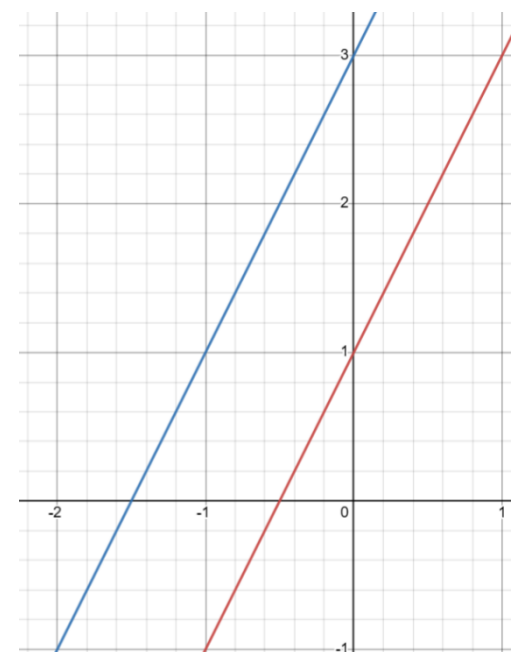
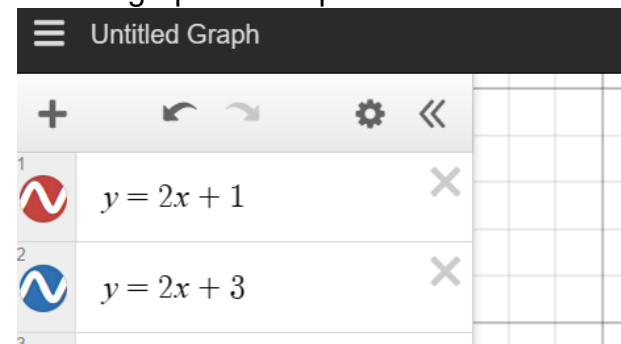
On the left, type in $y = 2x + 1$

This will graph the equation. Use the mouse wheel to zoom in and out and hold down left click to move around.



Below the first equation, type $y = 2x + 3$

This will graph both equations in different colours.



You can use the **X** button to remove the graphs.

Using DESMOS, draw a sketch and explain the similarities and differences between the set of lines.

Use these words to describe the lines: **gradient, parallel, increasing/decreasing, y-intercept**

Use these words to describe the equations: **coefficient, constant term**

1

$$y = 4x + 1$$

$$y = 4x - 3$$

$$y = 4x$$

These lines are all because the of x are the same.

parallel, coefficient

The lines have different because the constant terms are different.

y-intercept

2

$$y = 2x + 2$$

$$y = 4x + 2$$

$$y = 7x + 2$$

The lines have the same because the are the same.

y-intercept, constant term

The lines have different because the of x are not the same.

gradient, coefficient

3

$$y = 3x + 1$$

$$y = -3x + 1$$

The lines both have the same

y-intercept

because

the constant terms are the same

The lines have different

gradient

because

the coefficients of x are different

4

$$y = 2x$$

$$y = -2x$$

$y = 2x$ is an increasing graph because the of x is positive.

coefficient

$y = -2x$ is a graph because

decreasing, coefficient of x is negative

5

$$y = -2x + 2$$

$$y = -2x + 1$$

$$y = -2x - 3$$

The lines are all because they have the same of x .

parallel, coefficient

The lines are all decreasing because their coefficients are

negative

The lines have different because are different.

y-intercept, constant terms are different

6 Complete each statement:

a All equations of a line have the form:

$y = mx + c$

b m is the c..... of x and represents the g....., or steepness of the line.

coefficient, gradient

c c is the c..... t..... and represents the ...-....., or where the line cuts the-axis.

constant term, y-intercept, y

d The equations of parallel lines have the same

coefficient of x

e Equations with the same have graphs with the same y-intercept.

constant term

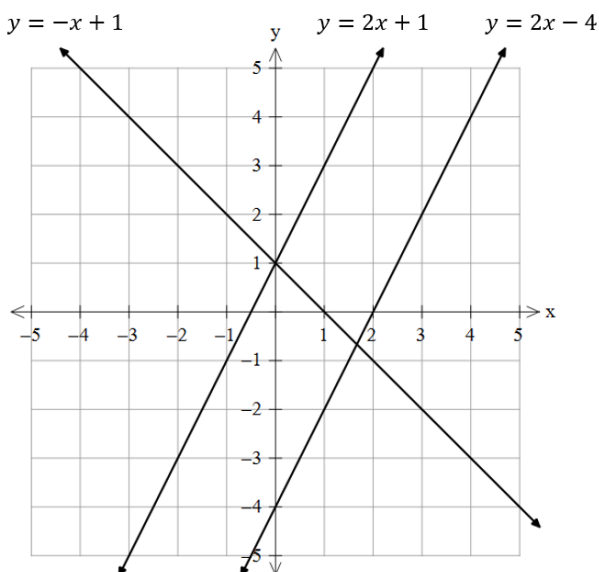
f Linear equations with a positive coefficient of x have graphs that are

increasing

g Linear equations with a coefficient of x have graphs that are decreasing.

negative

7 Consider these three lines.



- a Which two lines are parallel?
What part of the equation tells you this?

$$y = 2x + 1, y = 2x - 4$$

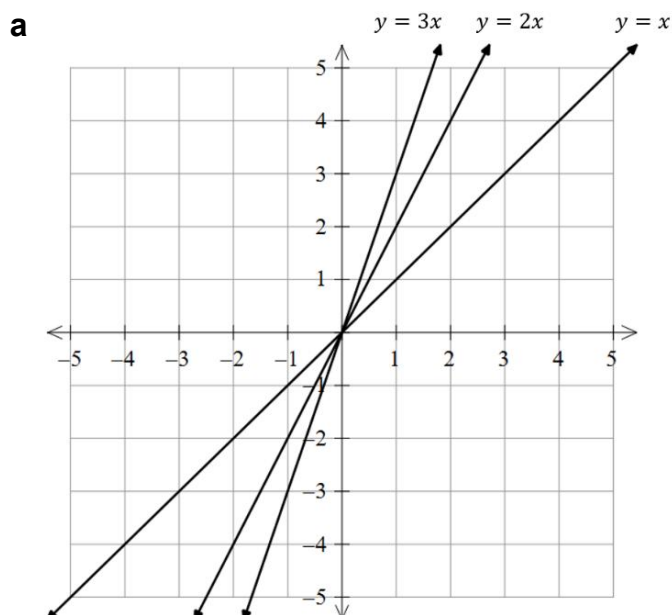
Coefficients of x are the same

- b Which two lines have the same y -intercept? What part of the equation tells you this?

$$y = -x + 1, y = 2x + 1$$

Constant terms are the same

8 Consider these lines.



Which line is the steepest?

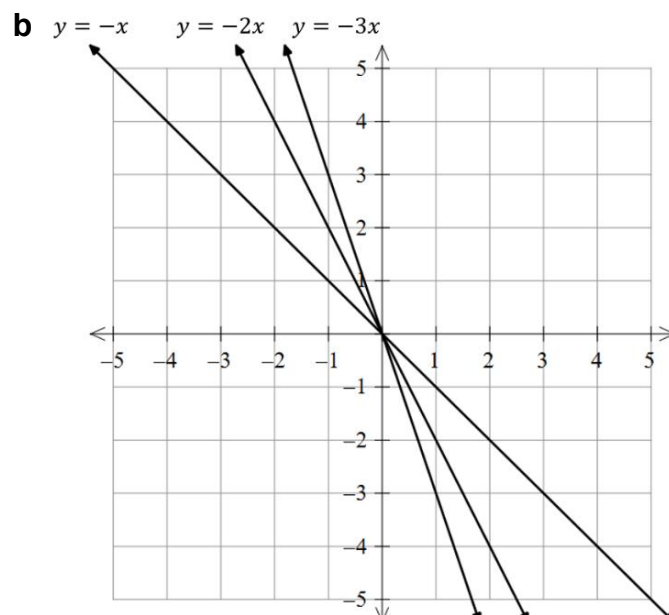
$$y = 3x$$

Which line is the least steep?

$$y = x$$

How does the coefficient of x tell you the steepness of a graph?

The larger the magnitude of the coefficient of x , the steeper

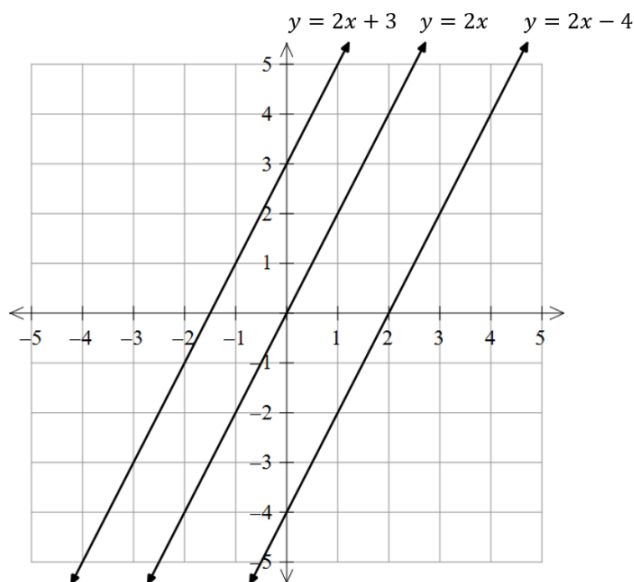


All the graphs in part a are increasing.

All the graphs shown here are decreasing.

What part of the equation tells you whether a graph is increasing or decreasing?

The sign of the coefficient of x

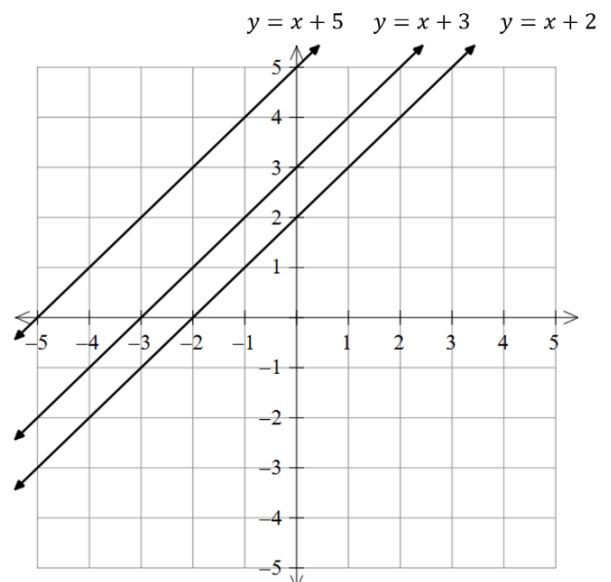
c

All of the lines are parallel.
What part of the equation tells you whether two or more lines will be parallel?

The coefficients of x are the same

Write down the equation of any other line that is parallel to these lines.

Any other line in the form $y = 2x + c$

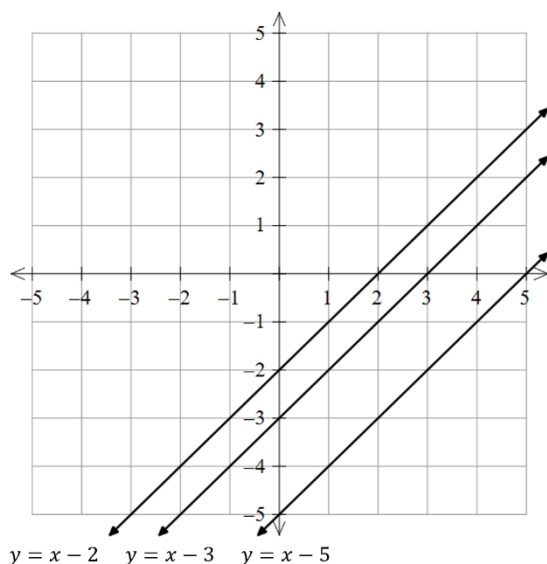
d

All these lines have a positive y -intercept.
Describe what this means.

The lines meet the y -axis above the x -axis.

What part of the equation tells you the y -intercept will be positive?

The constant terms are positive

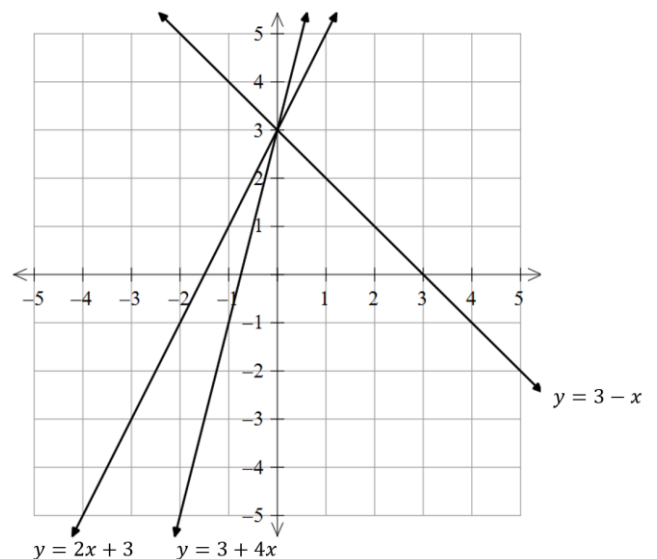
e

All these lines have a negative y -intercept.
Describe what this means.

The lines meet the y -axis below the x -axis.

What part of the equation tells you the y -intercept will be negative?

The constant terms are negative

f

All of the lines have the same y -intercept.
Explain what part of the equation tells you this.

The constant terms are the same

Write down the equation of any other line that has the same y -intercept.

Any other line in the form $y = mx + 3$

9 Consider the lines:

$y = 2x + 3, \quad y = 2x - 4, \quad y = -x + 1, \quad y = 4x + 3$

a Which lines are parallel?

$y = 2x + 3, y = 2x - 4.$

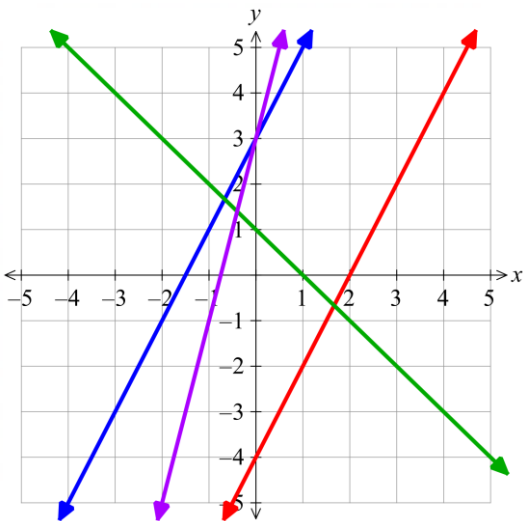
b Which lines are increasing?

$y = 2x + 3, y = 2x - 4, y = 4x + 3$

c Which lines have the same y-intercept?

$y = 2x + 3, y = 4x + 3$

d Label the four lines shown using the equations above.



Green: $y = -x + 1$
Blue: $y = 2x + 3$
Purple: $y = 4x + 3$
Red: $y = 2x - 4$

10 Consider the graph of $y = x + 2$. Using DESMOS, or the rules, describe what is the same and what is different between that graph and these graphs.

Comparison	Same	Different
$y = x + 2$ $y = x + 1$	Gradient	y-intercept
$y = x + 2$ $y = 2x + 2$	y-intercept	Gradient
$y = x + 2$ $y = -x + 2$	y-intercept	Gradient
$y = x + 2$ $y = x$	Gradient	y-intercept
$y = x + 2$ $y = x - 2$	Gradient	y-intercept

Solving Equations Graphically

Review

- ☐ Verify whether a value is a solution to an equation

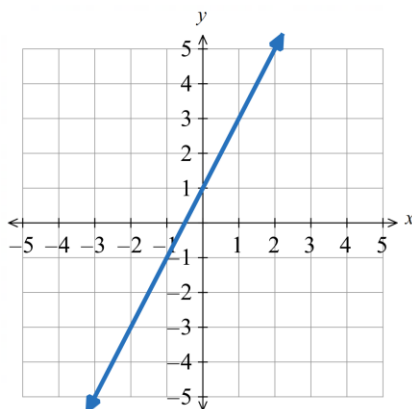
a Is $x = 3$ a solution of $2x + 1 = 7$?

b Is $x = 2$ a solution of $2x + 1 = 7$?

c Is $x = 2, y = 5$ a solution to $y = 2x + 1$?

- ☐ Read a graph

The graph shows $y = 2x + 1$.



What is the value of y when $x = 1$?

What is the x value when $y = 3$?

① Every Point on a Line is a Solution to the Equation

We draw a line using points formed by the equation.

Therefore, every point (x, y) that lies on a line makes the equation true.

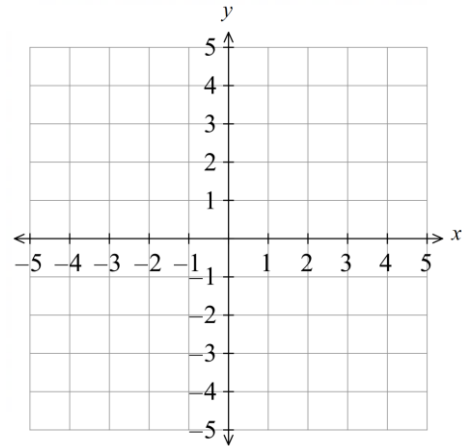
We say that every point on the line **satisfies** the equation.

② Investigation Every point on a line is a solution to the equation

Consider the equation $y = 2x + 1$.

Construct a table of values and sketch the line.

x	0	1	2	3	4
y					



Consider the point $(1, 3)$

Does $(1, 3)$ lie on the line?

Show that $x = 1, y = 3$ is a solution to
 $y = 2x + 1$.

Consider the point $(2, 4)$

Does $(2, 4)$ lie on the line?

Show that $x = 2, y = 4$ is not a solution to
 $y = 2x + 1$

Complete the sentence:

Every point that lies on the line is a solution to the equation.

Using the graph, what is the x value when
 $y = 5$?

Solve $2x + 1 = 5$ algebraically

Using the graph, what is the x value when
 $y = -3$?

Solve $2x + 1 = -3$ algebraically

? Investigation Every point on a line is a solution to the equation

Consider the equation $y = 2x + 1$.

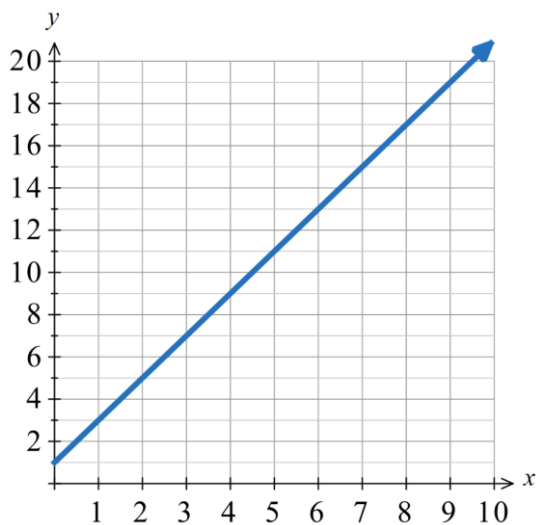
This represents this pattern:



x represents number of

y represents number of

This graph shows $y = 2x + 1$.



If we are asked to solve $13 = 2x + 1$

This means find how many triangles can be made with 13 matchsticks. Use the graph to solve this.

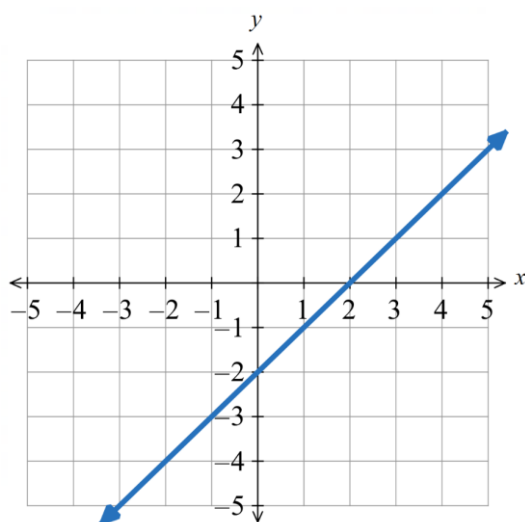
If we are asked to solve $17 = 2x + 1$

This means

.....

Use the graph to solve this.

This graph shows $y = x - 2$



Solve $3 = x - 2$ using the graph.

Solve $-3 = x - 2$ using the graph

① Solving Equations Graphically

When you are asked to solve $mx + c = k$

You need the graph of $y = mx + c$

And you need a horizontal line at $y = k$

② Interpret Solving an equation graphically

Solve $x + 4 = 9$

We need the line $y = x + 4$

We need the horizontal line at $y = 9$

Solve $2x + 4 = 9$

We need the line $y = 2x + 4$

We need the horizontal line at $y = 9$

Identify the two lines you need to draw to solve this problem.

a Solve $5x + 2 = 12$

We need the line

We need the horizontal line at

b Solve $x - 7 = 3$

We need the line

We need the horizontal line at

c Solve $-3x + 1 = 2$

We need the line

We need the horizontal line at

d Solve $3x + 4 = -5$

We need the line

We need the horizontal line at

e Solve $7x - 2 = 5$

We need the line

We need the horizontal line at

f Solve $6 = 4x - 3$

We need the line

We need the horizontal line at

i Solving Equations Graphically

To solve $mx + c = k$:

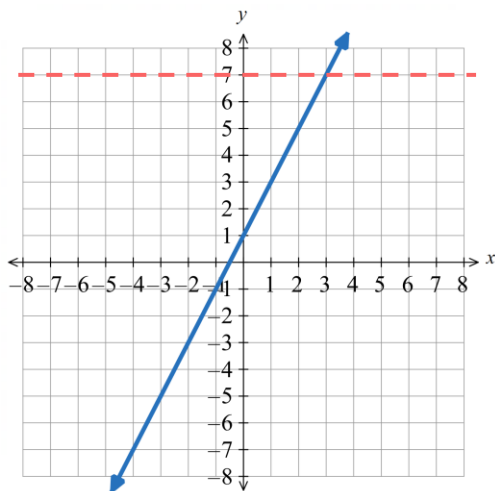
1. Draw a horizontal line from the y value indicated by the question.
2. Identify point of intersection.
3. Read off the x value.

Example Solve equations graphically

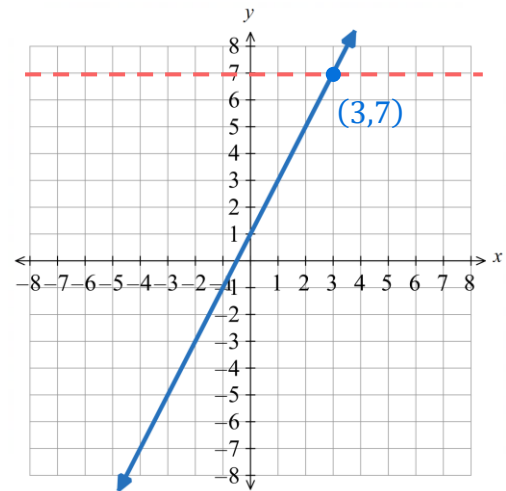
The graph of $y = 2x + 1$ is shown.

Use it to solve $7 = 2x + 1$.

1. Horizontal line



2. Point of intersection

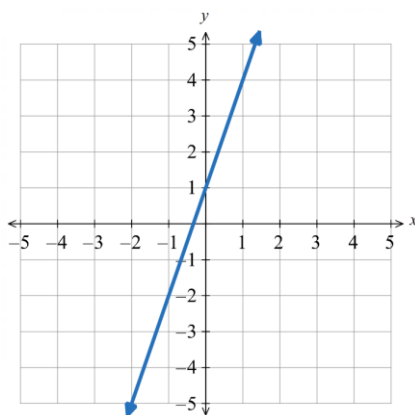


3. The solution is $x = 3$

Guided Practice

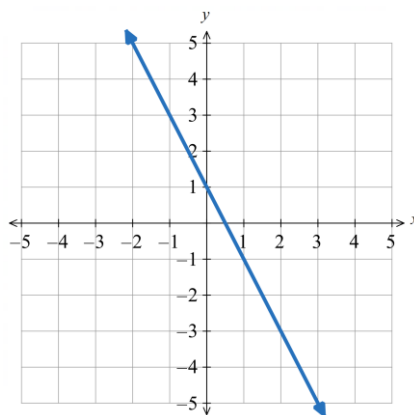
Solve using the given graph.

a $3x + 1 = 4$



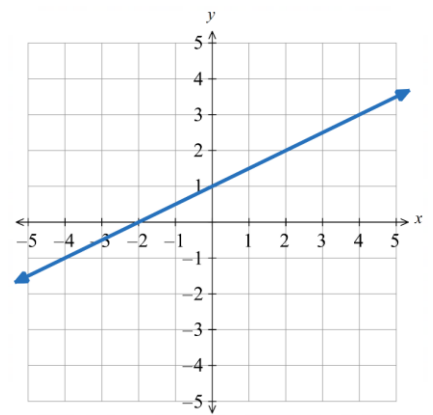
$x = 1$

b $3 = -2x + 1$



$x = -1$

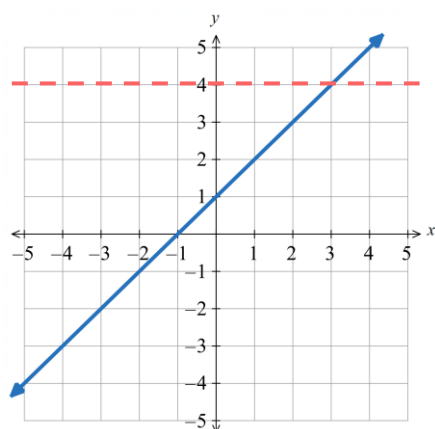
c $\frac{x}{2} + 1 = 3$



$x = 4$

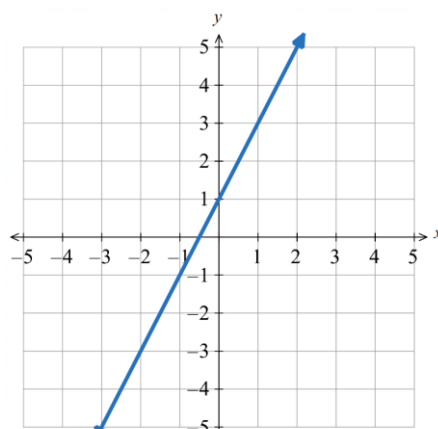
1 Solve using the given graph. Show your working.

a $x + 1 = 4$



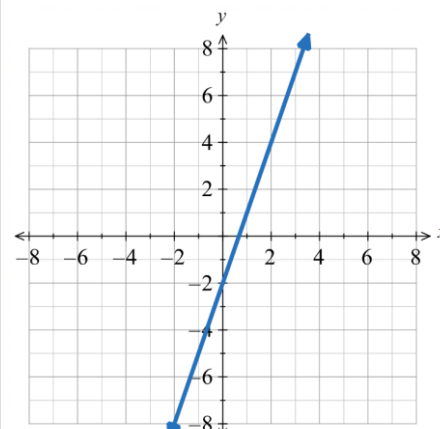
$x = 3$

b $2x + 1 = 5$



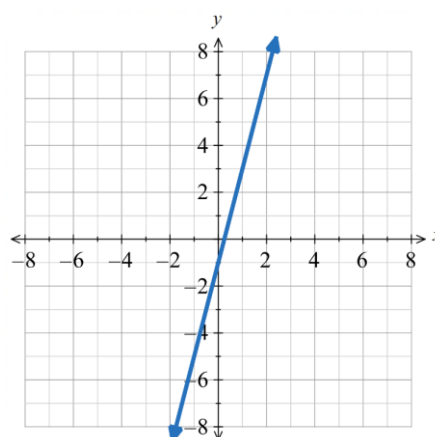
$x = 2$

c $3x - 2 = 7$



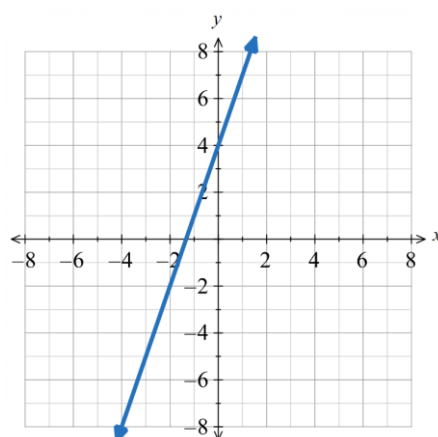
$x = 3$

d $7 = 4x - 1$



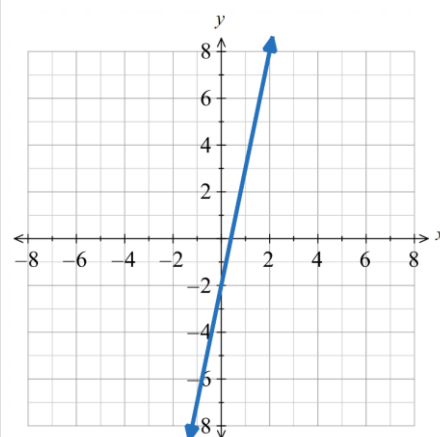
$x = 2$

e $3x + 4 = -5$



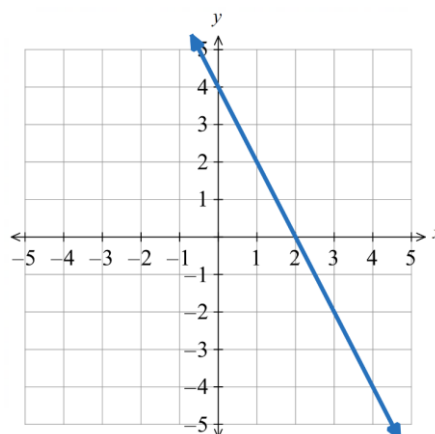
$x = -\frac{5}{3}$

f $5x - 2 = 3$



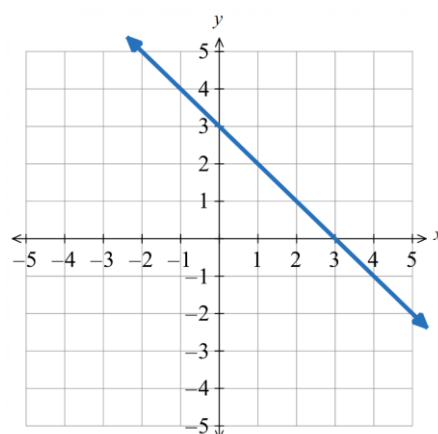
$x = 1$

g $-2x + 4 = 1$



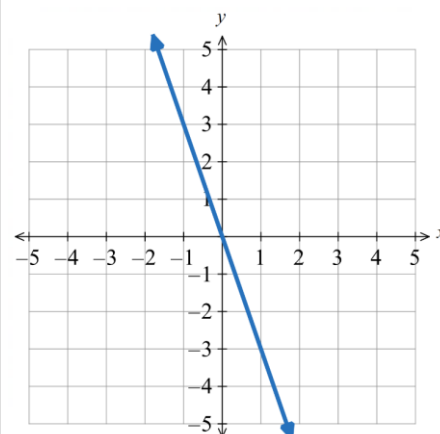
$x = 1.5$

h $0 = -x + 3$



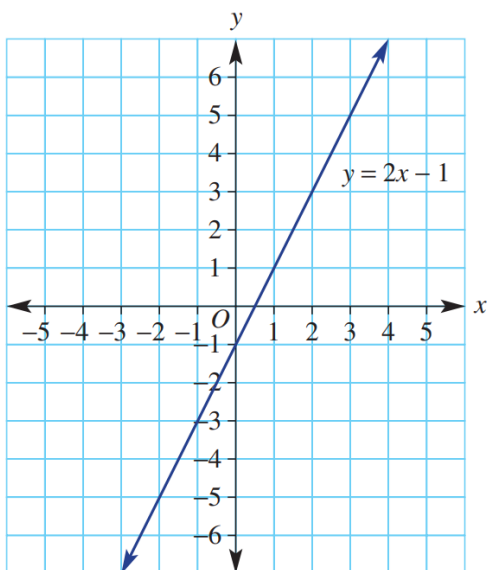
$x = 3$

i $-3x = 3$



$x = 0$

- 2 Use the graph of $y = 2x - 1$ to find the solution to each of these equations.



a $2x - 1 = 3$ $x = \dots\dots\dots$

$x = 2$

b $2x - 1 = 0$ $x = \dots\dots\dots$

$x = \frac{1}{2}$

c $2x - 1 = 5$ $x = \dots\dots\dots$

$x = 3$

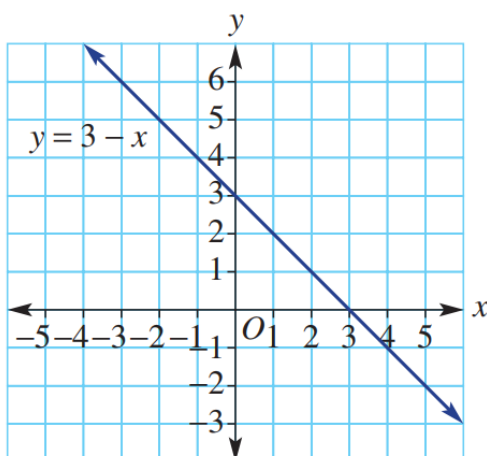
d $2x - 1 = -6$ $x = \dots\dots\dots$

$x = -\frac{5}{2}$

e $2x - 1 = -4$ $x = \dots\dots\dots$

$x = -\frac{3}{2}$

- 3 Use the graph of $y = 3 - x$, shown here, to solve each of the following equations.



a $3 - x = 6$ $x = \dots\dots\dots$

$x = -3$

b $3 - x = 0$ $x = \dots\dots\dots$

$x = 3$

c $3 - x = 3.5$ $x = \dots\dots\dots$

$x = -0.5$

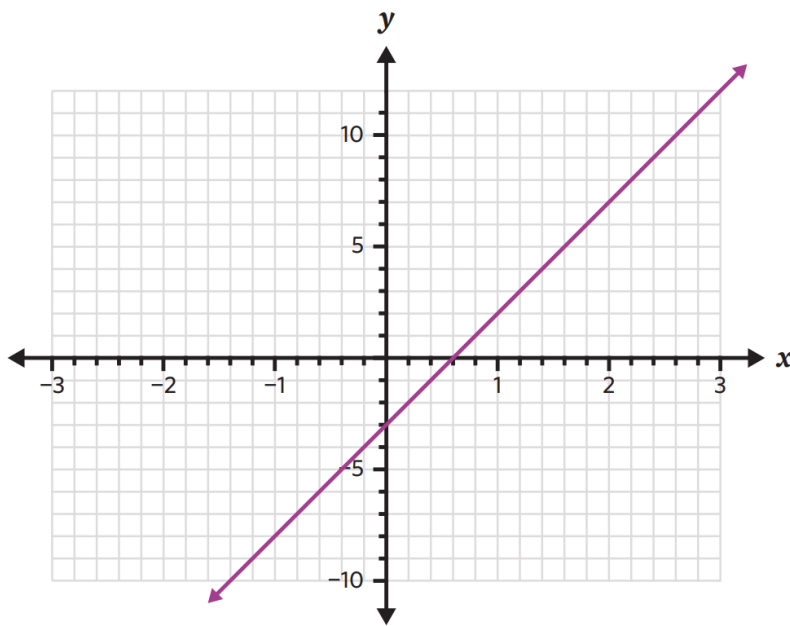
d $3 - x = -1$ $x = \dots\dots\dots$

$x = 4$

e $3 - x = -2$ $x = \dots\dots\dots$

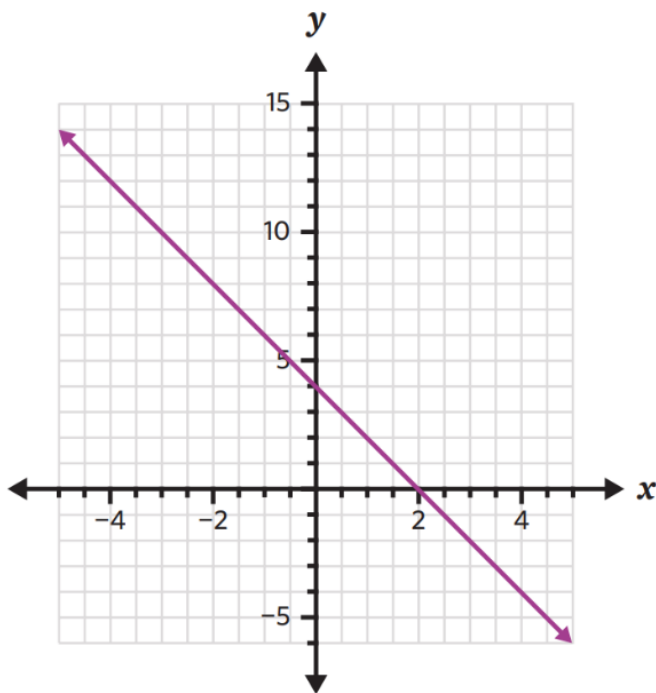
$x = 5$

4 Use the given graph of $y = 5x - 3$ to solve these equations.
Pay attention to the axis scale.



- | | | | |
|----------|---------------|-------|------------|
| a | $5x - 3 = 2$ | | $x = 1$ |
| b | $5x - 3 = 7$ | | $x = 2$ |
| c | $5x - 3 = 12$ | | $x = 3$ |
| d | $5x - 3 = -3$ | | $x = 0$ |
| e | $5x - 3 = 4$ | | $x = 1.4$ |
| f | $5x - 3 = -7$ | | $x = -0.8$ |
| g | $5x - 3 = 0$ | | $x = 0.6$ |

5 Use the given graph of $y = -2x + 4$ to solve these equations. Pay attention to the axis scale.



- | | | | |
|----------|----------------|-------|------------|
| a | $-2x + 4 = 2$ | | $x = 1$ |
| b | $-2x + 4 = 0$ | | $x = 2$ |
| c | $-2x + 4 = 3$ | | $x = 0.5$ |
| d | $-2x + 4 = -2$ | | $x = 3$ |
| e | $-2x + 4 = 6$ | | $x = -1$ |
| f | $-2x + 4 = 4$ | | $x = 0$ |
| g | $-2x + 4 = 12$ | | $x = -4$ |
| h | $-2x + 4 = 13$ | | $x = -4.5$ |

Identifying the Simultaneous Solution

? Investigation Simultaneous solution

1 Consider $y = x + 1$.

What values of x and y satisfy this equation?

x				
y				

2 Consider $y = -x + 5$.

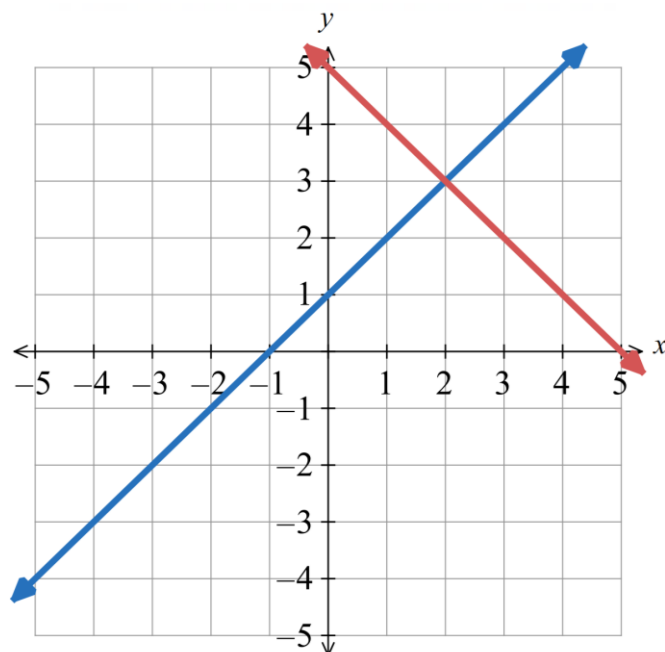
What values of x and y satisfy this equation?

x				
y				

3 What are the values of x and y that will satisfy both equations at the same time?

If we graph **both** linear equations on the same axis,
the coordinates of the **point of intersection**
satisfy both equations at the same time.

$x = 2, y = 3$



① Identifying the Simultaneous Solution

Solving equations simultaneously means finding the x and y values that make two equations true at the same time.

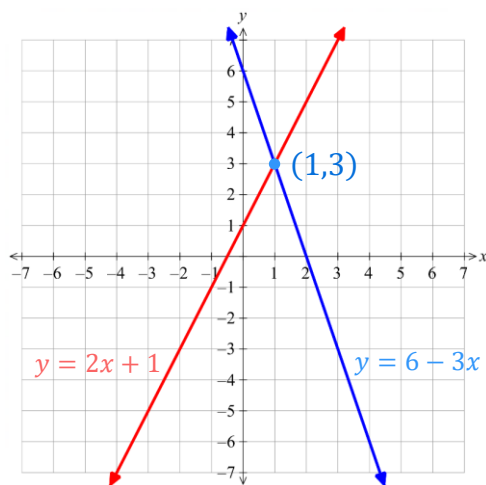
The x and y coordinates where the two lines meet solves both equations.

To find the **simultaneous solution**:

1. Identify point of intersection.
2. State the x and y values at the point of intersection.

② Identify simultaneous solution

- a** Find the values of x and y that solve $y = 6 - 3x$ and $y = 2x + 1$ simultaneously

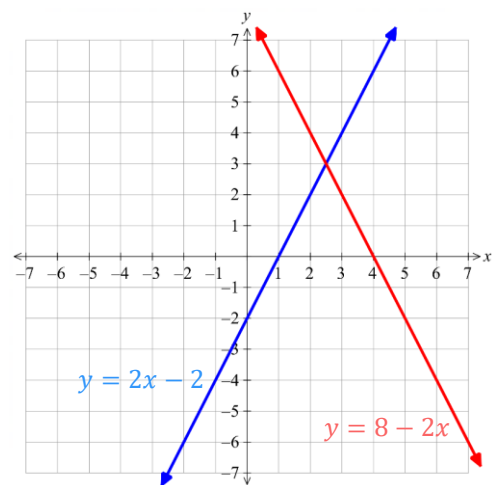


The point of intersection is:

The solution to both equations is:

$x = \dots\dots\dots y = \dots\dots\dots$

- b** Find the values of x and y that solve $y = 2x - 2$ and $y = 8 - 2x$ simultaneously



The point of intersection is:

The solution to both equations is:

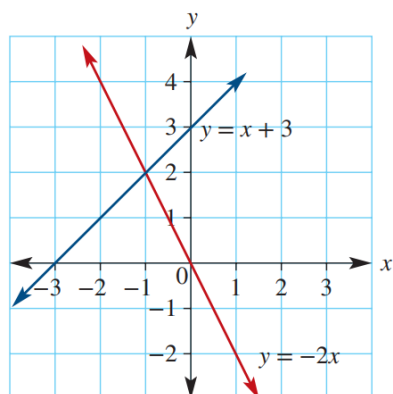
$x = \dots\dots\dots y = \dots\dots\dots$

💡 Key Ideas

1. To solve two equations simultaneously means to find the values of x and y that make both equations at the same time.
2. Every point on a line is a to the equation.
3. The point of of two lines solves both equations, as it lies on the lines

1 Determine the point of intersection and hence solve these equations simultaneously.

a



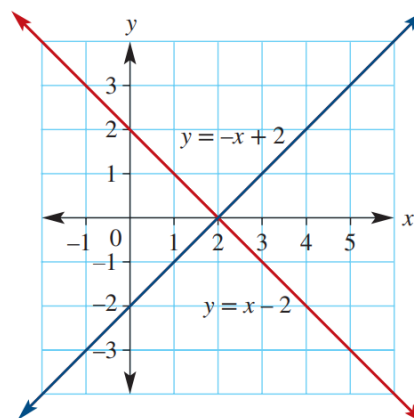
The point of intersection is:

The solution to both equations is:

$x = \dots\dots$ $y = \dots\dots$

$x = -1, y = 2$

b

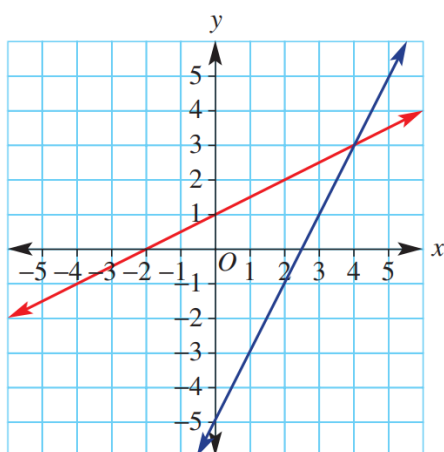


The point of intersection is:

The solution to both equations is:

$x = 2, y = 0$

c

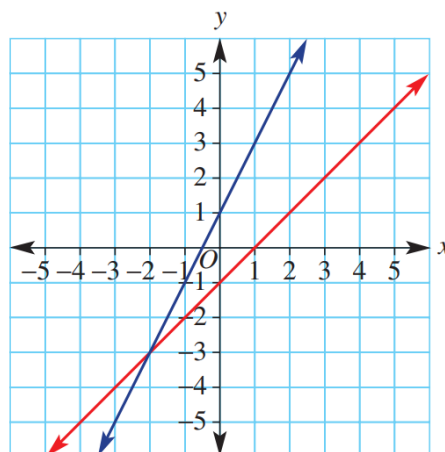


The point of intersection is:

The solution to both equations is:

$x = 4, y = 3$

d

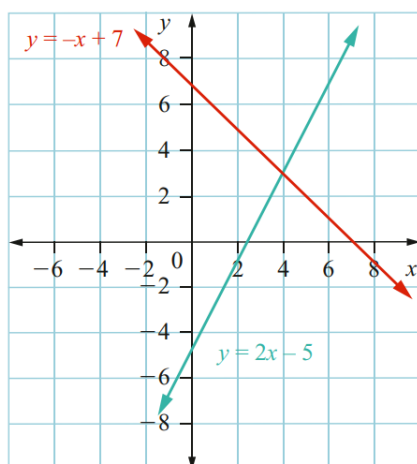


The point of intersection is:

The solution to both equations is:

$x = -2, y = 1$

e

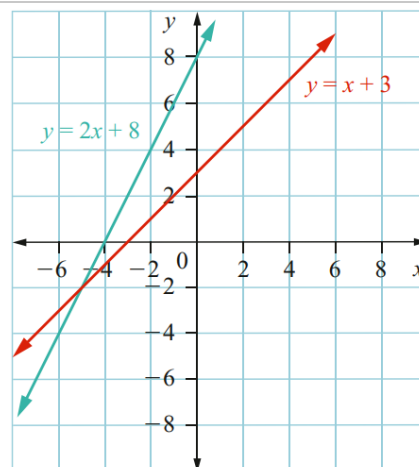


The point of intersection is:

The solution to both equations is:

$x = 4, y = 3$

f



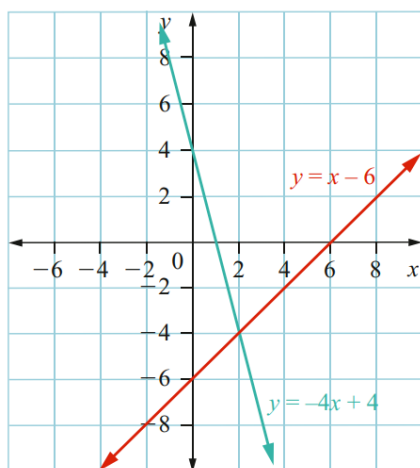
The point of intersection is:

The solution to both equations is:

$x = -5, y = -2$

2 Verify that the point of intersection solves both equations simultaneously.

a



Point of intersection is:

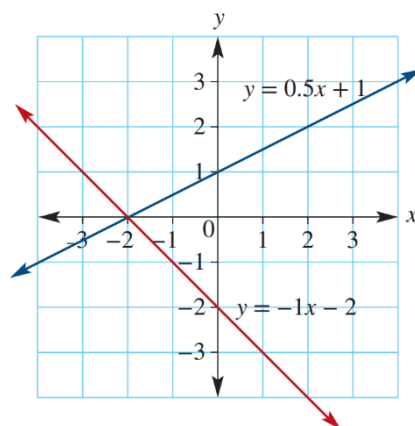
(2, -4)

Verify:

$$y = x - 6$$

$$y = -4x + 4$$

b



Point of intersection is:

(-2, 0)

Verify:

$$y = 0.5x + 1$$

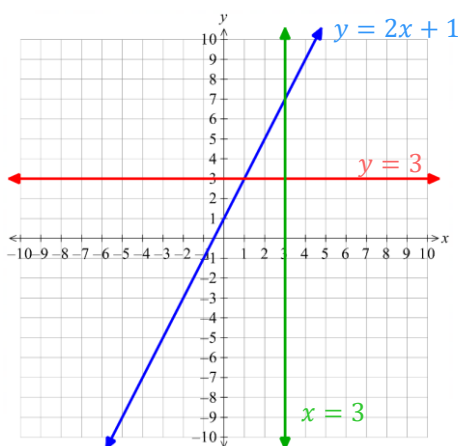
$$y = -x - 2$$

3 Here are the graphs of

$$y = 2x + 1$$

$$x = 3$$

$$y = 3$$



a Identify the coordinates of the point of intersection of $y = 2x + 1$ and $x = 3$

(3, 7)

b Substitute $x = 3$ into the equation $y = 2x + 1$ and evaluate y .

$y = 7$

c Identify the coordinates of the point of intersection of $y = 2x + 1$ and $y = 3$

(1, 3)

d Substitute $y = 3$ into the equation $y = 2x + 1$ and solve for x .

$x = 1$

4 Consider the straight lines A, B and C shown on this graph.

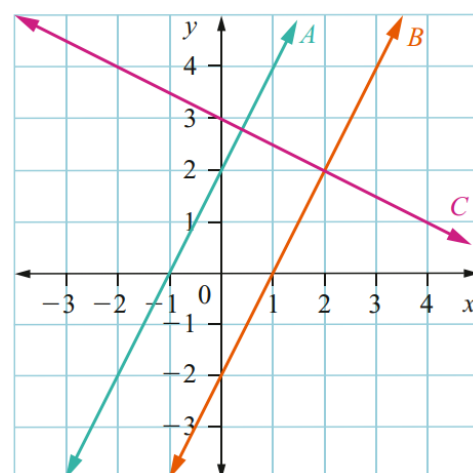
a Write the coordinates of the point of intersection of lines B and C and hence give the solution to the simultaneous equations for these lines.

b Use the graph to solve the equations for lines A and C simultaneously (give an estimate).

c Can the equations for lines A and B be solved simultaneously? Explain why or why not.

(2,2)

$\approx (0.4, 2.8)$



No, they are parallel so will never meet. There's no pair of x and y values that will solve both equations.

5 When two different lines have the same gradient, there will be no intersection point. Use this idea to decide if these pairs of simultaneous equations will have a solution.

a $y = 3x - 1$
 $y = 3x + 1$

No solution

b $y = 7x + 2$
 $y = 7x + 6$

No solution

c $y = 2x - 6$
 $y = -2x + 1$

Solution possible

d $y = -x + 7$
 $y = 2x - 1$

Solution possible

e $2x - y = 4$
 $y = 2x + 1$

No solution

f $7x - y = 1$
 $y = -7x + 2$

Solution possible

g $2y - 3x = 0$
 $y = -\frac{3x}{2} - 1$

Solution possible

h $y = \frac{x}{2} + 3$
 $\frac{x - 2y}{4} = 0$

No solution

6 For what value of k will these pairs of simultaneous equations have no solution?

a $y = -4x - 7$
 $y = kx + 2$

$k = -4$

b $y = kx + 4$
 $3x - 2y = 5$

$k = \frac{3}{2}$

c $kx - 3y = k$
 $y = 4x + 1$

$k = 12$